

Annex — Formula, Explanation & Implementation Compendium

Weather-Aware SIKD over Tropical Satellite Free-Space Optical Channels

The complete reference behind the results, the defense, and the interactive study site: every formula and coefficient, all 450 questions with answers and explanations, and the thesis annexes.

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Abstract

This annex is the exhaustive companion to the report and the defense. Part I gives the master parameter set and every governing equation with its derivation and coefficients. Part II is the 30-quantity glossary (value, typical range, rationale, cross-field comparison). Part III reproduces, in full, the **450 questions** of the interactive study site — 90 active-recall exercises with worked answers and detailed explanations, and 360 multiple-choice questions across 11 professional perspectives with answers and one-line theory frameworks. Part IV transcribes the thesis annexes: reproducibility and code structure, the full numerical-robustness detail (correlation, significance, autocorrelation, handover cost, optimality benchmark, correction history), the extended SIKD signal model, and the equatorial-season and publication notes. Sources: 05_Code_v3/modules, 10_Thesis_Shorten_v2, Tu-hoc/SIKD_Satellite_QKD, and the site's `symbols.json/exercises.json/questions.json` (<https://tatcataitn.github.io/for-ttn-USTH/Thesis-M1>).

Contents

I	Formulas, Coefficients and Derivations	3
1	Master parameter and coefficient reference	3
2	SIKD signal and FSO channel model	3
2.1	Geometric spreading loss	4
2.2	Atmospheric attenuation (Beer–Lambert)	4
2.3	Turbulence-induced fading (weak, log-normal)	4
3	Receiver: noise, sifting, QBER and secret key rate	4
4	Three-state weather model and availability	5
5	Coverage geometry and city-pair feasibility	5
6	Weather-aware scheduling and pair-key allocation	5
7	Inter-satellite-link relay and key/data power-split	6
II	Glossary of Physical Quantities	6
8	Glossary	6

III The 450 Study Questions, with Answers and Explanations	10
9 Active-recall exercises (90), by section	10
9.1 Channel physics — calculate	10
9.2 Geometry & orbits — calculate	13
9.3 Weather & availability	14
9.4 Key numbers & terms — fill in	16
9.5 Committee & literature — match	17
9.6 Reasoning, derivations & defense answers	19
10 Multiple-choice questions (360), by professional perspective	21
10.1 P1 — Quantum Physicist / QKD Theorist	21
10.2 P2 — FSO Optical-Channel Engineer	24
10.3 P3 — RF / Receiver / Detection Engineer	27
10.4 P4 — Orbital-Mechanics / Aerospace Analyst	29
10.5 P5 — Meteorologist / Climate-Data Scientist	32
10.6 P6 — Network / Scheduling / Operations Engineer	34
10.7 P7 — Statistician / Data-Quality Auditor	38
10.8 P8 — Systems Architect / Integrator	40
10.9 P9 — Thesis Examiner / Reviewer Angle	43
10.10 P10 — Applied / Operations / Policy & EDA Breadth	46
10.11 P11 — Figure, Diagram & Video Reading	47
IV Thesis Annexes (transcribed)	52
11 Reproducibility and code structure	52
11.1 Core simulation modules	53
12 Extended SIKD signal model	53
13 Numerical results: full robustness detail	53
13.1 Inter-city correlation	53
13.2 Joint availability vs. the closest prior model	54
13.3 Statistical significance and autocorrelation	54
13.4 Handover-cost characterization	54
13.5 ALG-2 optimality benchmark	54
13.6 Development history of the weather-awareness estimate	55
13.7 Physical-layer corrections	55
14 Equatorial-season robustness and publications	55
Sources	55

Part I

Formulas, Coefficients and Derivations

1 Master parameter and coefficient reference

Parameter	Symbol	Value
Wavelength	λ	1550 nm (telecom C-band)
Satellite altitude	H_S	550 km (Starlink Shell-1)
Transmit power	P_T	1 W (30 dBm)
QKD modulation index	m_K	0.15 (corrected v3 baseline)
Data modulation index	m_D	0.5
RF bandpass filter isolation	I_{so}	15 dB
Bit rate	R_b	1 Gbps ($\Delta f = R_b/2$)
Receiver aperture radius	a_R	0.05 m
Beam divergence half-angle	θ_C	10 μ rad
Effective atmospheric height	H_{atm}	20 km
Mean Earth radius	R_E	6371 km
High-altitude wind (H–V)	W	21 m/s
Ground turbulence	$C_n^2(0)$	$1.7 \times 10^{-14} \text{ m}^{-2/3}$
Rytov variance (operational ζ)	σ_R^2	0.057 to 0.203
Responsivity (InGaAs PIN)	R_e	0.9 A/W
Receiver temperature	T	280 K
Load resistance	R_L	1 k Ω
Background optical power	P_{bg}	0 W (night-time)
Rain-attenuation coefficients	α, ρ	0.509, 0.63 (Koné 2024, Abidjan)
Rain-hour fraction	$f_{rain}(c, m)$	0.047 to 0.620 (mean 0.308)
Cloud-outage threshold		85% cloud cover
Rain-hour threshold		0.1 mm/h
Reanalysis record		ERA5/ERA5-Land, 2015–2024 (10 yr), 8 cities
Elevation mask (geometric / secure)	ε	30° / 40°
BB84 security threshold	$QBER_{max}$	11%
Ground stations	$ \mathcal{G} $	8
Constellation subset	$ \mathcal{S} $	1,019 (Shell-1)
DT threshold-scale coefficient	κ	2.0
Gauss–Hermite order	N	20
Handover penalty (calibrated)	c_h	1,841,142 kbps-equivalent (30 s)
Full-enumeration window		620 days (Jan + Jul, 2015–2024)
ISL per-hop latency		30 s (conservative)

2 SIKD signal and FSO channel model

Received SCM photocurrent (direct detection):

$$i_r(t) = \frac{1}{2} R_e P_T h(t) [1 + m_K s_K(t) + m_D s_D(t)] + n(t)$$

One 1550 nm carrier multiplexes the quantum key (f_1, m_K) and classical data (f_2, m_D) by subcarrier multiplexing; both share one photodetector, so the data term s_D returns in the key receiver as cross-talk after imperfect RF filtering.

Composite channel: $h = h_g \cdot h_l \cdot h_a$ (geometric $\times$ atmospheric $\times$ turbulence), independent multiplicative factors that add in decibels.

2.1 Geometric spreading loss

$$w_0 = \frac{\lambda}{2\theta_C}, \quad w_L = w_0 \sqrt{1 + \left(\frac{L\lambda}{\pi w_0^2}\right)^2}, \quad h_g = [\text{erf}(\nu_R)]^2, \quad \nu_R = \sqrt{\frac{\pi}{2}} \frac{a_R}{w_L}$$

Spherical slant range (elevation $\varepsilon = 90^\circ - \zeta$):

$$L = \sqrt{R_E^2 \sin^2 \varepsilon + 2R_E H_S + H_S^2} - R_E \sin \varepsilon$$

Derivation: a Gaussian beam of waist w_0 spreads to radius w_L at range L ; the fraction of power captured by a circular aperture of radius a_R is $[\text{erf}(\nu_R)]^2$. v3 corrections: the ν_R coefficient $\sqrt{2} \rightarrow \sqrt{\pi/2}$ (read directly from Vu et al. 2022) and the slant range flat-Earth $H_S / \cos \zeta \rightarrow$ spherical (992.8 km, not 1100 km, at $\varepsilon = 30^\circ$). Because $w_L \approx 4\text{--}6 \text{ m} \gg a_R = 5 \text{ cm}$, $\nu_R \ll 1$, so $h_g \approx \nu_R^2 \propto 1/L^2$ and exceeds 30 dB alone — the dominant link-budget term. Worked value at 45° : $L = 749 \text{ km}$, $w_L = 4.77 \text{ m}$, $\nu_R = 0.0131$, $h_g = -36.6 \text{ dB}$.

2.2 Atmospheric attenuation (Beer–Lambert)

$$h_l = \exp[-(\sigma_{\text{clear}}(V) + \sigma_{\text{rain}}(R)) L_{\text{atm}}], \quad L_{\text{atm}} = \frac{H_{\text{atm}} - H_U}{\cos \zeta}$$

Clear air (Kruse/Kim): $\sigma_{\text{clear}} = \frac{3.912}{V} \left(\frac{\lambda}{550}\right)^{-q(V)}$, with $q(V) = 1.6$ ($V > 50$); 1.3 ($6 < V \leq 50$); $0.16V + 0.34$ ($1 < V \leq 6$); $V - 0.5$ ($0.5 < V \leq 1$); 0 else.

Rain (tropical): $\sigma_{\text{rain}} = \alpha R^\rho / 4.343$, $\alpha = 0.509$, $\rho = 0.63$; the factor $4.343 = 10 / \ln 10$ converts dB/km to Nepers/km for the exponential. Two independent extinction mechanisms add in the exponent. The wavelength factor $(\lambda/550)^{-q} < 1$ at 1550 nm means telecom light scatters less than visible. Visibility is derived $V = 15 - 10P_{\text{cloud}}$ (no ERA5 visibility variable). Worked values at $\zeta = 30^\circ$, $V = 10 \text{ km}$: clear $h_l = -10.2 \text{ dB}$; with $R = 30 \text{ mm/h}$ it drops to $\approx -110 \text{ dB}$, effectively cutting the link.

2.3 Turbulence-induced fading (weak, log-normal)

Hufnagel–Valley 5/7 profile:

$$C_n^2(h) = 0.00594 \left(\frac{W}{27}\right)^2 (10^{-5}h)^{10} e^{-h/1000} + 2.7 \times 10^{-16} e^{-h/1500} + C_n^2(0) e^{-h/100}$$

Rytov and log-amplitude variance:

$$\sigma_R^2 = 2.25 k^{7/6} \sec^{11/6}(\zeta) \int_{H_U}^{H_S} C_n^2(h) (h - H_U)^{5/6} dh, \quad \sigma_X^2 = \sigma_R^2 / 4, \quad k = 2\pi / \lambda$$

Fading: $h_a = e^{2X}$, $X \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$ so that $E[h_a] = 1$; log-normal while $\sigma_R^2 < 0.3$, else Gamma–Gamma. v3 correction: Rytov coefficient $0.56 \rightarrow 2.25$ (standard plane-wave slant-path value), which understated scintillation fourfold. The ground layer $C_n^2(0) e^{-h/100}$ decays within $\sim 500 \text{ m}$, so high-altitude wind W (jet-stream term) dominates the LEO slant-path integral; changing $C_n^2(0)$ by $17\times$ moves σ_R^2 only $\sim 2.8\%$. The operational range stays weak ($\sigma_R^2 = 0.057\text{--}0.203$), so log-normal holds throughout.

3 Receiver: noise, sifting, QBER and secret key rate

Noise budget ($\Delta f = R_b/2$):

$$\sigma_N^2 = \underbrace{2qR_e \left(\frac{1}{2} P_T h_g h_l\right) \Delta f}_{\text{shot}} + \underbrace{\frac{4k_B T}{R_L} \Delta f}_{\text{thermal}} + \underbrace{2qR_e P_{bg} \Delta f}_{\text{background}} + \underbrace{\frac{1}{8} (R_e P_T m_D h_g h_l)^2 10^{-I_{\text{so}}/10}}_{\text{cross-talk}}$$

v3 corrections: shot noise uses the total DC photocurrent (no m_K factor); the cross-talk denominator $2 \rightarrow 8$ (the photocurrent's leading $\frac{1}{2}$ makes the leaking sinusoid amplitude $\frac{1}{2}R_e P_T m_D h_g h_l$, whose variance is amplitude²/2). Cross-talk $\propto$ received power squared, so it dominates a strong link ($0.5\text{--}290\times$ thermal across elevation) — “thermal-dominated” holds only near 30° .

Dual-threshold decision ($i_{\text{mean}} = \frac{1}{2}R_e G_k P_T m_K h_g h_l$, scale $\kappa = 2$):

$$d_0 = -i_{\text{mean}} - \kappa\sigma_N, \quad d_1 = +i_{\text{mean}} + \kappa\sigma_N$$

Samples in the guard zone $[d_0, d_1]$ are discarded; wider κ lowers errors but sifts out more samples.

Sift probability and QBER (20-point Gauss–Hermite average over log-normal fading, $h_{a,k} = e^{2\sqrt{2\sigma_X^2}x_k + 2\mu_X}$):

$$P_{\text{sift}} = \frac{1}{2}(P_{00} + P_{11} + P_{10} + P_{01}), \quad \text{QBER} = \frac{\frac{1}{2}(P_{10} + P_{01})}{P_{\text{sift}}}$$

Secret key rate (Shor–Preskill / Devetak–Winter, ideal reconciliation $f_{EC} = 1$):

$$\text{SKR}_{\text{norm}} = P_{\text{sift}}[1 - (1 + f_{EC})H_2(\text{QBER}) - \chi_E] = P_{\text{sift}}[1 - 2H_2(\text{QBER}) - \chi_E], \quad \text{SKR} = \text{SKR}_{\text{norm}}R_b$$

with $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$, $= 0$ at $\text{QBER} \approx 11\%$; baseline $\chi_E = 0$. v3 correction: the earlier formula omitted the error-correction leakage $f_{EC}H_2(\text{QBER})$ and reached zero only at $\text{QBER} = 50\%$; the corrected form vanishes at the 11% BB84 threshold. Worked value at 45° : $\text{QBER} < 0.02\%$, $\text{SKR} \approx 61.4$ Mbps.

Classical data BER: $\text{BER}_{CC} = \frac{1}{\sqrt{\pi}} \sum_k w_k Q\left(\frac{A_0 h_{a,k}}{\sigma_{N,D}}\right)$, $A_0 = G_d R_e P_T m_D h_g h_l$, with reverse cross-talk $\propto m_K^2$ in $\sigma_{N,D}$.

4 Three-state weather model and availability

$$p_{\text{rain}} = \min(f_{\text{rain}}, 1 - P_{\text{cloud}}), \quad p_{\text{clear}} = \max(0, 1 - P_{\text{cloud}} - p_{\text{rain}}), \quad p_{\text{cloud}} = P_{\text{cloud}}$$

$$\text{SKR}_{\text{eff}} = p_{\text{clear}}\text{SKR}_c + p_{\text{rain}}\text{SKR}_r, \quad A = p_{\text{clear}} + p_{\text{rain}} = 1 - P_{\text{cloud}}$$

Rain rate from monthly total: $R_{\text{mm/h}} = R_{\text{month}}/(30 \cdot 24 \cdot f_{\text{rain}})$. Cloud is a binary outage; availability A is invariant to f_{rain} , so it is governed by cloud alone — spanning 10.7% (Kuala Lumpur, Nov) to 96.5% (Bangkok, Feb). Statistics: P_{cloud} = fraction of hours with cloud cover $\geq 85\%$; f_{rain} = fraction with precip ≥ 0.1 mm/h; inter-city Pearson correlation of daily cloud; joint-clear probability from the joint record. Best combined FSO window: mid-morning $\sim 05:00\text{--}10:00$ (in daylight, the day/night tension).

5 Coverage geometry and city-pair feasibility

$$R_{\text{cov}} = R_E \left[\arccos\left(\frac{R_E}{R_E + H_S} \cos \varepsilon\right) - \varepsilon \right], \quad \text{DUAL if } d \leq 2R_{\text{cov}} \text{ else store-and-forward}$$

Great-circle distance (haversine): $d = 2R_E \arcsin \sqrt{\sin^2 \frac{\Delta\varphi}{2} + \cos \varphi_1 \cos \varphi_2 \sin^2 \frac{\Delta\lambda}{2}}$.

Off-nadir angle: $\sin \eta = R_E \cos \varepsilon / (R_E + H_S)$. Slant range: $\sqrt{(R_E + H_S)^2 - (R_E \cos \varepsilon)^2} - R_E \sin \varepsilon$.

Directed store-and-forward latency: median time from a pass over i to the same satellite's next pass over j (asymmetric, heading-dependent). $R_{\text{cov}} = 793$ km (30°), 573 km (40°); DUAL threshold $2R_{\text{cov}} = 1587/1146$ km. The 28 ASEAN pairs split 14 DUAL / 14 store-and-forward — a geographic ceiling no scheduler can lift. The paper's earlier “950 km radius” was actually the slant range.

6 Weather-aware scheduling and pair-key allocation

Level 1 — station–satellite matching (Hungarian, per step t):

$$\max_x \sum_{g,s} w_{g,s}(t) x_{g,s}(t) - c_h \sum_g \mathbf{1}[\text{handover}], \quad \sum_s x_{g,s} \leq 1, \quad \sum_g x_{g,s} \leq 1, \quad x_{g,s} = 0 \text{ if not visible}$$

weight $w_{g,s} = \text{SKR}_{\text{eff}}$ (encodes the diurnal weather cycle).

Level 2 — pair key (trusted relay broadcasts $k_A \oplus k_B$):

$$K_q = \min \left(\sum_{p: \text{station}(p)=A} k_p y_{p,q}, \sum_{p: \text{station}(p)=B} k_p y_{p,q} \right), \quad k_p = w \cdot (t_{\text{set}} - t_{\text{rise}})$$

Strategies: ALG-0 (sticky elevation-priority baseline; ALG-0-feasible resolves double-booking by greedy tie-break); ALG-1 (weather-aware Hungarian matching, blind vs aware); ALG-2 (pass-to-pair allocation: greedy deficit rule, MILP/HiGHS, or max-min water-filling). Fair comparison at equal resources, full enumeration of 620 real days: matching gain +2.47/ +2.48%; weather-info gain −0.29/ −0.23%; pair-starvation 91.6–96.5%. The $\min(\cdot)$ caps the gain; a real 30 s handover is carried across segments (median segment 3.3 s).

7 Inter-satellite-link relay and key/data power-split

ISL multi-hop relay. BFS hop distance on the static “+Grid” mesh (RAAN-based plane clustering); a drop-off pass over j is feasible if $t_{\text{rise}} \geq t_{\text{pick}} + n_{\text{hops}} \cdot 30$ s. *time-optimal* = earliest feasible; *capacity-optimal* = highest-elevation (highest-SKR) drop-off within $3 \times$ the time-optimal latency. Replaces single-hop store-and-forward (worst case ~ 711 min) with 2.9–4.1 min.

Key/data power-split. Two-way cross-talk $\sigma_{\text{CT}}^2 = (R_e P_T m_{\text{leak}} h_g h_l)^2 10^{-I_{\text{so}}/10} / 8$ ($m_{\text{leak}} = m_D$ into key, m_K into data). Data goodput = $R_b(1 - \text{BER})$; constraint $m_K + m_D \leq 1$. Pareto frontier over (m_K, m_D) ; adaptive split takes the smallest m_K with $\text{SKR} \cdot 10^3 \cdot T_{\text{pass}} \geq K_{\text{req}}$. Raising m_D lifts data throughput but $\sigma_{\text{CT}}^2 \propto m_D^2$ raises key QBER — the central SIKD coupling: the two channels are not independent.

Part II

Glossary of Physical Quantities

8 Glossary

For each quantity: meaning, this project’s value, the typical range and why it is chosen, and how that range differs across telecom / satellite / quantum work versus other fields.

- **λ — Optical wavelength.** The carrier light’s wavelength; sets diffraction, atmospheric scattering, and detector response.
This project: 1550 nm (telecom C-band) throughout this project. **Typical range:** Free-space optics: ~ 800 nm, ~ 1064 nm, or 1550 nm. This project fixes 1550 nm.
Why: 1550 nm is eye-safe at higher power, sits in the atmosphere’s low-loss window, scatters less off aerosols than visible light (Kruse/Kim exponent), and reuses mature, cheap telecom lasers/detectors/fibre.
Across fields: Telecom/satellite FSO almost always pick 1550 nm for those reasons. General optics/astronomy span UV→IR freely; lidar often uses 532/1064 nm. Quantum labs sometimes prefer shorter wavelengths where single-photon detectors are most efficient — a tension this direct-detection design sidesteps.
- **θ_{C} — Beam divergence half-angle.** How fast the transmitted laser beam spreads; smaller means a tighter, farther-reaching beam.
This project: 10 μrad . **Typical range:** Satellite FSO: ~ 1 –50 μrad .
Why: 10 μrad keeps geometric spreading loss manageable at 550 km, but a tighter beam demands finer pointing (PAT), so it trades against pointing budget.
Across fields: Satellite laser-comm lives in the microradian regime because links are thousands of km. RF antennas think in degrees (milliradians+); a laser pointer is ~ 1 mrad — $100\times$ wider than this.
- **w_0 — Beam waist.** The beam’s narrowest radius, at the transmitter: $w_0 = \lambda/(2\theta_{\text{C}})$.
This project: ≈ 0.0775 m (7.75 cm). **Typical range:** Centimetre-scale for these parameters.

Why: Fixed by λ and θ_C ; a smaller divergence gives a larger waist and slower far-field spreading.

Across fields: A derived quantity specific to Gaussian-beam optics; RF engineering has no direct analogue.

- **w_L — Beam radius at the receiver.** The spot radius after propagating the slant range L; grows ~linearly with distance.
This project: ≈ 3.5 m (zenith) to ≈ 6.3 m (30° elevation). **Typical range:** Metres, at LEO ranges.
Why: Because w_L (metres) $\gg$ aperture a_R (5 cm), only $\sim(a_R/w_L)^2$ of the power is collected — the origin of the >30 dB geometric loss.
Across fields: This metre-scale spot over a centimetre aperture is why FSO link budgets are dominated by geometry, unlike fibre (no spreading) or RF (much wider beams but larger antennas).
- **a_R — Receiver aperture radius.** Radius of the ground telescope collecting the beam.
This project: 5 cm (10 cm diameter). **Typical range:** Satellite FSO ground optics: a few cm (CubeSat-class, e.g. TBIRD ~ 0.2 m) up to ~ 1 m (Micius ground station).
Why: A small aperture is cheap/light and lets you deploy many stations, but collects a smaller fraction of the already-weak key signal — an unresolved SWaP trade-off here.
Across fields: Astronomy uses metre-to-10-metre apertures for faint light; this project deliberately picks the small, cheap end to argue a multi-station network.
- **h_g — Geometric spreading loss.** Fraction of transmitted power reaching the aperture, from beam spreading: $h_g = [\text{erf}(v_R)]^2$.
This project: ≈ -34 dB (zenith) to ≈ -39 dB (30° elevation). **Typical range:** Tens of dB (very lossy) for LEO FSO.
Why: It scales $\sim 1/L^2$, so lower passes (longer range) are much worse; it is the single largest term in the link budget.
Across fields: Uniquely severe in free-space optics; fibre has none, and RF free-space path loss, while large, is offset by high antenna gains.
- **h_l — Atmospheric transmission.** Beer-Lambert transmission through the air: $h_l = \exp(-\sigma_{\text{total}} \cdot L_{\text{atm}})$.
This project: ≈ -9 dB (clear, zenith) to ≈ -18 dB (clear, 30°); ~ -100 dB in heavy rain. **Typical range:** A few dB (clear) to link-killing (rain/fog).
Why: Set by visibility and rain rate along the slant path; low elevation lengthens the path ($1/\cos \zeta$).
Across fields: Remote sensing / atmospheric science model exactly this extinction. In this project cloud is treated as a separate binary outage, not folded into h_l .
- **σ_{clear} — Clear-air extinction coefficient.** Per-km attenuation from aerosols/molecules (Kruse/Kim): $\sigma = (3.912/V)(\lambda/550)^{-q}$.
This project: $\approx 0.10 \text{ km}^{-1}$ at $V = 10 \text{ km}$, 1550 nm . **Typical range:** $\sim 0.05 \text{ km}^{-1}$ (very clear) to several km^{-1} (fog).
Why: Inversely tied to meteorological visibility V ; the wavelength factor makes 1550 nm scatter less than visible.
Across fields: A staple of atmospheric optics / lidar; identical machinery to remote-sensing atmospheric correction.
- **σ_{rain} — Rain extinction coefficient.** Per-km attenuation from raindrops: $\beta = \alpha R^\rho$ (dB/km), $\sigma_{\text{rain}} = \beta/4.343$.
This project: $\alpha = 0.509$, $\rho = 0.63$ (tropical, Koné 2024, Abidjan). **Typical range:** 0 (no rain) to $\sim 1.4 \text{ km}^{-1}$ at 50 mm/h .
Why: The sub-linear exponent ($\rho < 1$) reflects Mie scattering by large drops; calibrated on real tropical measurements, not textbook constants.
Across fields: Tropical FSO needs region-specific coefficients; temperate ITU rain models differ. RF rain fade (Ka-band) uses a related but distinct power law.
- **R — Rain rate.** Instantaneous rainfall intensity during raining hours.
This project: Converted from monthly totals: $R = R_{\text{month}}/(30 \cdot 24 \cdot f_{\text{rain}})$. **Typical range:** Light drizzle $\sim 0.1 \text{ mm/h}$ to tropical downpour $30\text{--}50+$ mm/h.
Why: Tropical rain is intermittent, so monthly totals must be concentrated into the hours it actually rains (f_{rain}).
Across fields: Meteorology/hydrology quantity; FSO and Ka-band RF both care because both suffer rain fade, unlike lower RF bands.
- **V — Meteorological visibility.** Horizontal distance at which contrast drops to a threshold; a proxy for

aerosol load.

This project: Derived here as $V = 15 - 10 \cdot P_{\text{cloud}}$ (no historical visibility record exists). **Typical range:** ≥ 10 –50 km (clear) down to < 1 km (fog).

Why: It directly sets σ_{clear} ; a derived estimate is used because ERA5 has no visibility variable.

Across fields: A standard weather-station output; here it is inferred, a stated modelling limitation.

- **H_S — Satellite altitude.** Height of the LEO satellite above the ground.

This project: 550 km (Starlink Shell-1). **Typical range:** LEO ~ 300 –2000 km; this project fixes 550 km.

Why: The channel model is parameterised for one altitude; mixing shells would need altitude-dependent tables.

Across fields: 550 km is the modern mega-constellation sweet spot (low latency, manageable drag). GEO comms sit at 35 786 km; the ISS at ~ 400 km.

- **ζ / ϵ — Zenith / elevation angle.** ζ is measured from straight up; elevation $\epsilon = 90^\circ - \zeta$ is measured from the horizon.

This project: Operating range $\epsilon = 30^\circ$ – 90° ($\zeta = 0^\circ$ – 60°). **Typical range:** Elevation masks of 20° – 40° are common.

Why: Low elevation lengthens both the geometric range and the atmospheric path; a 30° (or safer 40°) mask bounds the loss and QBER.

Across fields: Universal to any ground-to-space link (comms, GNSS, radio astronomy). GNSS often accepts 5 – 15° ; QKD wants higher for signal quality.

- **R_{cov} — Ground-coverage radius.** Radius on the ground within which a station can see the satellite above the mask.

This project: 793 km at 30° , 573 km at 40° . **Typical range:** Hundreds of km for LEO.

Why: Sets which city pairs (within $2 \cdot R_{\text{cov}}$) can share one pass — the DUAL/SF split.

Across fields: A coverage-geometry staple; the same footprint idea underlies cellular NTN planning.

- **σ_{R^2} — Rytov variance (scintillation).** Strength of turbulence-induced intensity fluctuations along the path.

This project: < 0.3 across the operating range (weak turbulence). **Typical range:** < 0.3 weak (log-normal fading) ; $\gtrsim 1$ strong (Gamma-Gamma).

Why: The slant path to LEO stays weak, so log-normal fading applies; the v_3 coefficient is 2.25 (plane-wave slant path).

Across fields: Central to astronomy's 'seeing' and adaptive optics, and to FSO. Fibre and RF do not have optical scintillation.

- **C_{n²} — Refractive-index structure parameter.** Local turbulence strength versus altitude (Hufnagel-Valley profile).

This project: Ground value $C_n^2(0) = 1.7 \times 10^{-14} \text{ m}^{-2/3}$ (moderate). **Typical range:** Weak 10^{-17} · moderate $\sim 10^{-14}$ · strong $10^{-13} \text{ m}^{-2/3}$.

Why: The ground layer decays within ~ 500 m, so high-altitude wind (not $C_n^2(0)$) dominates the LEO slant-path integral.

Across fields: A term shared by astronomy (site seeing) and FSO; its exotic $\text{m}^{-2/3}$ units come from Kolmogorov turbulence theory.

- **P_T — Transmit optical power.** Average optical power leaving the transmitter.

This project: 1 W (30 dBm). **Typical range:** Satellite laser-comm: ~ 0.1 –5 W.

Why: 1 W is a realistic small-payload figure; higher power eases the link budget but costs SWaP and raises cross-talk (which grows as P^2).

Across fields: Quantum single-photon QKD works at the femtowatt/photon level; this direct-detection SIKD design instead uses a strong classical-style carrier, which is why power is in watts, not photons.

- **m_K — Key-channel modulation index.** Depth of the quantum-key subcarrier's modulation.

This project: 0.15 (corrected from 0.05 in v_3). **Typical range:** ~ 0.1 –0.2 feasible at 15 dB filter isolation.

Why: The source SIKD manuscript's feasible region opens for $m_K \gtrsim 0.1$; 0.15 drops QBER below even the 10^{-3} target while leaving budget for data.

Across fields: A subcarrier-multiplexing parameter specific to this SIKD architecture; classical SCM systems tune it purely for SNR, here it also trades against the key's security.

- m_D — Data-channel modulation index.** Depth of the classical data subcarrier's modulation.
This project: 0.5. **Typical range:** 0.5–0.9 typical; constrained by $m_K + m_D \leq 1$.
Why: Higher m_D raises data throughput but leaks more cross-talk into the key receiver ($\sigma_{CT^2} \propto m_D^2$) — the core SIKD tension.
Across fields: In plain SCM you would push m_D high for capacity; here it is capped because it directly attacks the quantum channel's QBER.
- Iso — Filter isolation.** How well the RF/optical filter separates the key and data subcarriers.
This project: 15 dB. **Typical range:** ~10–30 dB for practical filters.
Why: Finite isolation lets the data channel leak in as cross-talk; better isolation directly lowers QBER ($\sigma_{CT^2} \propto 10^{-(Iso/10)}$).
Across fields: A filter-engineering figure; its impact here is amplified because the leaked signal lands on a security-critical quantum channel.
- R_b — Raw bit rate.** Symbol/bit rate of the link; sets the noise bandwidth $\Delta f = R_b/2$.
This project: 1 Gbps. **Typical range:** Satellite FSO downlinks: Mbps to hundreds of Gbps (TBIRD hit 200 Gbps).
Why: It scales the absolute secret key rate ($SKR_{bps} = SKR_{norm} \cdot R_b$) and the receiver noise bandwidth.
Across fields: Comms-wide quantity; FSO reaches far higher rates than RF thanks to optical bandwidth.
- σ_N — Receiver noise (std. dev.).** Total electrical noise: shot + thermal + background + cross-talk.
This project: Sum of four variances; cross-talk dominates when the link is strong. **Typical range:** Set by detector, temperature, bandwidth, and link strength.
Why: It sets the decision thresholds and hence QBER; the v3 cross-talk fix (/8) cut the dominant term four-fold.
Across fields: Every receiver has shot+thermal noise; the cross-talk term is special to this shared-detector SIKD design and, unusually, grows with signal power ($\propto P^2$).
- σ_{CT^2} — Cross-talk noise variance.** Data-channel power leaking into the key receiver: $\propto (R_e P_T m_D h_g h_l)^2 \cdot 10^{-(Iso/10)}/8$.
This project: Ranges from ~0.5× to ~290× thermal noise across elevation. **Typical range:** Grows as received power squared (P^2).
Why: Because it scales as P^2 , it dominates exactly when the link is strongest (high elevation) — a counter-intuitive, defining SIKD result.
Across fields: Ordinary noise falls or is flat with signal; this term rising with signal is the signature quirk of multiplexing data and key on one detector.
- QBER — Quantum bit error rate.** Fraction of sifted key bits where Bob disagrees with Alice; the master security indicator.
This project: < 0.02% across 30°–90° (v3, $m_K = 0.15$). **Typical range:** Must stay below 11% (BB84) to distil any secret key; stricter protocols want $\leq 10^{-3}$.
Why: Eavesdropping and noise both raise QBER; below threshold a positive secret key survives after error correction and privacy amplification.
Across fields: A QKD-specific metric with a hard security threshold — unlike a classical BER, where 10^{-3} is merely 'poor', here 11% is a wall.
- SKR — Secret key rate.** Rate of provably-secret key bits: $SKR_{norm} = P_{sift}[1 - 2H_2(QBER) - \chi_E]$.
This project: ≈ 61.4 Mbps at 45° elevation (v3). **Typical range:** Zero at QBER $\approx 11\%$; up to tens of Mbps here.
Why: The mutual information minus error-correction leakage minus Eve's information; it collapses steeply near the QBER threshold.
Across fields: The headline output of any QKD system. Real satellite QKD (Micius) delivered kbps; this direct-detection design models Mbps because it uses a strong carrier, not single photons.
- P_{sift} — Sift probability.** Probability a received sample is confidently decoded (falls outside the dual-threshold guard zone).
This project: Averaged over turbulence by 20-point Gauss-Hermite quadrature. **Typical range:** 0 to 0.5 (half the samples are the key basis).
Why: A wider guard zone lowers errors but sifts out more samples — a yield-vs-error trade set by the threshold scale

$\zeta_{\text{scale}} = 2$.

Across fields: Sifting is intrinsic to QKD (basis reconciliation / thresholding); no classical-comms analogue.

- **χ_E — Eve’s Holevo information.** The maximum information an eavesdropper can hold about the key.
This project: 0 in the baseline (no eavesdropping assumed). **Typical range:** 0 (no attack) up to values that zero the SKR.
Why: It is what privacy amplification must remove; setting it to 0 characterises performance, not adversarial security.
Across fields: A quantum-information quantity (Holevo bound); has no classical counterpart.
- **H_2 — Binary entropy function.** $H_2(p) = -p \cdot \log_2 p - (1-p) \cdot \log_2 (1-p)$; the information cost of a bit-flip probability p .
This project: $H_2(0.11) \approx 0.4999$. **Typical range:** 0 (at $p=0$ or 1) to 1 (at $p=0.5$).
Why: It appears twice in the SKR bound (mutual information and error-correction leakage), which is why SKR hits zero at QBER $\approx 11\%$.
Across fields: A Shannon-information staple used across coding theory; here it directly sets the QKD security threshold.
- **P_{cloud} / A — Cloud probability & availability.** P_{cloud} is the fraction of hours with thick cloud (link off); availability $A = 1 - P_{\text{cloud}}$.
This project: A ranges 10.7% (Kuala Lumpur, Nov) to 96.5% (Bangkok, Feb). **Typical range:** Wildly location/season dependent in the tropics.
Why: Thick cloud blocks the optical beam entirely, so availability is governed by cloud, not rain; A is invariant to the rain fraction.
Across fields: Optical links treat cloud as total blockage; RF satcom passes through cloud almost unaffected — the fundamental reason tropical optical QKD is hard.
- **c_h — Handover / acquisition cost.** Time lost re-acquiring a satellite when a station switches links.
This project: 30 s charged per handover. **Typical range:** Real FSO acquisition: $\sim 10\text{--}30$ s.
Why: Every re-assignment loses delivered key; charging it fairly is why weather-aware scheduling shows a small net cost.
Across fields: FSO needs seconds to re-acquire a narrow beam; RF beam-steering is far faster, so this penalty is an optical-link-specific concern.

Part III

The 450 Study Questions, with Answers and Explanations

This part reproduces, in full, every question implemented on the interactive study site: the 90 active-recall exercises (Section) and the 360 multiple-choice questions (Section). Numeric answers were computed from the formulas of Part I; the explanations are the same theory frameworks shown on the site.

9 Active-recall exercises (90), by section

9.1 Channel physics — calculate

E1 [calculate] Beam waist $w_0 = \lambda / (2 \cdot \theta_C)$, $\lambda = 1550$ nm, $\theta_C = 10$ urad. Compute w_0 in metres.

Answer: 0.0775 m

$w_0 = 1550 \times 10^{-9} / (2 \times 10^{-6}) = 0.0775$ m.

In detail: Method: the beam waist is fixed purely by wavelength and divergence, $w_0 = \lambda / (2 \cdot \theta_C) = 1550 \times 10^{-9} / (2 \times 10^{-6}) = 0.0775$ m. Insight: a SMALLER divergence gives a LARGER waist, which then spreads more slowly in the far field — so w_0 alone hints at how tight the beam stays. Connection: this is the very first number in the whole channel chain; everything downstream (w_L , ν_R , h_g)

is built on it.

E2 [calculate] Spherical slant range at elevation 30 deg (zeta=60), $H_S=550$ km, $R_E=6371$ km. Give L in km.

Answer: 992.778383 km

eps=30: $L = 992.8$ km (NOT the flat-Earth $H_S/\cos(\text{zeta})=1100$ km).

In detail: Method: use the spherical-Earth law $L = \sqrt{R_E^2 \sin^2(e) + 2 R_E H_S + H_S^2} - R_E \sin(e)$ with $e = 30$ deg, giving 992.8 km. Why it matters: the pre-v3 code used the flat-Earth $L = H_S/\cos(\text{zeta}) = 1100$ km — an 11% error that fed straight into wL and hg . Insight: at low elevation the curvature of the Earth genuinely shortens the geometric path versus the naive flat model, so the corrected loss is a little smaller.

E3 [calculate] Slant range at elevation 90 deg (satellite at zenith). Give L in km.

Answer: 550.0 km

eps=90: $L = H_S = 550$ km.

In detail: Method: at the zenith (elevation 90 deg) the satellite is directly overhead, so the slant range collapses to the plain altitude, $L = H_S = 550$ km. This is the best-case geometry: shortest path, smallest spot, lowest loss. Use it as a sanity anchor — any slant-range formula MUST return 550 km here.

E4 [calculate] Beam radius wL at the receiver for elevation 45 deg. Give wL in metres.

Answer: 4.769604 m

$L(45)=749$ km $\rightarrow wL = 4.77$ m.

In detail: Method: $wL = w_0 \sqrt{1 + (L \lambda / (\pi w_0^2))^2}$; with $L = 749$ km at 45 deg elevation this gives ~ 4.77 m. Insight: because the argument is $\gg 1$, the square root is nearly linear in L , so the spot grows roughly proportionally to distance. Consequence: a 4.77 m spot over a 5 cm aperture is why almost none of the transmitted power is captured.

E5 [calculate] Corrected normalized aperture $\text{nu}_R = \sqrt{\pi/2} a_R / wL$ at elevation 90 deg ($a_R=5$ cm). Give nu_R .

Answer: 0.017893 (dimensionless)

$\text{nu}_R = 1.2533 \cdot 0.05 / 3.502 = 0.0179$.

In detail: Method: $\text{nu}_R = \sqrt{\pi/2} a_R / wL = 1.2533 \cdot 0.05 / 3.502 = 0.0179$ at the zenith. The v3 fix replaced the old $\sqrt{2}$ coefficient with $\sqrt{\pi/2}$, read directly from Vu et al. 2022; the old value inflated hg by $(1.4142/1.2533)^2 \sim 1.27\times$, i.e. +1.05 dB of imaginary gain. $\text{nu}_R \ll 1$ tells you the aperture captures only the tiny central tip of a huge Gaussian spot.

E6 [calculate] Geometric loss hg at elevation 45 deg, in dB (v3 corrected).

Answer: -36.580447 dB

$hg(45 \text{ deg elev}) = -36.6$ dB.

In detail: Method: $hg = [\text{erf}(\text{nu}_R)]^2$; with nu_R tiny, $\text{erf}(x) \sim (2/\sqrt{\pi}) x$, so $hg \sim \text{nu}_R^2 \sim 1/wL^2 \sim 1/L^2$. Numerically $hg(45 \text{ deg}) = -36.6$ dB. Insight: geometric spreading alone costs over 30 dB — it is the single largest term in the link budget, dwarfing atmosphere and turbulence. Design levers are aperture ($hg \sim a_R^2$) and divergence.

E7 [calculate] Clear-air extinction σ_{clear} at $V=10$ km, 1550 nm (Kruse/Kim, $q=1.3$). Give km^{-1} .

Answer: 0.101728 km^{-1}

$q(10)=1.3 \rightarrow \sigma_{\text{clear}}=0.1017 \text{ km}^{-1}$.

In detail: Method: Kruse/Kim gives $\sigma_{\text{clear}} = (3.912/V)(\lambda/550)^{-q(V)}$; at $V = 10$ km, $q = 1.3$, so $\sigma_{\text{clear}} = 0.1017 \text{ km}^{-1}$. The wavelength factor $(\lambda/550)^{-1.3}$ is < 1 for 1550 nm, meaning telecom light scatters LESS off aerosols than visible light — a quiet advantage of the C-band choice. This is the same extinction machinery a remote-sensing atmospheric correction uses.

E8 [calculate] Rain extinction σ_{rain} at $R=30$ mm/h (tropical Kone 2024: $\beta=0.509 R^{0.63}$, $\sigma_{\text{rain}}=\beta/4.343$). km^{-1} .

Answer: 0.998882 km^{-1}

$\beta=0.509 \cdot 30^{0.63}=4.34$ dB/km; $\sigma_{\text{rain}}=0.999 \text{ km}^{-1}$.

In detail: Method: tropical rain uses $\beta = \alpha R^\rho$ dB/km with $\alpha = 0.509$, $\rho = 0.63$ (Kone 2024, Abidjan), then $\sigma_{\text{rain}} = \beta/4.343$; at $R = 30$ mm/h this is $\sim 1.0 \text{ km}^{-1}$. Insight: $\rho < 1$ makes attenuation SUB-linear — 5x more rain gives only $5^{0.63} \sim 2.8\times$ more loss, a Mie-scattering signature of large drops. The $4.343 = 10/\ln(10)$ converts dB to natural-log (Nepers) so it fits the exp() in Beer-Lambert.

E9 [calculate] Atmospheric transmission hl in dB at zeta=30 deg, V=10 km, clear (R=0). $L_{\text{atm}}=H_{\text{atm}}/\cos(\text{zeta})$.

Answer: -10.202881 dB

$L_{\text{atm}}=23.09$ km; $hl=\exp(-0.1017*23.09)=0.0954 \rightarrow -10.2$ dB.

In detail: Method: $L_{\text{atm}} = H_{\text{atm}}/\cos(\text{zeta}) = 20/\cos(30) = 23.09$ km; $hl = \exp(-\sigma_{\text{total}}*L_{\text{atm}}) = \exp(-0.1017*23.09) = 0.0954$, i.e. -10.2 dB. This is a clear-sky, moderate-visibility loss — modest next to the >30 dB of geometric spreading. Note the $1/\cos(\text{zeta})$: every degree lower in elevation lengthens the air path and compounds the loss.

E10 [calculate] hl in dB at zeta=30 deg, V=10 km, WITH heavy rain R=30 mm/h. Show how much rain adds.

Answer: -110.3868 dB

$\sigma_{\text{total}}=1.101$; $hl=\exp(-1.101*23.09) \rightarrow \sim -110$ dB.

In detail: Method: add the rain term — $\sigma_{\text{total}} = 0.1017 + 0.999 = 1.101 \text{ km}^{-1}$, so $hl = \exp(-1.101*23.09) \sim -110$ dB. That is ~100 dB WORSE than clear sky: heavy rain effectively severs the optical link. This is exactly why the weather model treats heavy precipitation and thick cloud as near-outage rather than a gentle fade.

E11 [calculate] Atmospheric slant path L_{atm} at zeta=60 deg (elevation 30), $H_{\text{atm}}=20$ km. Give km.

Answer: 40.0 km

$L_{\text{atm}}=20/\cos 60=40$ km (double the zenith value).

In detail: Method: $L_{\text{atm}} = (H_{\text{atm}} - H_U)/\cos(\text{zeta}) = 20/\cos(60) = 40$ km at 30 deg elevation, double the 20 km zenith path. The $1/\cos(\text{zeta})$ geometry is the reason low passes suffer far more attenuation AND more turbulence. It is also why a higher elevation mask (40 deg) trades coverage for a cleaner channel.

E12 [calculate] Binary entropy $H_2(0.11)$. Give the value (bits).

Answer: 0.499916 bits

$H_2(0.11)=0.4999$.

In detail: Method: $H_2(0.11) = -0.11 \log_2(0.11) - 0.89 \log_2(0.89) = 0.4999$. Why this number matters: it is almost exactly 0.5, so $1 - 2*H_2(0.11) \sim 0$ — the reason the secret-key rate vanishes right at QBER = 11%. Binary entropy is the information cost of a bit-flip probability; here it doubles as the security cliff.

E13 [calculate] Normalized SKR = $\text{Psift}*[1-2*H_2(\text{QBER})]$ at $\text{Psift}=0.5$, $\text{QBER}=0.0002$ (v3, $\text{fEC}=1$, $\text{chiE}=0$). Give bit/s/Hz.

Answer: 0.497254 bit/s/Hz

$H_2(2e-4) \sim 0.0027$; $1-2*0.0027=0.9946$; $*0.5=0.497$.

In detail: Method: the v3 secret-key rate is $\text{SKR}_{\text{norm}} = \text{Psift}*[1 - (1+\text{fEC})H_2(Q) - \text{chiE}]$; at $\text{fEC} = 1$, $\text{chiE} = 0$ it becomes $\text{Psift}*[1 - 2*H_2(Q)]$. With $Q = 0.0002$, $H_2 \sim 0.0027$, so $1 - 2*0.0027 \sim 0.9946$, times $\text{Psift} = 0.5$ gives ~ 0.497 . Insight: at the corrected sub-0.02% QBER almost every sifted bit becomes secret key — the opposite of the pre-v3 story where low elevations breached threshold.

E14 [calculate] Absolute SKR (Mbps) = $\text{SKR}_{\text{norm}} * R_b$, with $\text{SKR}_{\text{norm}}=0.497$ and $R_b=1$ Gbps. Give Mbps.

Answer: 497.0 Mbps

$0.497*1000 = 497$ Mbps (illustrative; table value 61.4 uses the real Psift/QBER).

In detail: Method: the normalized rate (bit/s/Hz) is scaled by the raw bit rate: $\text{SKR}_{\text{bps}} = \text{SKR}_{\text{norm}}*R_b = 0.497*1e9 = 497$ Mbps (illustrative). The real 61.4 Mbps table value uses the true Psift and QBER at 45 deg, but the scaling idea is identical. This is how an abstract information-theoretic fraction becomes an engineering throughput.

E15 [calculate] Shot-noise DC signal power $P = 0.5*P_T*hg*hl$ (v3, NO m_K), $P_T=1$ W, $hg=3.9e-4$, $hl=0.0954$. Give W.

Answer: 1.9e-05 W

$0.5*1*3.9e-4*0.0954 = 1.86e-5$ W.

In detail: Method: shot noise is driven by the TOTAL DC photocurrent, $P = 0.5*P_T*hg*hl$ (NOT scaled by m_K). The v3 fix removed a stray $m_K = 0.05$ factor that had understated shot noise $\sim 20x$. Physically the shared photodetector sees the full received power; the key and data are only separated AFTER detection by the RF filter, so no single subcarrier's depth sets the shot noise.

E16 [calculate] Cross-talk suppression factor $10^{-(\text{Iso}/10)}$ at $\text{Iso}=15$ dB. Give the linear factor.

Answer: 0.031623 (linear)

$10^{(-1.5)}=0.0316$.

In detail: Method: $10^{-(\text{Iso}/10)} = 10^{(-1.5)} = 0.0316$ — the fraction of leaked data power that survives the 15 dB filter. It multiplies the cross-talk variance, so every extra 10 dB of isolation cuts cross-talk tenfold. Because cross-talk is the dominant noise at high elevation, filter isolation is a direct QBER lever, not a second-order detail.

E17 [calculate] Rytov scaling: $\sec^{(11/6)}(60 \text{ deg}) / \sec^{(11/6)}(30 \text{ deg})$. Give the ratio.

Answer: 2.737544 (ratio)

$(2/1.1547)^{(11/6)}=2.74$.

In detail: Method: the Rytov variance carries a $\sec^{(11/6)}(\text{zeta})$ factor, so the ratio between 60 and 30 deg zenith is $(2/1.1547)^{(11/6)} = 2.74$. The integral part is unchanged when only zeta varies, which is a nice self-check that the angular and vertical dependencies factorise. Practically: turbulence-induced scintillation worsens steeply as the satellite drops toward the horizon.

E18 [calculate] Convert $h_g=2.6\text{e-}4$ (linear) to dB.

Answer: -35.850267 dB

$10*\log_{10}(2.6\text{e-}4) = -35.8 \text{ dB}$.

In detail: Method: $\text{dB} = 10*\log_{10}(\text{linear})$; $10*\log_{10}(2.6\text{e-}4) = -35.85 \text{ dB}$. Master this conversion — the whole link budget is bookkept in dB because the multiplicative losses ($h_g*hl*ha$) become an easy SUM in dB. A factor of 2 is +3 dB, a factor of 10 is +10 dB; internalising those makes link budgets mental arithmetic.

9.2 Geometry & orbits — calculate

E19 [calculate] Ground-coverage radius R_{cov} at 30 deg mask, $H_S=550 \text{ km}$. Give km.

Answer: 793.496646 km

$R_{\text{cov}}(30 \text{ deg})=793 \text{ km}$.

In detail: Method: $R_{\text{cov}} = R_E*\psi$ with $\psi = \arccos(R_E/(R_E+h)*\cos(e)) - e$; at a 30 deg mask this gives 793 km. Note: the old paper mislabelled 950 km (the SLANT range) as the ground radius — the true footprint on the ground is smaller. This radius decides which cities a single pass can physically reach.

E20 [calculate] Ground-coverage radius at the 40 deg security mask. Give km.

Answer: 573.417261 km

$R_{\text{cov}}(40 \text{ deg})=573 \text{ km}$.

In detail: Method: same formula at a 40 deg mask gives 573 km. The stricter security mask shrinks the footprint by ~28% in radius (~48% in area) versus 30 deg. That is the concrete price of insisting on higher-elevation, cleaner-QBER passes — fewer city pairs can be served at once.

E21 [calculate] Max station separation for a DUAL (shared-pass) city pair at 30 deg mask = $2*R_{\text{cov}}$. Give km.

Answer: 1586.993293 km

$2*793 = 1587 \text{ km}$.

In detail: Method: two stations can share one pass only if both footprint circles overlap, i.e. their separation is at most $2*R_{\text{cov}} = 1587 \text{ km}$ at the 30 deg mask. This single threshold is what splits the 28 ASEAN city pairs into DUAL versus store-and-forward. Geometry, not scheduling, sets this ceiling.

E22 [calculate] Max DUAL separation at the 40 deg mask = $2*R_{\text{cov}}(40)$. Give km.

Answer: 1146.834521 km

$2*573 = 1146 \text{ km}$.

In detail: Method: at the 40 deg mask the DUAL threshold tightens to $2*573 = 1146 \text{ km}$. So raising the mask not only shrinks coverage but also RECLASSIFIES some previously-DUAL pairs as store-and-forward. It is a clean example of how a security choice cascades into the network topology.

E23 [calculate] Off-nadir angle η at the satellite for elevation 30 deg. $\sin \eta = R_E \cos e / (R_E + h)$. Give deg.

Answer: 52.863913 deg

$\sin \eta = 6371*\cos 30/6921 = 0.797 \rightarrow \eta = 52.8 \text{ deg}$.

In detail: Method: the transmit-side pointing angle obeys $\sin(\eta) = R_E \cos(e)/(R_E + h)$; at 30 deg elevation $\eta = 52.9 \text{ deg}$. It is the angle from nadir the satellite must slew to. Larger off-nadir angles (low passes) stress the pointing/acquisition/tracking system and are part of why low elevations are costly.

E24 [calculate] Off-nadir angle at elevation 90 deg (zenith). Give deg.

Answer: 0.0 deg

$\eta = 0 \text{ deg}$.

In detail: Method: at the zenith (elevation 90) the ground station lies directly below the satellite, so the line of sight IS the nadir direction and $\eta = 0$. This is the pointing-easy, loss-minimal geometry. It anchors the off-nadir formula the same way $L = H_S$ anchors the slant range.

E25 [calculate] Great-circle distance Hanoi (21.03N,105.85E) to HCMC (10.82N,106.63E) by haversine. Give km.

Answer: 1138.34957 km

~1140 km.

In detail: Method: the haversine formula gives the great-circle distance Hanoi-HCMC ~1138 km. It depends only on the two lat/lon pairs and is symmetric, so it is used to CLASSIFY pairs (DUAL/SF). Contrast with store-and-forward LATENCY, which is directional because it depends on the satellite's ascending/descending heading.

E26 [calculate] Is Hanoi-HCMC (dist above) DUAL at 30 deg mask? Enter the max DUAL separation it is compared against (km).

Answer: 1586.993293 km

Threshold 1587 km; 1140 < 1587 -> DUAL.

In detail: Method: compare the 1138 km distance to the DUAL threshold $2 \cdot R_{\text{cov}}(30) = 1587$ km; since $1138 < 1587$, Hanoi-HCMC is a DUAL pair. This is exactly the test `build_pairwise_matrices` runs for all 28 pairs. It shows the two anchor cities of the study can be served by one satellite pass.

E27 [calculate] Elevation angle if zenith angle $\zeta = 55$ deg. Give deg.

Answer: 35 deg

$90 - 55 = 35$ deg.

In detail: Method: elevation and zenith are complementary, $\text{elevation} = 90 - \text{zenith} = 35$ deg. Keeping the two straight avoids a common sign error, since the physics formulas mix them (attenuation uses ζ , coverage uses elevation e). Always state which one you mean.

E28 [calculate] Pass frequency per day if 125 passes are seen in a 2-hour window. $\text{freq} = \text{count} \cdot (24/\text{hours})$.

Answer: 1500.0 passes/day

$125 \cdot 12 = 1500$ passes/day.

In detail: Method: extrapolate the observed count over the window, $\text{freq} = \text{count} \cdot (24/\text{hours}) = 125 \cdot 12 = 1500$ passes/day/station. The huge pass frequency (with only ~3 satellites simultaneously visible and ~204 s per pass) reflects how fast LEO satellites sweep past low-latitude ASEAN. Frequent-but-short passes shape the scheduling problem.

E29 [calculate] Number of unique city pairs among 8 ground stations = $C(8,2)$. Give the count.

Answer: 28 pairs

$8 \cdot 7 / 2 = 28$.

In detail: Method: pairs among 8 stations = $C(8,2) = 8 \cdot 7 / 2 = 28$. This tiny combinatorial fact is the root of pair-starvation: 8 stations must feed 28 city pairs, so on most days at least one pair is starved regardless of algorithm. Capacity, not cleverness, is the binding constraint.

E30 [calculate] Footprint area reduction going 30->40 deg mask: $1 - (R_{\text{cov}40}/R_{\text{cov}30})^2$, as a percentage.

Answer: 47.778261 %

$1 - (573/793)^2 \approx 48\%$ area (about 28% radius).

In detail: Method: coverage AREA scales as radius^2 , so going 30->40 deg cuts area by $1 - (573/793)^2 \approx 48\%$ (about 28% in radius). This quantifies the security-vs-coverage trade: a safer QBER margin costs roughly half the instantaneous footprint. It is a headline number for justifying the 30 deg operating choice.

9.3 Weather & availability

E31 [calculate] Link availability A at Kuala Lumpur, November ($P_{\text{cloud}} = 0.893$). $A = 1 - P_{\text{cloud}}$.

Answer: 0.107 (fraction)

$A = 1 - 0.893 = 0.107$ (10.7%).

In detail: Method: availability is simply $A = 1 - P_{\text{cloud}}$; Kuala Lumpur in November has $P_{\text{cloud}} = 0.893$, so $A = 0.107$ (10.7%) — the WORST cell in the dataset. Insight: A is invariant to the rain fraction (rain only moves probability between the clear and rain states), so cloud alone governs it. This is the empirical core of the 'cloud, not rain' finding.

E32 [calculate] Link availability at Bangkok, February ($P_{\text{cloud}} = 0.035$). Give fraction.

Answer: 0.965 (fraction)

$A = 0.965$ (96.5%).

In detail: Method: $A = 1 - 0.035 = 0.965$ (96.5%) at Bangkok in February — the BEST cell. Together with the KL extreme it bookends the headline 10.7%-96.5% range. That both extremes are tropical ASEAN cities in the same constellation shows how strongly local climate, not orbit, dictates optical availability.

E33 [calculate] Hourly rain rate from $R_{\text{month}}=313.2$ mm, $f_{\text{rain}}=0.15$. $R=R_{\text{month}}/(30*24*f_{\text{rain}})$. mm/h.

Answer: 2.9 mm/h

$$313.2/108 = 2.90 \text{ mm/h.}$$

In detail: Method: convert a monthly total to an instantaneous rate by spreading it over the raining hours only, $R = R_{\text{month}}/(30*24*f_{\text{rain}}) = 313.2/108 = 2.90$ mm/h at $f_{\text{rain}} = 0.15$. If you divided by all hours you would badly underestimate the intensity, since tropical rain is intermittent and heavy. The f_{rain} fraction is the bridge between climatology and channel physics.

E34 [calculate] Same conversion with the real $f_{\text{rain}}=0.30$ instead of 0.15. mm/h.

Answer: 1.45 mm/h

$$313.2/216 = 1.45 \text{ mm/h.}$$

In detail: Method: with the real $f_{\text{rain}} = 0.30$ the same monthly total gives $R = 313.2/216 = 1.45$ mm/h — half the value. Larger f_{rain} means the rain is spread over more hours, so each raining hour is gentler. This is why the flat 0.15 assumption was replaced by real per-city/month fractions from ERA5.

E35 [calculate] $p_{\text{rain}}=\min(f_{\text{rain}}, 1-P_{\text{cloud}})$ with $f_{\text{rain}}=0.30$, $P_{\text{cloud}}=0.893$. Give p_{rain} .

Answer: 0.107 (fraction)

$$\min(0.30, 0.107)=0.107.$$

In detail: Method: $p_{\text{rain}} = \min(f_{\text{rain}}, 1 - P_{\text{cloud}}) = \min(0.30, 0.107) = 0.107$ at Kuala Lumpur in November. The $\min()$ caps rain by the non-cloud time available: if it is cloudy 89% of the time, there is at most 11% left for any rain state. This coupling keeps the three probabilities consistent.

E36 [calculate] $p_{\text{clear}}=\max(0, 1-P_{\text{cloud}}-p_{\text{rain}})$ with $P_{\text{cloud}}=0.5$, $f_{\text{rain}}=0.30$. Give p_{clear} .

Answer: 0.2 (fraction)

$$p_{\text{rain}}=0.30; p_{\text{clear}}=1-0.5-0.30=0.20.$$

In detail: Method: $p_{\text{clear}} = \max(0, 1 - P_{\text{cloud}} - p_{\text{rain}}) = 1 - 0.5 - 0.30 = 0.20$. The three states partition the month: clear + rain + cloud = 1. Clear gets whatever is left after cloud (link off) and rain (attenuated) are removed.

E37 [calculate] Effective SKR = $p_{\text{clear}}*SKR_{\text{clear}} + p_{\text{rain}}*SKR_{\text{rain}}$, $p_{\text{clear}}=0.20$, $SKR_{\text{clear}}=61$, $p_{\text{rain}}=0.30$, $SKR_{\text{rain}}=5$ (Mbps).

Answer: 13.7 Mbps

$$0.20*61+0.30*5 = 13.7 \text{ Mbps.}$$

In detail: Method: $SKR_{\text{eff}} = p_{\text{clear}}*SKR_{\text{clear}} + p_{\text{rain}}*SKR_{\text{rain}} = 0.20*61 + 0.30*5 = 13.7$ Mbps; the cloud state contributes zero. This weighted average is how a single-link clear-sky rate becomes a realistic delivered rate. It also shows why availability (which state you are in) matters more than the peak SKR.

E38 [fill in] What ERA5 cloud-cover percentage threshold defines a 'cloudy hour' (link outage)? Give the number with % sign.

Answer: 85% %

In detail: The 85% total-cloud-cover threshold (CLOUD_OUTAGE_THRESHOLD) defines a 'cloudy hour' where the optical link is treated as OFF. It is a modelling choice, not a law of nature, so Annex A sweeps it to show the headline conclusions are robust to reasonable changes. Choosing a threshold is exactly the kind of assumption a careful reviewer probes — hence the sensitivity analysis.

E39 [fill in] How many years of ERA5/ERA5-Land reanalysis calibrate the weather model? Give the number.

Answer: 10 years

In detail: Ten years (2015-2024) of hourly ERA5/ERA5-Land reanalysis calibrate the weather model. A decade is long enough to average out interannual variability (the 6-year version differed by up to 0.12 in P_{cloud}), which is why the record was extended. Real reanalysis, not hand-tuned numbers, is what lets the availability claims stand up to a remote-sensing examiner.

E40 [fill in] In which local-hour window is the combined cloud+rain 'best FSO window'? (e.g. 05-10h)

Answer: 05-10h local

In detail: The best combined cloud+rain window is mid-morning, roughly 05:00-10:00 local — in DAYLIGHT. This is because total cloud peaks near dawn/overnight and dissipates by mid-morning, while convective rain peaks in the afternoon; the two minima overlap mid-morning. It creates the unresolved day/night tension: the clearest sky is exactly when a QKD receiver most fears background sunlight.

E41 [fill in] Which weather variable (not rain) governs tropical optical availability?

Answer: cloud

In detail: Cloud, not rain, governs tropical optical availability, because at optical wavelengths thick cloud is total blockage while rain only attenuates. This is the single most important meteorological finding of the thesis and the reason RF satcom (which passes through cloud) and optical QKD behave so differently in the tropics. Availability $A = 1 - P_{\text{cloud}}$ makes the dependence explicit.

E42 [fill in] The tropical rain-attenuation law ($\alpha=0.509$, $\rho=0.63$) was measured in which city?

Answer: Abidjan

In detail: The rain-attenuation coefficients $\alpha = 0.509$, $\rho = 0.63$ come from real FSO measurements in Abidjan, Cote d'Ivoire (Kone et al. 2024) — a genuinely tropical site. Using measured tropical constants rather than temperate ITU defaults is a deliberate, defensible choice a geophysics/remote-sensing examiner will respect. It anchors the rain model to data.

9.4 Key numbers & terms — fill in

E43 [fill in] v3 secret key rate at 45 deg elevation (give value with unit, e.g. '61.4 Mbps').

Answer: 61.4 Mbps

In detail: The v3 secret key rate at 45 deg elevation is ~61.4 Mbps (up from the pre-v3 13.3 Mbps). The jump came mainly from the cross-talk /8 fix, which quartered the dominant noise, plus resetting m_K to 0.15. Remember this as the headline single-link number — but always follow it with the caveat that delivered NETWORK key is far lower once weather and scheduling are applied.

E44 [fill in] v3 matching gain (ALG-1 vs ALG-0) in the dry season, as a percentage (e.g. +2.47%).

Answer: +2.47%

In detail: Coordinating the 8 stations (ALG-1 vs the naive ALG-0 baseline) gains +2.47% in the dry season, over a fair full-enumeration of 620 real days. It is deliberately modest and honest: the min()-bottleneck along each pair path caps how much any matching can help. State the baseline in the same breath as the gain — that fairness is the point.

E45 [fill in] Sign of the weather-information gain (aware vs blind): positive or negative?

Answer: negative

In detail: The weather-information gain is NEGATIVE (about -0.29%/-0.23%): adding weather-awareness on top of coordination slightly COSTS delivered key. Mechanism: a weather-aware scheduler re-assigns stations more often, and each re-assignment burns a real 30 s acquisition handover that outweighs the benefit of chasing a clearer satellite. It is a measured effect (handovers were counted directly), robust in sign across seasons and m_K values — a genuine, non-obvious finding, not a bug.

E46 [fill in] Fraction of days with at least one starved city pair (range, e.g. 91.6-96.5%).

Answer: 91.6-96.5%

In detail: On 91.6-96.5% of days at least one of the 28 city pairs receives ZERO usable key, regardless of algorithm. This pair-starvation is structural — 8 stations cannot feed 28 pairs — and stayed essentially unchanged through every physics correction, which is exactly what confirms it is a capacity limit, not a formula artefact. The engineering fix is more ground stations, not smarter scheduling.

E47 [fill in] Coverage radius at the 30 deg mask (km).

Answer: 793 km

In detail: The ground-coverage radius at the 30 deg mask is 793 km. Do not confuse it with the ~950 km SLANT range the original paper text mislabelled — the footprint on the ground is smaller than the line-of-sight distance. This radius, doubled, is the DUAL/SF threshold.

E48 [fill in] Coverage radius at the 40 deg mask (km).

Answer: 573 km

In detail: At the 40 deg security mask the coverage radius drops to 573 km. The 30->40 deg tightening is the coverage-vs-security trade: ~28% less radius for a QBER margin that keeps low passes secure. Knowing both numbers lets you argue either operating point.

E49 [fill in] How many DUAL city pairs (of 28) can share one satellite pass?

Answer: 14 pairs

In detail: 14 of the 28 pairs are DUAL (both cities fit under one satellite pass); the other 14 need store-and-forward relay. The clean 14/14 split is set purely by geography and the coverage radius, and it is a ceiling no scheduler can lift. It frames the whole scheduling problem: half the pairs are 'easy', half need a trusted relay.

E50 [fill in] Number of real Starlink Shell-1 satellites used.

Answer: 1019 satellites

In detail: 1,019 real Starlink Shell-1 satellites are used, filtered from the live catalogue by inclination 53.15-53.22 deg. Using ONE homogeneous shell keeps the single-altitude channel model valid and the analysis reproducible; the trade-off is ignoring ~9,000 other satellites, which the thesis flags as future work. The honest defense answer is 'my model is parameterised for one altitude', not 'this shell is special to ASEAN'.

E51 [fill in] Shell-1 orbital inclination (deg).

Answer: 53 deg

In detail: Shell-1's inclination is ~53 deg, so its satellites reach a maximum latitude of 53 deg N/S — covering ~95% of the world population, including all of ASEAN. SpaceX chose 53 deg for launch efficiency from Cape Canaveral and market coverage, not for ASEAN specifically. At low ASEAN latitudes the satellites sweep fast overhead, giving many short passes.

E52 [fill in] Satellite altitude assumed throughout (km).

Answer: 550 km

In detail: The satellite altitude is fixed at 550 km throughout — the Shell-1 value and the modern mega-constellation sweet spot (low latency, manageable drag). The channel model is calibrated for this one altitude, which is precisely why multi-shell analysis would need altitude-dependent SKR tables. For reference, GEO comms sit at 35,786 km and the ISS at ~400 km.

E53 [fill in] QKD carrier wavelength (nm).

Answer: 1550 nm

In detail: The carrier wavelength is 1550 nm (telecom C-band). It is eye-safe at higher power, sits in a low-loss atmospheric window, scatters less off aerosols than visible light, and reuses cheap mature telecom optics — four independent reasons the whole field converges on it. A quantum lab optimising for single-photon-detector efficiency might pick shorter wavelengths, a tension this direct-detection design avoids.

E54 [fill in] Corrected key modulation index m_K used in the thesis.

Answer: 0.15

In detail: The corrected key modulation index is $m_K = 0.15$, reset from an earlier 0.05. The source SIKD manuscript's own feasible region opens for $m_K \geq 0.1$ at 15 dB isolation, and 0.15 pushes QBER below even the stricter 10^{-3} target while leaving power for the data channel. This single parameter change is one of the two reasons v_3 QBER is so much lower than pre- v_3 .

E55 [fill in] Filter isolation Iso between the key and data subcarriers (dB).

Answer: 15 dB

In detail: Filter isolation is 15 dB. It quantifies how well the RF/optical filter separates the key and data subcarriers; the finite value is exactly why the data channel leaks into the key receiver as cross-talk. Because cross-talk dominates at high elevation, this 15 dB is one of the most consequential system parameters in the design.

E56 [fill in] Real acquisition/handover cost charged per handover (seconds).

Answer: 30 s

In detail: Every handover is charged its real 30 s acquisition/re-tracking cost. This is what makes the weather-information comparison fair: a scheduler that re-assigns more often must pay for the key lost during each 30 s re-acquisition. Charging it honestly is why weather-awareness shows a small net COST rather than a free gain.

E57 [fill in] BB84 QBER security threshold above which no secret key can be distilled (%).

Answer: 11%

In detail: The BB84 security threshold is $QBER = 11\%$: above it, no positive secret key can be distilled. Mathematically $1 - 2 \cdot H_2(Q)$ hits zero there, which is why the error-correction leakage term had to be added to the v_3 SKR formula. Unlike a classical BER (where 10^{-3} is merely 'poor'), 11% here is a hard wall.

9.5 Committee & literature — match

E58 [match] Match each external examiner to their home research field.

Answer: Guillaume Patanchon (APC) → CMB detector noise & component separation; Stephane Jacquemoud (IPGP) → Radiative transfer & remote sensing (PROSPECT); Cyrille Rosset (IN2P3/APC) → Statistical significance (Planck isotropy)

In detail: Patanchon (APC) is a CMB detector-noise and component-separation expert; Jacquemoud (IPGP) is a radiative-transfer/remote-sensing authority (author of PROSPECT); Rosset (IN2P3/APC) works on Planck statistical significance. The strategy is to bridge each part of your thesis to the examiner's OWN field, since none of them are QKD specialists. Getting these mappings automatic lets you

tailor answers on the fly.

E59 [match] Match each examiner to the bridge phrase you should use with them.

Answer: Jacquemoud → 'Beer-Lambert = atmospheric correction / radiative transfer'; Patanchon → 'detector noise budget & cross-talk = component separation'; Rosset → 'paired significance on a full enumeration + block bootstrap'

In detail: For Jacquemoud, frame the atmosphere as 'Beer-Lambert = atmospheric correction / radiative transfer'; for Patanchon, frame the receiver as a 'noise budget with cross-talk = component separation'; for Rosset, frame the result as 'paired significance on a full enumeration + block bootstrap'. Speaking each examiner's mother tongue turns unfamiliar QKD into familiar territory. Rehearse these three one-liners until they are reflexive.

E60 [match] Match each examiner to their toughest predicted question.

Answer: Jacquemoud → Why is Beer-Lambert enough without multiple scattering?; Patanchon → Why does cross-talk dominate when the link is strong?; Rosset → Is +2.5% significant or within day-to-day scatter?

In detail: The three trap questions map one-to-one to the examiners: Jacquemoud will ask why Beer-Lambert suffices without multiple scattering; Patanchon why cross-talk dominates a STRONG link; Rosset whether +2.5% is significant or just scatter. Master the MECHANISM behind each answer, not the number — that is what a science panel tests. These three are the highest-leverage items to over-prepare.

E61 [match] Match each reference to its role in the thesis.

Answer: Vu et al. 2022 (vu2022dtdd) → Noise model & dual-threshold (BBM92 provenance); Kone et al. 2024 (kone2024tropical) → Tropical rain-attenuation calibration; Polnik et al. 2020 (polnik2020scheduling) → Nearest scheduling precedent; Liao et al. 2017 (liao2017satellite) → Micius: first satellite QKD from LEO

In detail: Vu et al. 2022 supplies the noise model and dual-threshold rule (its BBM92 provenance is the honest caveat); Kone 2024 supplies the tropical rain calibration; Polnik 2020 is the nearest scheduling precedent; Liao 2017 is Micius, the first satellite QKD from LEO. Knowing which paper anchors which layer lets you cite precisely under pressure. Precise citation is a credibility signal to a research panel.

E62 [match] Match each reference to its role (set 2).

Answer: Andrews & Phillips 2005 → Rytov variance & H-V turbulence profile; Leverrier 2015 → Composable CV-QKD proof (future work); Gruneisen et al. 2015 → Daytime sky-access for QKD downlink; Open-Meteo / ERA5 (open-meteo2023) → 10-year real weather climatology

In detail: Andrews & Phillips 2005 anchors the Rytov variance and H-V turbulence profile; Leverrier 2015 is the composable-security proof you cite as future work; Gruneisen 2015 is the daytime sky-access reference for the day/night tension; Open-Meteo/ERA5 is the 10-year weather source. These four cover the physics, security, daytime, and weather corners of the argument. Each is a ready answer to a 'where did that come from?' challenge.

E63 [match] Match each reference to its role (set 3 — SWaP, PAT & NTN).

Answer: Schieler et al. 2023 (TBIRD) → Small-aperture SWaP comparison (0.4 m RX); Gerard et al. 2025 (PAT) → Pointing/Acquisition/Tracking delay → 30 s handover; Farid & Hranilovic 2007 → Pointing-error term omitted (hg = lower bound); 3GPP TR 38.821 → Non-terrestrial network (NTN) standardisation

In detail: Schieler 2023 (TBIRD) justifies the small-aperture SWaP regime; Gerard 2025 (PAT) justifies charging a 30 s handover; Farid & Hranilovic 2007 covers the pointing-error term you deliberately omit (so hg is a lower bound); 3GPP TR 38.821 frames the non-terrestrial-network context. These are the engineering-choice anchors: each one defends a specific modelling decision. Have them ready for 'why did you assume X?' questions.

E64 [match] Match each reference to its role (set 4 — QKD landscape).

Answer: Bennett & Brassard 1984 → BB84 protocol (the SKR rate's basis); Scarani et al. 2009 → Security of practical QKD; Li et al. 2025 (Jinan-1) → Microsatellite real-time QKD (cheap sats); Toka et al. 2025 → Weather-aware LEO routing (classical data)

In detail: Bennett & Brassard 1984 is BB84 (the basis of your SKR rate); Scarani 2009 is the practical-security framework; Li 2025 (Jinan-1) is microsatellite real-time QKD (cheap sats); Toka 2025 is weather-aware LEO routing for classical data. Together they place your work in the QKD landscape: foundations, security, the cheap-sat trend, and the weather-aware precedent. This mapping answers 'how is this different from prior work?'.

E65 [match] Match each examiner to what they will most likely NOT probe deeply.

Answer: Son Tong Si (USTH generalist) → Deep QKD math (asks motivation/big picture); Jacquemoud (remote sensing) → The Hungarian algorithm internals; Rosset (statistics) → Beam-optics derivations

In detail: Son Tong Si (a USTH generalist) is unlikely to probe deep QKD math and will instead ask motivation and big-picture questions; Jacquemoud will not dwell on the Hungarian algorithm; Rosset will not chase beam-optics derivations. Knowing what each examiner will NOT focus on lets you pitch each answer at the right depth. Aim answers at strengths, keep the rest high-level.

E66 [fill in] Which examiner did the internal supervisor name as the likely primary reviewer? (surname)

Answer: Jacquemoud

In detail: The internal supervisor named Jacquemoud (IPGP) as the likely primary reviewer. As a radiative-transfer/remote-sensing expert he will scrutinise the atmospheric model most closely, so the Beer-Lambert, visibility, rain, and ERA5 parts must be airtight. Prepare his three likely questions (single-scattering, visibility model, ERA5 cloud threshold) first.

E67 [fill in] Which classic model is Jacquemoud the author of? (acronym)

Answer: PROSPECT

In detail: Jacquemoud is the author of PROSPECT (and PROSAIL), the classic leaf radiative-transfer models spanning 400-2500 nm. His home ground is optical extinction/attenuation through a scattering-absorbing medium — exactly the physics of your h_l term. Framing your atmosphere as 'the same radiative transfer you pioneered' is the strongest possible bridge to him.

E68 [fill in] Which satellite mission is the canonical 'first satellite QKD from LEO' you cite in the motivation?

Answer: Micius

In detail: Micius (Liao et al. 2017, Nature) is the canonical first satellite QKD from LEO — the existence proof that space QKD works. Cite it in the motivation to establish that the idea is proven, then pivot to your distinct question: a NETWORK of cheap tropical stations under real cloud, not a single clean high-power link. It legitimises the field while setting up your novelty.

E69 [fill in] How many real references are catalogued in the literature review?

Answer: 34 references

In detail: 34 real references are catalogued in the literature review, built directly from the thesis references.bib and mapped to both their role in the work and the examiner they speak to. Knowing the count and structure signals you have genuinely surveyed the field. It also means any 'have you considered X?' likely already has a slot in your map.

E70 [fill in] In one word, the strategy with this non-QKD panel: report results with ____ (integrity vs. spin).

Answer: integrity

In detail: With a non-QKD panel the winning strategy is INTEGRITY: report fairly-compared results, flag limitations proactively (the negative weather gain, the day/night tension, the borrowed BB84 threshold), and never oversell. A science committee rewards a mature, honest account over a big headline number. Leading with your own caveats disarms the toughest questions.

E71 [fill in] Complete Jacquemoud's answer: 'I treat cloud as a ____ outage, so multiple scattering only refines an already-lost state.'

Answer: binary

In detail: Jacquemoud's prepared answer: 'I treat cloud as a BINARY outage, so multiple scattering only refines an already-lost state.' You concede Beer-Lambert is a single-scattering, first-order model, then justify that for clear/rain a narrow line-of-sight beam is extinction-dominated, and for cloud the link is simply off. This turns a potential weakness into a defensible modelling choice.

E72 [fill in] Complete Rosset's answer: 'It is not a random sample — I ____ every day in two 10-year proxy months.'

Answer: enumerate

In detail: Rosset's prepared answer: 'It is not a random sample — I ENUMERATE every day in two 10-year proxy months (620 days), so there is no sampling error in the mean.' You then add the paired day-by-day test (same weather, only the algorithm changes) that isolates the ~2.5% effect, and the block bootstrap that handles autocorrelation. This is the statistically literate defense of a small number.

9.6 Reasoning, derivations & defense answers

E73 [order] Put the SIKD computation pipeline in the correct order.

Answer: channel (h_g , h_l , σ_X^2) → noise budget (4 terms) → dual-threshold (d_0, d_1) → Psift & QBER (Gauss-Hermite) → SKR (Shor-Preskill)

In detail: The SIKD pipeline runs channel (h_g , h_l , σ_X^2) → noise budget → dual-threshold (d_0, d_1) → Psift & QBER (Gauss-Hermite) → SKR. Each stage strictly feeds the next: you cannot set thresholds without the noise, or get QBER without the thresholds. Knowing the order lets you locate any question ('where does turbulence enter?') on the chain instantly.

E74 [order] Order the FSO channel factors as applied from transmitter to receiver.

Answer: Transmitted power P_T → Geometric spreading h_g → Atmospheric attenuation h_l → Turbulence fading h_a → Received power

In detail: The FSO channel factors apply in physical order from the transmitter: P_T → geometric spreading h_g → atmospheric attenuation h_l → turbulence fading h_a → received power. They multiply ($h = h_g * h_l * h_a$) because they are independent loss mechanisms acting in sequence along the beam. In dB they simply add, which is why link budgets are done in dB.

E75 [order] Order the three weather states by increasing severity of link impact.

Answer: Clear (attenuation only) → Rain (added attenuation) → Cloud (link OFF)

In detail: The three weather states rank by increasing severity: clear (attenuation only) → rain (added attenuation) → cloud (link OFF). Cloud is a binary outage, not a deeper fade, which is the modelling decision that makes availability $A = 1 - P_{\text{cloud}}$. Ordering them clarifies why cloud, not rain, dominates.

E76 [order] Order the two-level scheduling problem's steps.

Answer: Station-satellite matching (Hungarian) → Build matched pass table → Pair-key allocation (greedy/ILP/maxmin) → $K_q = \min(\text{sideA}, \text{sideB})$

In detail: Scheduling is two nested levels in order: station-satellite matching (Hungarian) → build the matched pass table → pair-key allocation (greedy/ILP/maxmin) → compute $K_q = \min(\text{sideA}, \text{sideB})$. Level 1 decides who tracks whom; level 2 routes the resulting key to city pairs on a shared trusted relay. The $\min()$ at the end is where the structural bottleneck lives.

E77 [order] Rank the v3 noise terms by when cross-talk dominates (weak link → strong link).

Answer: Thermal noise (floor, weak link) → Shot noise (grows with power) → Cross-talk ($\sim P^2$, strong link)

In detail: Rank the noise terms by when each dominates from weak to strong link: thermal noise (the floor, when the signal is weak) → shot noise (grows with power) → cross-talk ($\sim P^2$, dominant when the link is STRONGEST). The counter-intuitive part is that a stronger link brings proportionally MORE cross-talk, so 'thermal-dominated' is only true near 30 deg. This ordering is the essence of the SIKD noise story.

E78 [fill in] Complete the identity that makes log-normal fading energy-conserving: $\mu_X = \underline{\hspace{1cm}}$ (in terms of σ_X^2).

Answer: $-\sigma_X^2$

In detail: For log-normal fading $h_a = \exp(2X)$ with $X \sim N(\mu_X, \sigma_X^2)$, setting $\mu_X = -\sigma_X^2$ forces $E[h_a] = 1$. Derivation: $E[e^{2X}] = \exp(2\mu_X + 2\sigma_X^2)$, which equals 1 only when $\mu_X = -\sigma_X^2$. This is the energy-conservation condition — turbulence should redistribute intensity, not add or remove average power.

E79 [fill in] The secret-key formula reaches zero at QBER = $\underline{\hspace{1cm}}$ % ($f_{EC}=1$, $\chi_E=0$). Give the percentage.

Answer: 11%

In detail: The secret-key formula reaches zero at QBER = 11% because $1 - 2H_2(0.11) \sim 0$. The factor of 2 comes from subtracting BOTH the mutual-information deficit $H_2(Q)$ and the error-correction leakage $f_{EC}H_2(Q)$; the v3 fix added the missing leakage term so the rate correctly hits zero at the Shor-Preskill threshold instead of at 50%. It is the single most important line of the SKR derivation.

E80 [fill in] Cross-talk variance scales as received power to the power $\underline{\hspace{1cm}}$ (give the exponent).

Answer: 2

In detail: Cross-talk variance scales as received power SQUARED (P^2), because $\sigma_{CT}^2 \sim (h_g h_l)^2$ while the key signal scales as $h_g h_l$ (power 1). The consequence is profound: cross-talk dominates exactly when the link is strong, the opposite of ordinary noise. This single exponent is the defining quirk of multiplexing data and key on one detector.

E81 [fill in] The v3 cross-talk coefficient changed from $/2$ to $/\underline{\hspace{1cm}}$ (give the number).

Answer: 8

In detail: The v3 cross-talk coefficient changed from $/2$ to $/8$. Reason: the photocurrent carries a leading $1/2$, so the leaking data sinusoid has amplitude $(1/2) \cdot \text{Re} \{ P_T m_D h_g h_l \}$, and the variance of a sinusoid of amplitude A is $A^2/2$ — giving $(\dots)^2/8$, a factor of 4 (+6 dB) smaller than the old $/2$. This was the single highest-impact fix: it quartered the dominant noise and roughly quartered QBER.

E82 [fill in] Delivered pair key is a $\underline{\hspace{1cm}}()$ over the pair path (one word/function), capping the gain.

Answer: min

In detail: Delivered pair key is a $\text{MIN}()$ over the pair path: $K_q = \min(\text{sideA total}, \text{sideB total})$, under the trusted-relay model where the satellite XORs both stations' keys. A pair can only use as much key as its WORSE-served side received, so one weak link caps everything. This $\min()$ is the structural ceiling that limits how much any scheduling cleverness can help.

E83 [fill in] Complete the defense answer: 'Weather-awareness costs because it re-assigns more often and the extra $\underline{\hspace{1cm}}$ outweigh the benefit.'

Answer: handovers

In detail: Weather-awareness costs because it re-assigns stations more often, and the extra HANDOVERS (each a real 30 s re-acquisition) outweigh the benefit of chasing a clearer satellite. The mechanism was measured directly by counting handovers, not inferred, which is why the small negative gain is credible rather than a bug. Deliver this answer confidently, not defensively.

E84 [fill in] Complete: 'Pair-starvation is a ____-planning finding, not an algorithm failure.'

Answer: capacity

In detail: Pair-starvation is a CAPACITY-planning finding, not an algorithm failure: 8 ground stations physically cannot serve 28 city pairs, so some pair is starved on most days no matter how it is scheduled. The remedy is more infrastructure (stations, or de-correlated sites), which is a concrete engineering recommendation for an operator. Framing it as capacity planning turns a 'weakness' into a useful result.

E85 [fill in] Which algorithm solves the station-satellite matching EXACTLY? (name)

Answer: Hungarian

In detail: The station-satellite matching is solved EXACTLY by the Hungarian algorithm (scipy linear_sum_assignment) on the negated weight matrix, giving the globally optimal max-weight bipartite matching each step. 'Exact' matters: it means the modest coordination gain is not an artefact of a weak heuristic — even optimal matching helps only ~2.5%. That strengthens the structural-limit argument.

E86 [fill in] In one word, the trusted-relay assumption requires the satellite node to be ____.

Answer: trusted

In detail: The trusted-relay assumption requires the satellite node to be TRUSTED (honest, not compromised), because the relay learns both stations' keys before broadcasting their XOR. Security therefore holds only under that trust; a composable proof for this model is listed as future work. Stating this openly is part of the honest framing the panel rewards.

E87 [fill in] Complete the one-line thesis claim: 'Judge these networks at the ____ level, not by a single clear-sky link rate.'

Answer: network

In detail: The one-line thesis claim: judge these networks at the NETWORK level (fairly-compared scheduling outcomes under real weather and orbits), not by a single idealised clear-sky link rate. It reframes the unit of evaluation from a physics-lab number to an operational, network-wide one. Ending your defense on this sentence ties every result together.

E88 [fill in] Vu et al. 2022 implements which protocol (not BB84)? Give the acronym.

Answer: BBM92

In detail: Vu et al. 2022 implements BBM92 (entanglement-based), not BB84, with a design target QBER $\leq 10^{-3}$. Your model borrows its noise/threshold machinery but applies the wider 11% BB84 bound — an explicit, openly-stated assumption, not an inherited result. After the v3 corrections your QBER sits below even the stricter 10^{-3} , so the operating points are secure under either threshold, which turns the caveat into a strength.

E89 [order] Order the four v3 fixes by the stage they touch (channel → receiver).

Answer: erf coefficient $\sqrt{2} \rightarrow \sqrt{\pi/2}$ → spherical slant range → Rytov $0.56 \rightarrow 2.25$ → cross-talk $/2 \rightarrow /8$

In detail: Order the four v3 fixes by stage from channel to receiver: erf coefficient $\sqrt{2} \rightarrow \sqrt{\pi/2}$ → spherical slant range → Rytov $0.56 \rightarrow 2.25$ → cross-talk $/2 \rightarrow /8$. The first three are geometric/turbulence corrections in the channel; the last is the receiver cross-talk fix, which had the largest numerical impact. Knowing the sequence lets you explain how a single formula error cascades through the whole pipeline.

E90 [fill in] The daytime-background penalty is described as an 'order-of-____ estimate', not a proof. (one word)

Answer: magnitude

In detail: The daytime-background penalty is described as an 'order-of-MAGNITUDE estimate', not a proof. An FOV sweep suggests the penalty may be small for this PIN/direct-detection receiver (unlike a photon counter), but a full daytime-background simulation is explicitly future work. Labelling the confidence level honestly — estimate versus proof — is exactly the scientific care a panel looks for.

10 Multiple-choice questions (360), by professional perspective

10.1 P1 — Quantum Physicist / QKD Theorist

Q1 What does "information-theoretic security" mean for QKD?

A) The key's secrecy follows from physical law, independent of the eavesdropper's computing power **B)** The key is encrypted with AES-256 **C)** The key is secure only against classical attackers **D)** The key is hard to compute with today's computers

Answer: A. *Framework:* QKD security rests on quantum mechanics (no-cloning, measurement disturbance), not computational hardness — so it survives even a quantum computer, unlike RSA/ECC.

Q2 In BB84, what physical principle prevents an eavesdropper from copying the quantum states undetected?

- A) The no-cloning theorem plus measurement disturbance B) Conservation of energy C) Heisenberg uncertainty only D) Entanglement monogamy

Answer: A. *Framework:* An unknown quantum state cannot be copied (no-cloning), and any measurement to learn it disturbs it, raising the QBER — this is how Eve is detected.

Q3 QBER stands for:

- A) Quantum Bit Error Rate B) Quantum Beam Extinction Ratio C) Quantized Background Error Rate D) Quantum Baseline Error Reduction

Answer: A. *Framework:* QBER = fraction of sifted bits where Bob's decoded bit disagrees with Alice's; it is the master security indicator.

Q4 In the project's model, QBER is computed as:

- A) $P_{\text{error}} / P_{\text{sift}}$, where $P_{\text{error}} = \frac{1}{2}(P_{10} + P_{01})$ and P_{sift} includes all four detection outcomes B) $1 - P_{\text{sift}}$ C) P_{error} only D) The raw photodetector error

Answer: A. *Framework:* `sikd_performance.compute_Psift_QBER`: $QBER = P_{\text{error}}/P_{\text{sift}}$, $P_{\text{error}} = \frac{1}{2}(P_{10}+P_{01})$, $P_{\text{sift}} = \frac{1}{2}(P_{00}+P_{11}+P_{10}+P_{01})$.

Q5 The Shor-Preskill / GLLP asymptotic secret key rate used here is:

- A) $SKR = P_{\text{sift}}$ B) $SKR = 1 - QBER$ C) $SKR = P_{\text{sift}} \cdot [1 - (1+f_{\text{EC}}) \cdot H_2(QBER) - \chi_E]$ D) $SKR = P_{\text{sift}} \cdot QBER$

Answer: C. *Framework:* `compute_SKR`: mutual info $I(A;B)=1-H_2(Q)$, minus reconciliation leakage $f_{\text{EC}} \cdot H_2(Q)$, minus Eve's Holevo χ_E .

Q6 At what QBER does the corrected SKR formula reach zero (for $f_{\text{EC}} = 1$, $\chi_E = 0$)?

- A) ~11% B) 15% C) 25% D) 50%

Answer: A. *Framework:* $1 - 2 \cdot H_2(Q) = 0$ at $Q \approx 0.11$ — the classic BB84 Shor-Preskill threshold; this is exactly why the EC-leakage term had to be added.

Q7 Why was the earlier SKR formula (missing the reconciliation term) wrong?

- A) It used the wrong entropy base B) It double-counted χ_E C) It ignored P_{sift} D) It reached zero only at $QBER = 50\%$, contradicting the project's own "SKR = 0 above 11%" claim

Answer: D. *Framework:* Slepian-Wolf/Devetak-Winter: error correction leaks $f_{\text{EC}} \cdot H_2(Q)$ bits publicly even with no eavesdropper, so it must be subtracted.

Q8 In the baseline analysis, χ_E (Eve's Holevo information) is set to:

- A) 0 (no eavesdropping assumed for the performance baseline) B) 1 C) $H_2(QBER)$ D) QBER

Answer: A. *Framework:* `compute_SKR(chi_E=0.0)`: performance is characterised assuming no active eavesdropper; $\chi_E > 0$ would model an attack.

Q9 $H_2(p)$, the binary entropy, equals:

- A) $-p \cdot \ln(p)$ B) p^2 C) $-p \cdot \log_2(p) - (1-p) \cdot \log_2(1-p)$ D) $p \cdot (1-p)$

Answer: C. *Framework:* `binary_entropy`; returns 0 at $p=0$ or $p=1$; quantifies both mutual information and reconciliation cost.

Q10 The mutual information $I(A;B)$ between Alice and Bob is:

- A) P_{sift} B) $H_2(QBER)$ C) QBER D) $1 - H_2(QBER)$

Answer: D. *Framework:* Binary symmetric channel capacity with crossover probability QBER; the raw information before subtracting leakage.

Q11 Sifting in this dual-threshold scheme refers to:

- A) Basis reconciliation only B) Discarding samples that fall in the guard zone $[d_0, d_1]$ C) Error correction D) Privacy amplification

Answer: B. *Framework:* Samples inside $[d_0, d_1]$ are ambiguous and discarded; P_{sift} is the probability a sample is confidently decoded as 0 or 1.

Q12 The protocol whose noise model this project follows (Vu et al. 2022) actually implements:

- A) BBM92 entanglement-based, with design target $QBER \leq 10^{-3}$ B) Decoy-state BB84 C) BB84 single-photon D) E91

Answer: A. *Framework:* The provenance caveat — the noise/threshold machinery is borrowed from a BBM92 study; applying the BB84 11% bound is an explicit, stated assumption.

Q13 After the v3 corrections ($m_k = 0.15$), the QBER across 30° – 90° elevation is:

- A) Exactly 11% B) 10.3%–15.1% C) Below $\sim 0.02\%$, under even the stricter 10^{-3} BBM92 target D) Above 20%

Answer: C. *Framework:* `IMPACT_EVALUATION.md`: the cross-talk /8 fix and $m_k \rightarrow 0.15$ drop QBER far below threshold everywhere in the operating range.

Q14 Privacy amplification (not explicitly coded here but implied by the SKR bound) serves to:

- A) Correct bit errors B) Compress the sifted key to remove any partial information Eve may hold C) Detect eavesdropping D) Increase key length

Answer: B. *Framework:* The χ_E term in the SKR bound is the amount removed by privacy amplification; secrecy is bought by shrinking the key.

Q15 Why is 1550 nm chosen as the QKD carrier wavelength?

- A) It is visible light B) It maximises QBER C) Telecom C-band: eye-safe, low fibre/atmosphere loss, mature components D) It is the only wavelength lasers produce

Answer: C. *Framework:* System-parameter choice; 1550 nm also has weaker aerosol scattering than visible (Kruse/Kim exponent), aiding the link budget.

Q16 The "dual-threshold / direct-detection" (DT/DD) receiver differs from single-photon QKD in that:

- A) It counts individual photons B) It uses a PIN photodiode measuring a continuous photocurrent, with two decision thresholds C) It cannot produce a key D) It requires cryogenic cooling

Answer: B. *Framework:* `compute_thresholds`; DT/DD is a CV-QKD-style continuous detection, which changes how daytime background noise matters.

Q17 In the sift/QBER integral, the four conditional probabilities are averaged over turbulence fading using:

- A) Monte Carlo only B) 20-point Gauss-Hermite quadrature C) Simpson's rule D) A single mean value

Answer: B. *Framework:* `_GH_N = 20, hermgauss`; the log-normal fading is integrated as a Gaussian in log-amplitude, ideal for Gauss-Hermite.

Q18 The guard-zone width in the dual-threshold rule scales with:

- A) Only the signal mean B) $\zeta_{\text{scale}} \cdot \sigma_N$ (a chosen multiple of the noise standard deviation) C) QBER D) The wavelength

Answer: B. *Framework:* `compute_thresholds`: $d1 - d0 = 2 \cdot i_{\text{mean}} + 2 \cdot \zeta_{\text{scale}} \cdot \sigma_N$; larger ζ_{scale} widens the discard region (higher sifting loss, lower QBER).

Q19 Increasing the modulation index m_k (key channel) tends to:

- A) Have no effect B) Only affect the data channel C) Lower QBER (stronger key signal vs noise) and raise SKR, until the modulation budget binds D) Raise QBER and lower SKR

Answer: C. *Framework:* $i_{\text{mean}} \propto m_k$; the source SIKD manuscript's feasible region opens for $m_k \gtrsim 0.1$ at $\text{Iso} = 15$ dB — the reason m_k was reset to 0.15.

Q20 Why can't m_k simply be pushed arbitrarily high?

- A) It increases wavelength B) $m_k + m_a \leq 1$ (shared modulation budget on one carrier), and higher m_k steals power from the data channel C) It has no upper limit D) It would violate energy conservation

Answer: B. *Framework:* `evaluate_split` raises `ValueError` if $m_k + m_a > 1$; the SCM carrier's total modulation depth is bounded.

Q21 The secret key rate in absolute terms (bps) is:

- A) $\text{QBER} \cdot R_b$ B) SKR_{norm} alone C) $\text{SKR}_{\text{norm}} / R_b$ D) $\text{SKR}_{\text{norm}} \cdot R_b$ (normalized rate times raw bit rate)

Answer: D. *Framework:* `compute_SKR_bps`: $R_b = 1$ Gbps, so SKR_{norm} (bit/s/Hz) scales to absolute kbps/Mbps.

Q22 At 45° elevation, the corrected pipeline reports $\text{SKR} \approx$:

- A) 13,280 kbps B) 500 kbps C) 0 D) 61.4 Mbps

Answer: D. *Framework:* v3 Table 6.1 at $m_k=0.15$; up from the pre-correction 13.3 Mbps because the /8 cross-talk fix cut the dominant noise $4\times$.

Q23 The Holevo bound χ_E represents:

- A) The maximum information Eve can hold about the key given her quantum side-information B) The bit error rate C) The sifting loss D) The classical channel capacity

Answer: A. *Framework:* Devetak-Winter secure rate = $I(A;B) - \chi_E$; setting $\chi_E = 0$ is the no-eavesdropper baseline.

Q24 Why is a *composable* security proof listed as future work?

A) It removes the need for QKD B) It changes the wavelength C) Composable proofs are easier D) The current SKR is an asymptotic, non-composable bound that does not certify finite-key, trusted-relay end-to-end security

Answer: D. *Framework:* Leverrier (2015) composable CV-QKD; the trusted-relay XOR model needs a composable argument to be operationally secure end-to-end.

Q25 In the trusted-relay key model, a satellite holding keys k_A and k_B for two stations broadcasts:

A) Nothing B) $k_A \cdot k_B$ C) k_A only D) $k_A \oplus k_B$ (their bitwise XOR), so either ground node can recover the other's key

Answer: D. *Framework:* `schedulers docstring`; the pair's usable key is limited by the worse-served side, $K_q = \min(\text{sideA}, \text{sideB})$.

Q26 The security of the trusted-relay scheme fundamentally assumes:

A) Infinite key length B) Zero QBER C) No atmosphere D) The satellite node is trusted (not compromised)

Answer: D. *Framework:* "Trusted-relay" means the relay learns both keys; security holds only if the relay itself is honest — a known limitation of relay QKD.

Q27 If QBER exceeds 11% on a pass, the correct interpretation is:

A) No positive secret key can be distilled on that pass ($\text{SKR} = 0$) B) The data channel fails C) The satellite is lost D) The key is still fully secure

Answer: A. *Framework:* SKR bound goes non-positive; the pass yields raw bits but zero *secret* key.

Q28 The reason cross-talk noise is uniquely a QKD concern (not just a classical-link concern) here is:

A) It raises QBER directly, and QBER caps the secret key by a steep entropy function B) It changes the orbit C) It only affects data D) It is negligible

Answer: A. *Framework:* $H_2(Q)$ rises sharply near threshold, so a small QBER increase from cross-talk can zero the secret key while barely denting classical BER.

Q29 Gauss-Hermite quadrature is the natural choice for averaging over log-normal fading because:

A) In log-amplitude the variable is Gaussian, and GH quadrature integrates $\int e^{-(x^2)} f(x) dx$ exactly for polynomials B) The fading is uniform C) It avoids the noise model D) It is faster than everything

Answer: A. *Framework:* $X \sim N(-\sigma^2_X, \sigma^2_X)$; substitution $h_a = \exp(2\sqrt{(2\sigma^2_X)x} + 2\mu_X)$ turns the fade average into the GH weight form.

Q30 The single most important honesty point a QKD theorist should note about this thesis is:

A) It hides its QBER B) It uses no references C) It openly flags that the 11% BB84 threshold is borrowed from a BBM92-based model, an explicit assumption, not an inherited result D) It claims perfect security with no assumptions

Answer: C. *Framework:* Provenance caveat in Ch. 6 — scientific integrity: state which claims are assumptions vs. validated inheritances.

10.2 P2 — FSO Optical-Channel Engineer

Q31 The composite channel coefficient factorises as:

A) $h = h_g / h_l$ B) $h = h_g \cdot h_l \cdot h_a$ (geometric $\times$ atmospheric $\times$ turbulence) C) $h = h_l \cdot h_a$ only D) $h = h_g + h_l + h_a$

Answer: B. *Framework:* `channel_model` three-component multiplicative model; the factors are physically independent mechanisms.

Q32 Geometric spreading loss h_g arises because:

A) The atmosphere absorbs light B) A finite-divergence Gaussian beam widens with distance, so only a fraction hits the aperture C) Turbulence bends the beam D) The detector is noisy

Answer: B. *Framework:* $h_g = [\text{erf}(\nu R)]^2$; beam radius $w_L \gg$ aperture a_R , so most power spills past the receiver.

Q33 The beam waist is $w_0 =$:

A) $\lambda \cdot \theta C$ B) $\theta C / \lambda$ C) $2 \cdot \theta C / \lambda$ D) $\lambda / (2 \cdot \theta C)$

Answer: D. *Framework:* `compute_beam_waist`; smaller divergence $\theta C \rightarrow$ larger waist $\rightarrow$ slower spreading.

Q34 The Gaussian beam radius at distance L is $w_L = w_0 \cdot \sqrt{1 + (L\lambda/(\pi w_0^2))^2}$. At large L it grows:

A) As $\sqrt{L}$ B) Not at all C) Quadratically D) Linearly with L

Answer: D. *Framework:* `compute_beam_radius`; far-field spreading is linear in L , so slant range directly sets spot size.

Q35 The corrected slant range uses spherical geometry:

- A) $L = R_E + H_S$ B) $L = \sqrt{(R_E^2 \sin^2 \varepsilon + 2R_E \cdot H_S + H_S^2)} - R_E \cdot \sin \varepsilon$ C) $L = H_S / \cos \zeta$ (flat Earth) D) $L = H_S \cdot \sin \varepsilon$

Answer: B. *Framework:* `compute_hg` fix; at $\varepsilon=30^\circ$ this gives 992.8 km vs the flat-Earth 1100 km — an 11% error that fed straight into `wL` and `hg`.

Q36 The corrected normalized aperture radius is $\nu R =$:

- A) $\sqrt{(\pi/2)} \cdot aR / wL$ B) $\sqrt{2} \cdot aR / wL$ C) aR / wL D) $2 \cdot aR / wL$

Answer: A. *Framework:* `compute_hg` fix, verified against Vu et al. 2022; the old $\sqrt{2}$ made `hg` optimistic by $\sim 1.27 \times (+1.05 \text{ dB})$.

Q37 Geometric loss alone at operational elevations is roughly:

- A) 0 dB B) -100 dB C) Over -30 dB D) -1 dB

Answer: C. *Framework:* `wL` $\approx 4\text{--}6 \text{ m}$ vs `aR` = 5 cm $\rightarrow$ the beam is $\sim 100\times$ the aperture, so `hg` dominates the link budget.

Q38 Atmospheric attenuation `hl` follows:

- A) An inverse-square law B) Beer-Lambert: $hl = \exp(-\sigma_{\text{total}} \cdot L_{\text{atm}})$ C) A logarithmic law D) A linear law

Answer: B. *Framework:* `compute_hl`; extinction accumulates exponentially along the slant path $L_{\text{atm}} = (H_{\text{atm}} - H_U) / \cos \zeta$.

Q39 The clear-air extinction coefficient uses the Kruse/Kim model:

- A) $\sigma = \lambda / V$ B) $\sigma = V / 3.912$ C) $\sigma = 3.912 / V$ only D) $\sigma = (3.912 / V) \cdot (\lambda / 550)^{-q(V)}$

Answer: D. *Framework:* `compute_sigma_visibility`; V is meteorological visibility, $q(V)$ the size-distribution exponent.

Q40 At 1550 nm vs visible light, the wavelength factor $(\lambda/550)^{-q}$ makes clear-air scattering:

- A) Weaker (longer wavelength scatters less off small aerosols) B) Infinite C) Stronger D) Identical

Answer: A. *Framework:* Mie/Rayleigh scaling; a key reason telecom wavelength helps the link budget.

Q41 The tropical rain attenuation model is:

- A) $\sigma_{\text{rain}} = 3.912 / R$ B) $\sigma_{\text{rain}} = R$ C) $\sigma_{\text{rain}} = \alpha \cdot R^p / 4.343$, with $\alpha=0.509$, $p=0.63$ (Koné 2024, Abidjan) D) $\sigma_{\text{rain}} = R^2$

Answer: C. *Framework:* `compute_sigma_rain`; the power-law was calibrated on real tropical FSO measurements, not textbook constants.

Q42 Why divide the dB/km rain coefficient by 4.343 to get km^{-1} ?

- A) It is the rain rate B) It is the wavelength ratio C) Arbitrary scaling D) $4.343 = 10 / \ln(10)$ converts dB to natural-log (Nepers) for the exponential Beer-Lambert form

Answer: D. *Framework:* dB $\rightarrow$ linear: $10 \cdot \log_{10}(e) = 4.343$; needed because `hl` uses `exp()`, not $10^{\wedge}()$.

Q43 Clear-air and rain extinction are combined by:

- A) Multiplying the coefficients B) Taking the minimum C) Adding them: $\sigma_{\text{total}} = \sigma_{\text{clear}} + \sigma_{\text{rain}}$ (independent mechanisms) D) Taking the maximum

Answer: C. *Framework:* `compute_hl`; independent extinction processes add in the exponent (their transmissions multiply).

Q44 Turbulence fading `ha` is characterised first by the Rytov variance, corrected to:

- A) $\sigma^2_R = 0.56 \cdot k^{(7/6)} \cdot \dots$ B) $\sigma^2_R = k^2$ C) $\sigma^2_R = Cn^2$ D) $\sigma^2_R = 2.25 \cdot k^{(7/6)} \cdot \sec^{(11/6)}(\zeta) \cdot \int Cn^2(h) (h - H_U)^{(5/6)} dh$

Answer: D. *Framework:* `compute_rytov_variance` fix ($0.56 \rightarrow 2.25$), the standard plane-wave slant-path coefficient (Andrews & Phillips).

Q45 The log-amplitude variance relates to Rytov variance by:

- A) $\sigma^2_X = \sqrt{\sigma^2_R}$ B) $\sigma^2_X = \sigma^2_R / 4$ C) $\sigma^2_X = 4 \cdot \sigma^2_R$ D) $\sigma^2_X = \sigma^2_R$

Answer: B. *Framework:* `compute_rytov_variance`; the earlier double-division-by-4 bug understated scintillation $16\times$.

Q46 The $Cn^2(h)$ vertical profile used is:

- A) Constant with altitude B) Hufnagel-Valley 5/7 (three terms: jet-stream, mid-layer, ground) C) Linear decay D) Kolmogorov only

Answer: B. *Framework:* `compute_Cn2_HV`; note the $(10^{-5}h)^{10}$ term — using 1 would err by 10 orders of magnitude.

Q47 This link operates in which turbulence regime?

- A) No turbulence B) Saturated C) Weak ($\sigma^2_R < 1$, deep in the weak regime), so log-normal fading applies D) Strong ($\sigma^2_R \gg 1$)

Answer: C. *Framework:* `get_turbulence_model`; $\sigma^2_R < 0.3 \rightarrow$ log-normal; the satellite-downlink slant path stays weak.

Q48 Log-normal fading is modelled as $h_a = \exp(2X)$, $X \sim N(\mu_X, \sigma^2 X)$ with $\mu_X = -\sigma^2 X$ so that:

- A) $E[h_a] = \sigma^2 X$ B) $E[h_a] = 0$ C) $E[h_a] = 1$ (unit mean, energy-conserving fade) D) $E[h_a] = \infty$

Answer: C. *Framework:* `sample_ha_lognormal`; the $-\sigma^2 X$ mean shift enforces unit-mean multiplicative fading.

Q49 When would the Gamma-Gamma distribution be used instead of log-normal?

- A) For moderate-to-strong turbulence ($\sigma^2_R \geq 0.3\text{--}0.5$), where log-normal underestimates deep fades B) Never C) Always D) Only at night

Answer: A. *Framework:* `compute_gamma_gamma_params`; two-parameter model capturing large- and small-scale scintillation; not needed here.

Q50 The atmospheric slant path L_{atm} scales with elevation as:

- A) Independent of elevation B) $1/\cos \zeta$ (longer path at low elevation / high zenith angle) C) $\sin \zeta$ D) $\cos \zeta$

Answer: B. *Framework:* `compute_hl`; low passes traverse more atmosphere, compounding both attenuation and turbulence.

Q51 As elevation drops from 90° to 30° , h_g (geometric loss):

- A) Worsens (longer slant range $\rightarrow$ bigger spot $\rightarrow$ more loss) B) Becomes positive C) Stays constant D) Improves

Answer: A. *Framework:* $L \propto 1/\sin \varepsilon$ roughly; larger $L \rightarrow$ larger $wL \rightarrow$ smaller $\text{erf}(\nu R) \rightarrow$ more loss.

Q52 Why is H_{atm} set to 20 km, and what is the honest caveat?

- A) It is the beam waist B) It is exact C) It is the satellite altitude D) It is an effective Beer-Lambert path length; the physical aerosol scale height is only $\sim 1\text{--}2$ km, so 20 km lacks a fully justified provenance (flagged in Limitations)

Answer: D. *Framework:* `H_ATM_KM = 20.0`; a known open item — do not defend it as derived.

Q53 The receiver aperture radius a_R is:

- A) 1 m (like Micius ground station) B) 1 mm C) 5 cm (small, TBIRD-like, favouring cheap multi-station deployment) D) 50 cm

Answer: C. *Framework:* `a_R = 0.05`; the SWaP trade-off — small aperture collects less of an already-weak key signal.

Q54 The beam divergence half-angle θ_C is:

- A) $10 \mu\text{rad}$ B) 90° C) 1 mrad D) 1 rad

Answer: A. *Framework:* `theta_C_urad = 10`; tighter divergence would reduce h_g but demands finer pointing (PAT).

Q55 Pointing/Acquisition/Tracking (PAT) error is treated in this project how?

- A) Ignored entirely B) Only its *time* cost is folded into the handover penalty; the residual angular/SNR loss is NOT modelled, making h_g a lower bound on true loss C) Modelled as rain D) Fully modelled as beam wander

Answer: B. *Framework:* h_g assumes perfect boresight (Farid & Hranilovic 2007 for the omitted jitter term); a stated scope limit.

Q56 Off-nadir angle η at the satellite satisfies $\sin \eta = R_E \cdot \cos \varepsilon / (R_E + h)$. It is zero when:

- A) Always B) The satellite is at the station's zenith ($\varepsilon = 90^\circ$) C) At $\varepsilon = 30^\circ$ D) The satellite is at the horizon

Answer: B. *Framework:* `off_nadir_deg`; complements the receive-side zenith angle in the pass ledger.

Q57 Increasing visibility V (clearer air) affects h_l by:

- A) Decreasing transmission B) No effect C) Only affecting turbulence D) Increasing transmission (smaller $\sigma_{\text{clear}} \rightarrow$ larger h_l)

Answer: D. *Framework:* $\sigma_{\text{clear}} = 3.912/V(\dots)$; higher $V \rightarrow$ lower extinction $\rightarrow$ h_l closer to 1.

Q58 In this project, "cloud" is modelled as:

- A) A binary outage state (link fully OFF), not a partially transmitting medium B) Gradual attenuation added to h_l C) A rain term D) A turbulence term

Answer: A. *Framework:* Three-state weather model; at optical wavelengths thick cloud is total blockage, so no multiple-scattering transmission is modelled.

Q59 Why is single-scattering Beer-Lambert sufficient despite cloud involving multiple scattering?

- A) Cloud is handled as outage, not transmission; clear/rain states are dominated by extinction along a narrow line-of-sight beam **B)** The beam is very wide C) The wavelength is too long **D)** Multiple scattering does not exist

Answer: A. *Framework:* The Jacquemoud-style objection answered: cloud = binary OFF, so refining its transmission is moot.

Q60 The dominant term in the FSO link budget (largest dB loss) is:

- A) Thermal noise **B)** Turbulence C) Rain **D)** Geometric spreading hg (>30 dB), far exceeding atmospheric and turbulence terms in clear air

Answer: D. *Framework:* Link-budget accounting; this is why aperture and divergence choices dominate design.

10.3 P3 — RF / Receiver / Detection Engineer

Q61 The total receiver noise variance is composed of:

- A) Cross-talk only **B)** Shot noise only C) $\sigma^2_N = \sigma^2_{\text{shot}} + \sigma^2_{\text{thermal}} + \sigma^2_{\text{bg}} + \sigma^2_{\text{CT}}$ **D)** Thermal noise only

Answer: C. *Framework:* `compute_noise`; four independent additive contributions, each with a distinct physical origin.

Q62 The effective noise bandwidth Δf is:

- A) $2 \cdot R_b$ **B)** $R_b / 2$ C) R_b^2 **D)** R_b

Answer: B. *Framework:* `compute_noise`; $\Delta f = R_b/2$ sets the noise integration bandwidth for all photocurrent-driven terms.

Q63 Shot noise variance is (corrected):

- A) $4kT/RL \cdot \Delta f$ **B)** Independent of PT C) $2q \cdot Re \cdot (\frac{1}{2} \cdot PT \cdot m_k \cdot hg \cdot hl) \cdot \Delta f$ **D)** $2q \cdot Re \cdot (\frac{1}{2} \cdot PT \cdot hg \cdot hl) \cdot \Delta f$ — driven by the TOTAL DC photocurrent, NOT scaled by m_k

Answer: D. *Framework:* `compute_noise` fix; the shared photodetector sees full received power; the earlier m_k -scaling understated shot noise $\sim 20\times$.

Q64 Why is shot noise set by total photocurrent, not the key channel's m_k ?

- A) m_k affects only thermal noise **B)** m_k is always 1 C) Shot noise ignores signal **D)** Both channels are demultiplexed AFTER photodetection; the detector sees the full optical power regardless of subcarrier depths

Answer: D. *Framework:* Fig. 3.1 architecture (RF splitter + BPF after the PIN); verified against Vu et al. 2022 Eq. 8.

Q65 Thermal noise variance is:

- A) Proportional to $hg \cdot hl$ **B)** Zero at night C) $(4 \cdot kB \cdot T / RL) \cdot \Delta f$ — independent of optical power **D)** $2q \cdot Re \cdot P \cdot \Delta f$

Answer: C. *Framework:* `compute_noise`; Johnson-Nyquist noise sets a floor even with zero signal.

Q66 Background noise σ^2_{bg} is zero in the baseline because:

- A) There is no atmosphere **B)** Night-time operation is assumed ($P_{bg} = 0$), no scattered sunlight C) The detector is perfect **D)** Cross-talk cancels it

Answer: B. *Framework:* `PBG_DEFAULT = 0.0`; this is the crux of the unresolved day/night tension.

Q67 The cross-talk variance (corrected) is:

- A) $4kT/RL$ **B)** $(Re \cdot PT \cdot m_k \cdot hg \cdot hl)^2 / 8$ C) $(Re \cdot PT \cdot m_a \cdot hg \cdot hl)^2 \cdot 10^{(-Iso/10)} / 8$ **D)** $(Re \cdot PT \cdot m_a \cdot hg \cdot hl)^2 \cdot 10^{(-Iso/10)} / 2$

Answer: C. *Framework:* `compute_noise` fix ($/2 \rightarrow /8$); the photocurrent's leading $\frac{1}{2}$ makes the leaking sinusoid amplitude $\frac{1}{2} \cdot Re \cdot PT \cdot m_a \cdot hg \cdot hl$, variance $A^2/2$.

Q68 Cross-talk noise is unique because it scales as:

- A) $1/P$ **B)** Received power P (linearly) C) Received power squared, P^2 — so it dominates exactly when the link is strongest **D)** Independent of P

Answer: C. *Framework:* $\sigma^2_{CT} \propto (hg \cdot hl)^2$; the counter-intuitive result that a stronger link brings proportionally more cross-talk.

Q69 After the $/8$ correction, cross-talk relative to thermal noise ranges over elevation from:

- A) $\sim 0.5\times$ to $\sim 290\times$ thermal (dominant except near 30°) **B)** Always negligible C) Exactly equal **D)** Always $1000\times$

Answer: A. *Framework:* `IMPACT_EVALUATION.md`; "thermal-dominated" is only true near 30° — corrected in the thesis text.

Q70 Iso (filter isolation) of 15 dB means the leaked data power is suppressed by:

- A) A factor $10^{(-15/10)} \approx 0.032$ in power B) $15\times$ C) $150\times$ D) $1.5\times$

Answer: A. *Framework:* ISO_DEFAULT = 15.0; imperfect RF bandpass filtering between subcarriers f1 and f2.

Q71 The photodetector is an InGaAs PIN with responsivity $R_e =$:

- A) 90 A/W B) 0.09 A/W C) 9 A/W D) 0.9 A/W

Answer: D. *Framework:* RE_DEFAULT = 0.9; typical for 1550 nm InGaAs; converts optical W to photocurrent A.

Q72 The dual-threshold decision boundaries are:

- A) $d_0 = -\sigma_N$, $d_1 = \sigma_N$ B) $d_0 = 0$, $d_1 = \infty$ C) $d_0 = -i_{\text{mean}} - \zeta_{\text{scale}} \cdot \sigma_N$; $d_1 = +i_{\text{mean}} + \zeta_{\text{scale}} \cdot \sigma_N$ D) $d_0 = d_1$

Answer: C. *Framework:* compute_thresholds; $i_{\text{mean}} = \frac{1}{2} \cdot R_e \cdot G_k \cdot P_T \cdot m_k \cdot h_g \cdot h_l$ is the nominal key-signal amplitude.

Q73 A sample falling inside the guard zone $[d_0, d_1]$ is:

- A) Discarded (sifted out) as ambiguous B) Decoded as 1 C) Counted as an error D) Decoded as 0

Answer: A. *Framework:* DT rule; only confident samples (outside the zone) are kept, trading yield for lower error.

Q74 Widening ζ_{scale} (guard zone) trades:

- A) Nothing B) Higher QBER for higher P_{sift} C) Lower QBER (fewer errors) for lower P_{sift} (more discards) D) More background noise

Answer: C. *Framework:* zeta_scale = 2.0; a classic detection trade-off between error rate and sift yield.

Q75 The classical data channel BER is averaged over fading via:

- A) Gauss-Hermite quadrature over log-normal h_a , $\text{BER} = (1/\sqrt{\pi}) \sum w_k \cdot Q(A_0 \cdot h_{a_k} / \sigma_{\text{ND}})$ B) Monte Carlo C) A single SNR D) Beer-Lambert

Answer: A. *Framework:* compute_BER_CC; $A_0 = G_d \cdot R_e \cdot P_T \cdot m_a \cdot h_g \cdot h_l$ is the data-signal amplitude.

Q76 $Q(x)$, the Gaussian tail function used in detection, equals:

- A) $\frac{1}{2} \cdot \text{erfc}(x/\sqrt{2})$ B) $\exp(-x)$ C) $\text{erf}(x)$ D) $1 - x$

Answer: A. *Framework:* _q_function; probability a unit Gaussian exceeds x , the building block of all detection probabilities.

Q77 In the reverse direction, key-into-data cross-talk scales as:

- A) $m_k \cdot m_a$ B) m_k^2 (small, since $m_k \ll m_a$) C) Zero D) m_a^2

Answer: B. *Framework:* crosstalk_variance($m_{\text{leak}}=m_k$); the data receiver's own noise budget includes this small reverse leakage.

Q78 The data channel's own noise budget consists of:

- A) Only thermal B) Its own shot + thermal noise (driven by m_a) plus reverse cross-talk from m_k C) Background only D) Only cross-talk

Answer: B. *Framework:* _data_channel_noise_std; symmetric treatment of the two subcarriers.

Q79 Data throughput in the power-split analysis is modelled as:

- A) $R_b \cdot (1 - H_2(\text{BER}))$ B) $R_b \cdot \text{BER}$ C) $R_b \cdot (1 - \text{BER})$ — uncoded goodput D) R_b only

Answer: C. *Framework:* evaluate_split; a deliberate simple uncoded goodput choice, stated explicitly.

Q80 The Pareto frontier in power-split trades:

- A) QBER vs elevation B) SKR (key) vs data throughput over the (m_k, m_a) grid C) h_g vs h_l D) Rain vs cloud

Answer: B. *Framework:* pareto_frontier; non-dominated operating points where you cannot raise one without lowering the other.

Q81 Adaptive power-split chooses:

- A) $m_k = m_a$ always B) The largest m_k C) The smallest m_k such that $\text{SKR} \cdot 1000 \cdot T_{\text{pass}} \geq K_{\text{req}}$ (just enough key, leaving power for data) D) A random split

Answer: C. *Framework:* adaptive_split; demand-driven allocation per pass.

Q82 Why does raising the data modulation m_a hurt the key channel?

- A) It has no effect B) It changes the orbit C) It lowers h_g D) $\sigma_{\text{CT}}^2 \propto m_a^2$, so more data power leaks more cross-talk into the key receiver, raising QBER

Answer: D. *Framework:* The two channels are coupled through the shared detector — the core SIKD design tension.

Q83 If PT (transmit power) is 1 W, that corresponds to:

- A) 10 dBm B) 0 dBm C) 3 dBm D) 30 dBm

Answer: D. *Framework:* $10 \cdot \log_{10}(1000 \text{ mW}) = 30 \text{ dBm}$; the nominal Paper 2 transmit power.

Q84 The receiver temperature T is:

- A) 1000 K B) 280 K (near ambient) C) 4 K (cryogenic) D) 0 K

Answer: B. *Framework:* $T_{\text{DEFAULT}} = 280$; ambient operation — no cryogenic cooling, unlike single-photon detectors.

Q85 The load resistance RL is:

- A) 1 k Ω B) 1 Ω C) 1 M Ω D) 50 Ω

Answer: A. *Framework:* $RL_{\text{DEFAULT}} = 1e3$; appears in the thermal noise term $4kT/RL$.

Q86 Why does the daytime background question matter *less* for a PIN/DT-DD receiver than a photon counter?

- A) PIN detectors are immune to light B) Photon counters have no background C) The already-large signal power and P^2 -scaling cross-talk mean background may sit well below the dominant noise; an order-of-magnitude estimate suggests a small penalty (unproven) D) Daytime has no sunlight

Answer: C. *Framework:* `IMPACT_EVALUATION.md` FOV sweep; a *plausibility* argument, explicitly not a proof.

Q87 If Iso were improved from 15 dB to 25 dB, cross-talk variance would drop by:

- A) $100\times$ B) Unchanged C) $10\times$ D) A factor 10 ($10^{(-10/10)} = 0.1$)

Answer: D. *Framework:* $\sigma^2_{\text{CT}} \propto 10^{(-\text{Iso}/10)}$; better optical/RF filtering is a direct QBER lever.

Q88 The "signal" that sets the dual-threshold amplitude i_{mean} is proportional to:

- A) m_a B) Iso C) m_k (key channel modulation), times $\frac{1}{2} \cdot \text{Re} \cdot G_k \cdot PT \cdot hg \cdot hl$ D) σ_N

Answer: C. *Framework:* `compute_thresholds`; the key signal rides on m_k , the data on m_a .

Q89 Which noise term does NOT depend on the received optical power?

- A) Shot noise B) Background (when $P_{bg} > 0$) C) Thermal noise ($4kT/RL \cdot \Delta f$) D) Cross-talk

Answer: C. *Framework:* Thermal noise is a circuit property; it sets the floor when the link is weak.

Q90 The receiver-engineering headline of the v3 correction is:

- A) Thermal noise was removed B) Fixing the cross-talk factor ($/2 \rightarrow /8$) cut the dominant noise $4\times$, roughly quartering QBER and raising SKR to tens of Mbps C) QBER got worse D) The detector was changed

Answer: B. *Framework:* `IMPACT_EVALUATION.md`; the single largest-impact fix of the v3 round.

10.4 P4 — Orbital-Mechanics / Aerospace Analyst

Q91 The constellation used is:

- A) A single GEO satellite B) Real Starlink Shell-1, ~1,019 satellites, from live TLEs C) A synthetic Walker constellation D) GPS

Answer: B. *Framework:* Shell-1 filtered by inclination 53.15° – 53.22° from the active catalogue; real, reproducible data.

Q92 Orbits are propagated with:

- A) A straight line B) Kepler's two-body equations only C) SGP4 (which folds in J2 and drag perturbations) from TLEs D) Numerical N-body

Answer: C. *Framework:* `orbital_mechanics`; SGP4 is the standard TLE propagator, capturing perturbations Kepler ignores.

Q93 A TLE (Two-Line Element set) encodes:

- A) Only the satellite name B) Orbital elements (inclination, RAAN, eccentricity, mean anomaly, etc.) at a reference epoch C) The weather D) The ground station location

Answer: B. *Framework:* SGP4 input; accuracy degrades with time since epoch — a data-quality concern.

Q94 Satellite altitude H_S is:

- A) 550 km (Shell-1 LEO) B) 36,000 km (GEO) C) 400 km (ISS) D) 20,200 km (MEO)

Answer: A. *Framework:* $H_S = 550$; the channel model is parameterised for this single altitude.

Q95 Inclination 53° means the satellite reaches maximum latitude of:

- A) 23.5° B) 0° C) 90° D) 53° N and S

Answer: D. *Framework:* Inclination = angle of orbital plane to equator; 53° covers ~95% of world population (all of ASEAN).

Q96 Why does SpaceX use 53° for Shell-1 (not the student's choice)?

- A) Polar coverage B) Launch efficiency from Cape Canaveral ($\sim 28.5^\circ$ N) plus coverage of most populated latitudes C) To optimise ASEAN specifically D) Random

Answer: B. *Framework:* Launch delta-v and market coverage; the student merely *uses* this shell as a real proxy.

Q97 The ground-coverage radius formula is:

- A) $R_{\text{cov}} = 2 \cdot h$ B) $R_{\text{cov}} = h \cdot \tan \varepsilon$ C) $R_{\text{cov}} = R_E$ D) $R_{\text{cov}} = R_E \cdot \psi$, with $\psi = \arccos(R_E / (R_E + h) \cdot \cos \varepsilon) - \varepsilon$

Answer: D. *Framework:* `ground_coverage_radius_km`; the Earth central angle ψ subtended at the given elevation mask.

Q98 At a 30° elevation mask ($h = 550$ km), the ground-coverage radius is:

- A) 950 km B) 573 km C) 1587 km D) 793 km

Answer: D. *Framework:* `link_geometry`; the paper's earlier "950 km" was actually the slant range, not the ground radius — a corrected bug.

Q99 At a 40° elevation mask, the coverage radius shrinks to:

- A) 573 km B) 400 km C) 793 km D) 1000 km

Answer: A. *Framework:* Stricter mask $\rightarrow$ smaller footprint ($\sim 28\%$ loss), the security-vs-coverage trade-off.

Q100 Two ground stations can be served by one satellite pass simultaneously only if their separation is under:

- A) $2 \cdot R_{\text{cov}}$ (footprint circles overlap): 1587 km at 30° , 1146 km at 40° B) h C) R_E D) R_{cov}

Answer: A. *Framework:* `dual_downlink_max_separation_km`; the DUAL/SF classification threshold.

Q101 The great-circle distance between two cities is computed with:

- A) Euclidean distance B) Slant range C) The haversine formula D) Manhattan distance

Answer: C. *Framework:* `haversine_km`; symmetric, depends only on lat/lon.

Q102 A satellite pass is defined as:

- A) A downlink packet B) A contiguous window where elevation $\geq$ the mask C) Any moment the satellite is in orbit D) One full orbit

Answer: B. *Framework:* `extract_passes`; rise/set are the mask-crossing times.

Q103 Rise and set times are refined by:

- A) Rounding to the sample grid B) Linear interpolation between straddling elevation samples (30 s step) C) Ignoring them D) SGP4 alone

Answer: B. *Framework:* `_interp_crossing_time`; avoids grid bias, important for brief high-mask passes.

Q104 A typical Shell-1 pass over an ASEAN station lasts about:

- A) 5 minutes B) 30 seconds C) ~ 204 seconds (~ 3.4 min) D) 1 hour

Answer: C. *Framework:* LEO at 550 km moves fast; short passes despite high frequency.

Q105 Why are passes short but frequent over ASEAN specifically?

- A) Satellites slow down there B) ASEAN is near the equator, the "belly" of a 53° orbit where the ground-projected angular velocity is highest C) Fewer satellites D) ASEAN is at high latitude

Answer: B. *Framework:* Orbital geometry; low-latitude sites get many fast passes vs slow dwell near max latitude.

Q106 The number of satellites simultaneously visible (30° mask) averages about:

- A) 14–15 B) 100 C) ~ 3.1 – 3.5 (minimum 0) D) 1

Answer: C. *Framework:* Corrected count (the earlier "14–15" came from using dates outside valid TLE epoch — a data-quality lesson).

Q107 Off-nadir angle at the satellite grows as elevation:

- A) Increases B) Stays fixed C) Reaches 90° D) Decreases (lower passes $\rightarrow$ larger off-nadir pointing)

Answer: D. *Framework:* `off_nadir_deg`; $\sin \eta = R_E \cdot \cos \varepsilon / (R_E + h)$; zero at zenith.

Q108 The RAAN (Right Ascension of Ascending Node) drifts over time due to:

- A) Nothing B) The Moon only C) Solar wind only D) J2 perturbation (Earth's equatorial bulge) causing nodal precession

Answer: D. *Framework:* Why SGP4 (not Kepler) is needed; also why inclination shows a small 53.15–53.22° spread.

Q109 Store-and-forward relay (single-hop, no ISL) requires:

- A) A ground cable B) The SAME satellite to visit the pickup city, then later the drop-off city C) Any two satellites D) Simultaneous visibility

Answer: B. *Framework:* `citypair_feasibility`; no ISL means one satellite physically carries the key.

Q110 Single-hop SF relay latency is DIRECTIONAL ($i \rightarrow j \neq j \rightarrow i$) because:

- A) A satellite's ground-track heading (ascending vs descending) makes the visit order depend on the cities' latitudes B) Distance differs C) The Earth rotates D) Weather differs

Answer: A. *Framework:* `sf_latency_minutes`; measured $\sim 100\times$ asymmetry for one ASEAN pair in tests.

Q111 The worst-case single-hop SF latency measured was about:

- A) 1 hour B) 10 seconds C) 5 minutes D) ~ 711 minutes (the motivation for ISL multi-hop)

Answer: D. *Framework:* ISL relay pivot; single-hop can force a wait for the same satellite to come around.

Q112 ISL multi-hop relay finds delivery paths by:

- A) Kepler propagation B) BFS on the (static) inter-satellite link graph for hop distance C) Dijkstra on weighted edges D) Random walk

Answer: B. *Framework:* `bfs_hop_distance`; the "+Grid" mesh connects each satellite to plane-neighbours.

Q113 The per-hop ISL latency of 30 s represents:

- A) Conservative on-board trusted-node processing/re-keying overhead (propagation is only ms) B) Orbital period C) Handover time D) Laser propagation delay

Answer: A. *Framework:* `ISL_HOP_LATENCY_S = 30`; deliberately pessimistic so the result is credible.

Q114 Treating the ISL topology as static over a relay is justified because:

- A) Satellites never move B) ISL is instantaneous C) `n_hops`·30 s is a few minutes, far less than the ~ 95 -min orbital period, so the mesh barely reconfigures D) There are no orbits

Answer: C. *Framework:* `isl_relay` modeling assumption; timescale separation.

Q115 `time_optimal_relay` vs `capacity_optimal_relay` differ in that:

- A) time-optimal ignores hops B) They are identical C) time-optimal takes the earliest feasible drop-off; capacity-optimal trades a little latency for a higher-elevation (higher-SKR) drop-off D) capacity-optimal is slower always

Answer: C. *Framework:* `isl_relay`; latency budget factor $3\times$ lets capacity-optimal pick a much better-angled pass.

Q116 Why use only $\sim 1,019$ satellites (Shell-1) rather than all $\sim 10,000$?

- A) Laziness B) Shell-1 is special to ASEAN C) The channel model is parameterised for one altitude (550 km); mixing shells needs altitude-dependent SKR tables — a stated scope choice, not an ASEAN-specific reason D) Shell-1 is the only real data

Answer: C. *Framework:* The honest defense answer; multi-shell is future work but weather remains the bottleneck.

Q117 Which perturbation most directly degrades TLE accuracy at 550 km over time?

- A) Atmospheric drag (plus J2), so error grows with time since epoch B) Solar radiation pressure C) Lunar tides D) Magnetic torque

Answer: A. *Framework:* EDA Layer 6 (data quality); always plot error vs time-since-epoch before trusting downstream results.

Q118 "Snapshot bias" in constellation analysis means:

- A) A TLE format issue B) A blurry image C) A camera error D) Data taken at one instant may not represent the full orbital cycle / deployment phase / season

Answer: D. *Framework:* EDA Layer 6; a reason to use a homogeneous, near-epoch template.

Q119 Pass frequency per day is estimated by:

- A) Observed passes $\times (24 / \text{duration_hours})$ — extrapolating the enumerated window B) The orbital period C) SGP4 mean motion D) Counting one pass

Answer: A. *Framework:* `pass_frequency_per_day`; $\sim 1,500$ passes/day/station reported.

Q120 The elevation-mask trade-off (30° vs 40°) is fundamentally between:

- A) Rain and cloud B) Day and night C) Coverage footprint (larger at 30°) vs link security/quality (better QBER at 40°) D) Cost and mass

Answer: C. *Framework:* Geometry + performance coupling; ~28% footprint sacrificed for a safer QBER margin.

10.5 P5 — Meteorologist / Climate-Data Scientist

Q121 The weather model has how many states?

- A) One B) Two (clear, cloud) C) Four D) Three: clear, rain, cloud-outage

Answer: D. *Framework:* `weather_model`; clear and rain attenuate hl, cloud switches the link OFF.

Q122 The effective SKR under the three-state model is:

- A) $\text{SKR}_{\text{eff}} = p_{\text{clear}} \cdot \text{SKR}_{\text{clear}} + p_{\text{rain}} \cdot \text{SKR}_{\text{rain}}$ (cloud contributes 0) B) $\text{SKR}_{\text{clear}}$ only C) $\text{SKR}_{\text{clear}} \cdot \text{SKR}_{\text{rain}}$ D) $p_{\text{cloud}} \cdot \text{SKR}_{\text{clear}}$

Answer: A. *Framework:* `compute_effective_skr_3state`; a probability-weighted average with the cloud state zeroed.

Q123 Link availability A is defined as:

- A) SKR_{eff} B) $1 - p_{\text{rain}}$ C) $A = p_{\text{clear}} + p_{\text{rain}} = 1 - P_{\text{cloud}}$ D) p_{clear} only

Answer: C. *Framework:* `compute_reliability_metrics`; availability is invariant to `rain_fraction` — only the outage (cloud) matters.

Q124 The weather data source is:

- A) Satellite imagery B) A single year of station data C) Real ERA5 / ERA5-Land reanalysis (Open-Meteo), 2015–2024, 10 years D) Hand-estimated tropical averages

Answer: C. *Framework:* `scripts/01_fetch_weather_era5`; hourly cloud cover and precipitation, per city.

Q125 A "cloudy hour" (link outage) is defined by the threshold:

- A) Cloud cover $\geq 50\%$ B) Any cloud C) Rain > 0 D) Cloud cover $\geq 85\%$ (`CLOUD_OUTAGE_THRESHOLD`)

Answer: D. *Framework:* `weather_stats.CLOUD_OUTAGE_THRESHOLD = 85.0`; a modelling choice with a sensitivity analysis (Annex A).

Q126 Link availability across the 8 ASEAN cities ranges from:

- A) 10.7% (Kuala Lumpur, Nov) to 96.5% (Bangkok, Feb) B) 80% to 90% C) 50% to 60% D) 0% to 100%

Answer: A. *Framework:* `ASEAN_CLIMATE_DATA`; KL Nov $P_{\text{cloud}} = 0.893 \rightarrow A = 0.107$; Bangkok Feb $P_{\text{cloud}} = 0.035 \rightarrow A = 0.965$.

Q127 Availability is driven almost entirely by:

- A) Turbulence B) Cloud outage, not rain C) Rain attenuation D) Elevation angle

Answer: B. *Framework:* At optical wavelengths cloud is total blockage; rain merely attenuates. The headline meteorological finding.

Q128 The diurnal timing of PRECIPITATION peaks:

- A) In the afternoon (13–17h local), a convective thunderstorm signature (8/8 cities) B) Uniformly C) At dawn D) Overnight

Answer: A. *Framework:* `load_hourly_rain_climatology`; verified against the raw precip series.

Q129 The diurnal timing of TOTAL CLOUD COVER peaks:

- A) At noon B) In the afternoon C) Near dawn/overnight (radiative-cooling stratus), lowest mid-morning 9–12h (6/8 cities) D) Never

Answer: C. *Framework:* `weather_stats.docstring`; cloud and rain have DIFFERENT diurnal phases — a subtle, important distinction.

Q130 Therefore the best combined cloud+rain optical window is:

- A) Midnight B) Mid-morning (~05:00–10:00 local), in DAYLIGHT C) Evening D) Afternoon

Answer: B. *Framework:* The finding that creates the day/night tension: best weather is daytime, but SKR assumes night ($P_{\text{bg}}=0$).

Q131 This day/night tension is problematic because:

- A) Every reported SKR assumes night-time (zero background), yet the clearest sky is in daylight — an unresolved, openly stated conflict
 B) It affects only rain C) It changes the orbit D) It has no effect

Answer: A. *Framework:* Ch. 6/8 limitation; honesty over a genuine unresolved gap.

Q132 `rain_hour_fraction(city)` computes:

- A) Daily rain probability B) Cloud fraction C) Monthly rainfall total D) Fraction of hours per month with $\text{precip} \geq 0.1 \text{ mm/h}$ (real per-city/month values)

Answer: D. *Framework:* `weather_stats.rain_hour_fraction`; replaced the flat 0.15 assumption (real values 0.047–0.620, mean 0.308).

Q133 Why does replacing the flat 0.15 rain fraction NOT change reported availability?

- A) It does change it B) Rain has no effect C) $A = p_{\text{clear}} + p_{\text{rain}} = 1 - P_{\text{cloud}}$ identically for any `rain_fraction`; only `R_mm_h` / seasonal SKR / `K_day` shift D) Cloud is ignored

Answer: C. *Framework:* `compute_reliability_metrics`; a robustness property of the availability metric.

Q134 Monthly rainfall is converted to an hourly rate by:

- A) $R_{\text{mm}_h} = R_{\text{mm}_\text{month}} / (30 \cdot 24 \cdot \text{rain_fraction})$ B) $R_{\text{mm}_h} = R_{\text{mm}_\text{month}} \cdot 30$ C) $R_{\text{mm}_h} = R_{\text{mm}_\text{month}} / 24$ D) $R_{\text{mm}_h} = R_{\text{mm}_\text{month}}$

Answer: A. *Framework:* `rain_mm_month_to_mm_h`; tropical rain is intermittent, so intensity concentrates in the raining hours.

Q135 Visibility `V_km` in the dataset is:

- A) Derived as $V = 15 - 10 \cdot P_{\text{cloud}}$ (no historical visibility variable exists) B) A constant 10 km C) Directly measured by ERA5
 D) From satellite lidar

Answer: A. *Framework:* `weather_model`; a stated derived estimate, matching how the reference lab (Nguyen et al. 2023) also derives `V`.

Q136 Seasons (wet/dry) are classified by:

- A) Calendar (Jun–Nov wet) B) Relative ranking: the 6 rainiest months of each city = wet, 6 driest = dry C) Temperature D) Absolute threshold $R > 100 \text{ mm}$

Answer: B. *Framework:* `get_season`; a relative rule so year-round-rainy Singapore still has a "dry" half (an absolute threshold would leave it empty).

Q137 Inter-city cloud correlation is measured with:

- A) Pearson correlation of daily mean cloud cover over the 10-year record B) Cosine similarity C) Euclidean distance D) Mutual information

Answer: A. *Framework:* `daily_cloud_correlation / correlation_matrix` (8×8); relevant to site-diversity gain.

Q138 `joint_clear_probability` computes:

- A) P(all cities clear) B) P(at least one city clear) directly from the joint daily record (not an independence assumption) C) Rain probability D) Average cloud

Answer: B. *Framework:* `joint_clear_probability`; comparable to Dang 2023's 81.92/96.86/99.44%.

Q139 Why compute joint clear probability from the joint record, not by assuming independence?

- A) Independence is always true B) The record is short C) Cloud fields are spatially correlated; assuming independence would overstate diversity gain D) It is faster

Answer: C. *Framework:* Correlated tropical weather systems; a data-quality principle (check autocorrelation before assuming independence).

Q140 Which city/month has the WORST availability in the dataset?

- A) Kuala Lumpur, November ($P_{\text{cloud}} = 0.893$) B) Bangkok, February C) Manila, March D) Hanoi, January

Answer: A. *Framework:* `ASEAN_CLIMATE_DATA["kuala_lumpur"]`; $A = 0.107$.

Q141 Which city/month has the BEST availability?

- A) Bangkok, February ($P_{\text{cloud}} = 0.035$) B) Jakarta, January C) Kuala Lumpur, November D) Singapore, December

Answer: A. *Framework:* `ASEAN_CLIMATE_DATA["bangkok"]`; $A = 0.965$.

Q142 The reference tropical FSO rain model (α, ρ) comes from measurements in:

- A) Singapore B) Abidjan, Côte d’Ivoire (Koné et al. 2024) C) London D) Hanoi

Answer: B. *Framework:* `compute_sigma_rain`; a real tropical dataset, defensible under a remote-sensing examiner.

Q143 The reanalysis period was extended from 6 years to 10 years (2015–2024) because:

- A) To match a deadline B) It was arbitrary C) Data was missing D) Longer records capture interannual variability; P_{cloud} shifts up to 0.12 between the 6-yr and 10-yr versions — real variability, not error

Answer: D. *Framework:* `data/PROVENANCE.md`; a data-quality upgrade.

Q144 `diurnal_amplitude(city, month)` measures:

- A) Temperature swing B) Rain total C) max - min of the 24-hour P_{cloud} profile for that month D) Wind speed

Answer: C. *Framework:* `weather_stats.diurnal_amplitude`; quantifies how much the best-window advantage varies within a day.

Q145 The ground-station coordinates were verified against independent geocoders because:

- A) For fun B) To match the satellite C) TLEs need it D) The reference paper’s HCMC point was ~35–40 km off from the true city centre, which would bias the fetched weather

Answer: D. *Framework:* `weather_model` provenance note; wrong coordinates → wrong climatology.

Q146 Why is the single-state effective SKR ($\times (1 - P_{\text{cloud}})$) less accurate than the three-state model?

- A) It is more accurate B) If `SKR_clear` already includes rain attenuation, it underestimates true SKR (~340× in the wet season); the three-state model separates clear and rain C) It ignores cloud D) It has no rain term

Answer: B. *Framework:* `compute_effective_skr` note; motivates `compute_effective_skr_3state`.

Q147 Cloud is treated as complete optical blockage, citing:

- A) A guess B) Rain data C) Turbulence theory D) The physical fact that thick cloud optical depth blocks a narrow laser beam (referenced to Potter/JPL-style optical-blockage literature)

Answer: D. *Framework:* Binary outage assumption; distinct from RF, where cloud is nearly transparent.

Q148 From a meteorology standpoint, the correlation matrix matters for scheduling because:

- A) It changes QBER B) It sets the orbit C) Highly correlated cities cloud over together, so multi-station diversity helps less than if outages were independent D) It has no use

Answer: C. *Framework:* Site-diversity theory; correlated failures cap the benefit of adding stations.

Q149 The “measurable precipitation” hourly threshold (0.1 mm/h) differs from the daily “rain day” threshold (1.0 mm/day) because:

- A) They are the same B) Rain is binary C) Both are arbitrary D) One counts raining HOURS (for p_{rain}), the other flags rain DAYS — distinct physical quantities

Answer: D. *Framework:* `RAIN_HOUR_MM_MIN` vs `RAIN_DAY_MM_MIN`; a common confusion the code deliberately separates.

Q150 The meteorological headline a climate-data examiner should take away:

- A) Weather is negligible B) Rain is the main problem C) ERA5 is unreliable D) Cloud (not rain) governs tropical optical availability, availability swings 11%–96% by city/month, and the clearest window is daytime — creating the day/night tension

Answer: D. *Framework:* Synthesis of the weather chapter; the story a remote-sensing reviewer (Jacquemoud) would probe.

10.6 P6 — Network / Scheduling / Operations Engineer

Q151 The scheduling problem has how many levels?

- A) Four B) One C) Three D) Two: (1) station-satellite matching, (2) pass-to-pair key allocation

Answer: D. *Framework:* `schedulers`; a nested optimisation, not a single routing problem.

Q152 The two hard matching constraints are:

- A) No constraints B) Each station uses all satellites C) Each station uses ≤ 1 satellite AND each satellite serves ≤ 1 station (one optical head each) D) Unlimited assignments

Answer: C. *Framework:* `schedulers` $\sum_s x_{s,gs} \leq 1, \sum_g x_{s,gs} \leq 1$; the exclusivity that creates contention.

Q153 The per-step objective is:

- A) min QBER B) max satellites used C) min elevation D) $\max \sum w_{gs} \cdot x_{gs} - c_h \cdot (\text{number of handovers})$

Answer: D. *Framework:* `schedulers obj_step`; weight w = weather-aware SKR_eff, c_h penalises handovers.

Q154 ALG-0 (baseline) is:

- A) Optimal matching B) Sticky, weather-blind, per-station elevation-priority (keep current satellite until it sets) C) ILP D) Random assignment

Answer: B. *Framework:* `schedule_baseline`; deliberately independent per-station (no coordination).

Q155 Why is ALG-0 "sticky" rather than re-evaluating every step?

- A) To be lazy B) To match ALG-2 C) It is faster D) Per-step re-evaluation let the baseline opportunistically hop satellites, inflating the apparent gain of weather-aware schemes; stickiness isolates weather value from handover frequency

Answer: D. *Framework:* `schedule_baseline docstring`; a fairness fix (the +678% history lesson).

Q156 ALG-0's key weakness that ALG-1 fixes is:

- A) It is too slow B) It ignores weather C) Two stations can lock onto the SAME satellite (no coordination), which is physically impossible for one optical head per satellite D) It uses too much power

Answer: C. *Framework:* `match_weather_aware`; bipartite matching enforces one satellite per station jointly.

Q157 ALG-1 solves the matching with:

- A) The Hungarian algorithm (`scipy linear_sum_assignment`) on the negated weight matrix B) Greedy argmax C) Random search D) Gradient descent

Answer: A. *Framework:* `match_weather_aware`; globally optimal max-weight bipartite matching per step.

Q158 Infeasible (station, satellite) pairs are handled by:

- A) Padding with a very large cost (`_BIG_COST`) so they are chosen only if truly forced B) Ignoring the constraint C) Deleting them D) Setting weight 0

Answer: A. *Framework:* `match_weather_aware`; keeps the cost matrix rectangular for the Hungarian solver.

Q159 ALG-0-feasible exists because:

- A) ALG-0 was coded carelessly B) It is faster C) It is optimal D) To resolve ALG-0's double-booking into a conflict-free schedule so all three policies feed ALG-2's pair allocation on the SAME final metric

Answer: D. *Framework:* `schedule_baseline_feasible`; a fairness fix so the comparison is apples-to-apples.

Q160 ALG-0-feasible resolves collisions by:

- A) The Hungarian algorithm B) A simple greedy tie-break (higher elevation keeps the satellite) repeated to a fixed point — deliberately non-optimal C) Global re-optimisation D) Randomly

Answer: B. *Framework:* `schedule_baseline_feasible`; stays in the spirit of a naive baseline, distinct from ALG-1's optimality.

Q161 In the trusted-relay model, a city pair's usable key is:

- A) $K_q = \min(\text{side-A total}, \text{side-B total})$ B) The sum of both sides C) Side-A only D) The max of both sides

Answer: A. *Framework:* `pair_key_totals`; the pair is bottlenecked by its worse-served station.

Q162 The key delivered by one pass is:

- A) $k_p = w_{\text{(station,sat)}(t)} \cdot (t_{\text{set}} - t_{\text{rise}})$ — SKR times pass duration B) $\text{QBER} \cdot \text{duration}$ C) A constant D) Elevation

Answer: A. *Framework:* `schedulers`; longer, higher-elevation passes deliver more key.

Q163 ALG-2 (greedy) assigns each pass to:

- A) The pair with most key already B) The eligible pair with the largest deficit (the side furthest behind), processing largest-key passes first C) All pairs D) A random pair

Answer: B. *Framework:* `schedule_pairs_greedy`; deficit rule balances both sides toward a higher `min()`.

Q164 Why can't ALG-2 use a naive "marginal gain in K_q " rule at cold start?

- A) K_q is negative B) It uses too much memory C) It is too fast D) $\min(x, 0) = 0$ for any x , so every candidate looks useless until both sides have received a pass — the rule is degenerate

Answer: D. *Framework:* `schedule_pairs_greedy docstring`; deficit/capped-detection is used instead of gain.

Q165 ALG-2 (ILP) formulates pass-to-pair allocation as:

- A) A linear program B) A regression C) A shortest-path problem D) A MILP maximising $\sum K_q$ with $K_q \leq \text{side-A}$ and $K_q \leq \text{side-B}$ (linearising the min), solved by HiGHS

Answer: D. *Framework:* `schedule_pairs_ilp`; the `min()` is linearised by two upper-bound constraints.

Q166 Why might the ILP not always be run to proven optimality?

- A) Greedy is always better B) A full day's binary MILP can exceed 15 minutes without converging; `time_limit/mip_gap` return the best incumbent C) HiGHS is broken D) It is unstable

Answer: B. *Framework:* `schedule_pairs_ilp`; a practical optimality-vs-time trade-off, benchmarked.

Q167 ALG-2 (max-min) differs from throughput-greedy by:

- A) Maximising total key B) Nothing C) Ignoring fairness D) Always helping the currently WORST-OFF pair first (water-filling), raising the floor at the cost of total sum

Answer: D. *Framework:* `schedule_pairs_greedy_maxmin`; fairness-oriented objective `max min_q K_q`.

Q168 The handover penalty `c_h` in the matching objective is:

- A) A soft priority weight favouring keeping the previous satellite (acquisition cost folded in), NOT a direct key subtraction B) The orbital period C) Zero always D) A physical key subtraction

Answer: A. *Framework:* `match_weather_aware(handover_cost=...)`; the real key cost is accounted separately in `evaluate_segments`.

Q169 The real acquisition (handover) time charged per handover is:

- A) 30 s (the headline configuration; a real satellite re-acquisition delay) B) 5 min C) 0 s D) 5 s

Answer: A. *Framework:* `v3` uses 30 s; every re-assignment loses 30 s of delivered key.

Q170 A subtle handover-accounting bug (fixed 2026-07-23) was:

- A) Acquisition cost was clamped to the single segment after a handover; since segments (median 3.3 s) are shorter than 30 s, ~91% of handovers spanned multiple segments — a $\sim 2.7\times$ undercount B) Handovers were double-counted C) QBER was wrong D) Handovers were free

Answer: A. *Framework:* `_segments_with_handover_cost`; the budget now carries across consecutive same-satellite segments.

Q171 Why are segments cut so finely (median 3.3 s)?

- A) To slow the code B) Arbitrary C) `sweep_segments` cuts a new segment at EVERY rise/set event anywhere in the 8-station network, not just this station's handovers D) Because of weather

Answer: C. *Framework:* `sweep_segments`; event-driven; explains why acquisition cost spans many segments.

Q172 A second scheduling bug (Task 17) involved:

- A) Wrong weather B) A satellite passing the same station multiple times in a window was mis-attributed by a flat (station,sat) lookup; fixed by time-containment matching C) Missing satellites D) Wrong SKR

Answer: B. *Framework:* `_find_pass_for_segment`; 43% of (station,sat) keys collide; resolve by which pass's time window contains the segment.

Q173 The three scheduling strategies must be compared:

- A) On random samples B) On the SAME stations, satellites, and time window, at equal resources, stating the baseline in the same sentence C) On different networks D) On different metrics

Answer: B. *Framework:* `schedulers` docstring; the fair-comparison discipline (07-4 lesson).

Q174 The coordination (matching) gain, ALG-1 vs ALG-0, at equal resources over 620 days is:

- A) +50% B) -10% C) +678% D) +2.47% (dry) / +2.48% (wet)

Answer: D. *Framework:* `v3 IMPACT_EVALUATION.md`; a modest but real gain.

Q175 The weather-information gain (ALG-1-aware vs ALG-1-blind) is:

- A) +10% B) NEGATIVE: -0.29% (dry) / -0.23% (wet) — weather-awareness costs slightly C) +5% D) Zero

Answer: B. *Framework:* `v3`; weather-aware re-assigns more, and the extra handovers outweigh the clearer-satellite benefit.

Q176 Why is the weather-information gain negative?

A) The orbit changed B) Weather has no effect C) A bug D) A measured mechanism: more re-assignments → more 30-s handovers → net key lost exceeds the gain from picking a clearer satellite

Answer: D. *Framework:* Handover counting; "I counted the handovers directly" — a real, small, systematic cost.

Q177 Why is the overall scheduling gain fundamentally capped?

A) The satellites are slow B) QBER is too high C) Delivered key is a min() over the pair path, so one weak link caps the improvement regardless of matching cleverness D) The code is slow

Answer: C. *Framework:* `pair_key_totals`; the min() bottleneck is the structural ceiling.

Q178 Pair-starvation (≥ 1 pair getting zero usable key) occurs on what fraction of days?

A) 10% B) 50% C) 91.6%–96.5% of days, regardless of algorithm D) 0%

Answer: C. *Framework:* `v3`; a structural limit, not an algorithmic failure.

Q179 The root cause of pair-starvation is:

A) 8 stations cannot serve 28 city pairs; a hub station's fully-clouded day zeroes every pair through it B) Weak lasers C) Wrong orbit D) Bad scheduling

Answer: A. *Framework:* Capacity planning: the network is under-provisioned, not mis-scheduled.

Q180 This makes pair-starvation a finding of what type?

A) An algorithm bug B) A capacity-planning finding (add stations, not cleverer scheduling) C) A weather error D) A coding mistake

Answer: B. *Framework:* Engineering takeaway; the fix is more infrastructure.

Q181 There are how many city pairs among 8 cities?

A) 64 B) 28 (= $C(8,2)$) C) 8 D) 16

Answer: B. *Framework:* Combinatorics; $8 \cdot 7/2 = 28$ pairs to serve.

Q182 Geometry splits the 28 pairs into:

A) 20 + 8 B) All DUAL C) All SF D) 14 DUAL (one shared pass) + 14 SF (store-and-forward relay)

Answer: D. *Framework:* `classify_pair` at the 30° mask; a clean geographic split.

Q183 "This project does link scheduling, not network routing" means:

A) It allocates passes to pairs on a shared trusted-relay satellite; no multi-hop path selection at the network layer (per the supervisor's framing) B) It routes packets C) It ignores satellites D) It uses BGP

Answer: A. *Framework:* `schedulers` docstring; "selection/scheduling, not routing" (Dr. Minh's guidance).

Q184 `run_matching_over_time` counts a handover when:

A) Any change B) A station becomes assigned from unassigned C) Every step D) A station goes from one satellite to a DIFFERENT satellite (not from unassigned→assigned)

Answer: D. *Framework:* `run_matching_over_time`; only real satellite changes count.

Q185 The event-driven sweep (`sweep_segments`) is centralised because:

A) It was previously duplicated across scripts with drifting differences; one canonical implementation prevents divergence B) To use more memory C) It is prettier D) It is faster

Answer: A. *Framework:* `sweep_segments`; single-source-of-truth engineering.

Q186 `build_matched_pass_rows` fixes what phantom-key bug?

A) Missing passes B) None C) ALG-2 was fed the raw unconstrained pass list, crediting stations key for satellites they never won under the matching; ~65% of ALG-0's time had 2–3 stations claiming one satellite D) Wrong weather

Answer: C. *Framework:* `build_matched_pass_rows`; ALG-2 must use the matched segments, not all passes.

Q187 The demand parameter in `schedule_pairs_greedy` is used to:

A) Compute QBER B) Break ties toward pairs not yet meeting their key target C) Hard-constrain allocation D) Set the orbit

Answer: B. *Framework:* `schedule_pairs_greedy(demands=...)`; a soft preference, not a hard cap.

Q188 From an operations view, adding more ground stations would primarily:

A) Do nothing B) Change the orbit C) Reduce pair-starvation (the structural bottleneck), more than any scheduling change D) Raise QBER

Answer: C. *Framework:* Capacity-planning implication; the highest-leverage intervention.

Q189 The max-min variant cannot always raise the starvation floor above zero because:

- A) It is buggy B) Weather is fine C) If a hub station has zero real channel capacity on a fully-clouded day, no allocation can deliver key through it — a physical limit D) It ignores fairness

Answer: C. *Framework:* `schedule_pairs_greedy_maxmin` docstring; fairness cannot manufacture capacity that weather removed.

Q190 The network-engineering headline is:

- A) Coordination helps ~2.5%, weather-awareness slightly costs, and starvation (~92–96% of days) is structural — so the real lever is more stations, not smarter scheduling B) Scheduling solves everything C) Weather is irrelevant D) Routing is the answer

Answer: A. *Framework:* Synthesis of the scheduling chapter; the operations takeaway.

10.7 P7 — Statistician / Data-Quality Auditor

Q191 The algorithm comparison uses:

- A) A random sample of days B) One illustrative day C) Full enumeration of every day in two 10-year proxy months (620 days total) D) Monte Carlo

Answer: C. *Framework:* `11d_algorithm_comparison_fullenum`; no sampling error in the mean itself.

Q192 Full enumeration (vs sampling) matters because:

- A) It is slower B) It is required by law C) There is no sampling uncertainty in the population mean — the 620 days ARE the population being compared D) It changes the orbit

Answer: C. *Framework:* Census vs sample; the reported means are exact for that window.

Q193 The difference between algorithms is tested with:

- A) An unpaired t-test B) A PAIRED test day-by-day (same day, same weather, same geometry, only the scheduler changes) C) No test D) Chi-square

Answer: B. *Framework:* Pairing removes between-day variance, which would otherwise swamp a ~2.5% effect.

Q194 Why is pairing essential for a 2.5% effect?

- A) Day-to-day key varies far more than 2.5%; only within-day pairing (holding weather/geometry fixed) makes the small systematic difference detectable B) It is not C) It removes the orbit D) It changes the mean

Answer: A. *Framework:* Paired-difference statistics; variance reduction by matching.

Q195 Weather autocorrelation between consecutive days would, if ignored:

- A) Have no effect B) Fix the orbit C) Overstate significance (treating 620 correlated days as 620 independent samples inflates effective N) D) Lower the mean

Answer: C. *Framework:* Effective sample size < nominal N under positive autocorrelation.

Q196 Autocorrelation is addressed by:

- A) Ignoring it B) A single mean C) Removing weather D) A BLOCK bootstrap that resamples multi-day blocks, preserving temporal correlation

Answer: D. *Framework:* `autocorrelation_block_bootstrap`; the qualitative conclusion survives; the CI widens as expected.

Q197 The sign of the weather-information effect (negative) is robust because:

- A) It flips randomly B) It depends on the day C) It is a single measurement D) It is the same sign in both dry and wet seasons and across all three feasible m_k values swept — consistent with a systematic cost, not noise

Answer: D. *Framework:* Sensitivity sweep $m_k \in \{0.10, 0.15, 0.20\}$; robustness by replication.

Q198 A block bootstrap resamples blocks rather than individual days to:

- A) Save time B) Preserve the autocorrelation structure that i.i.d. resampling would destroy C) Remove seasonality D) Increase the mean

Answer: B. *Framework:* Standard time-series bootstrap; blocks retain short-range dependence.

Q199 EDA Layer 6 (data-quality/provenance) recommends doing what FIRST?

- A) A data-provenance audit before trusting any downstream number B) Publish C) Fit a model D) Draw every plot

Answer: A. *Framework:* Một vài vấn đề...md; provenance-first discipline.

Q200 The “TLE staleness” check plots:

- A) QBER vs elevation B) Propagation error vs time-since-epoch C) SKR vs m_k D) Rain vs cloud

Answer: B. *Framework:* EDA Layer 6; SGP4 error grows with Δt from epoch.

Q201 The infamous “14–15 visible satellites” error was caused by:

- A) Bad weather B) A typo C) Wrong wavelength D) Using dates OUTSIDE the valid TLE epoch window (a snapshot/epoch mismatch), later corrected to ~3.1–3.5

Answer: D. *Framework:* Data-quality lesson; always validate the epoch window.

Q202 Why report bootstrap confidence intervals instead of just the mean?

- A) To look sophisticated B) It changes the mean C) To quantify uncertainty; a mean without a CI hides whether a small effect is distinguishable from zero D) It is required

Answer: C. *Framework:* EDA Layer 10 (uncertainty); resampling over point estimates.

Q203 The provenance discipline (save_provenance / save_verify_numbers) exists to:

- A) Encrypt data B) Slow the code C) Compress files D) Log parameters, sources, and checksums per run so results are reproducible and past mistakes (epoch mismatch, snapshot bias, overclaims) are not repeated

Answer: D. *Framework:* `utils.py` convention; reproducibility over convenience.

Q204 The retracted “+678%” result was invalid because:

- A) The math was wrong B) It compared an 8-station network against a 1-station one WITHOUT controlling for the number of stations — conflating the value of adding stations with the value of weather-aware scheduling C) The weather was fake D) The orbit was wrong

Answer: B. *Framework:* N-vs-1 confound; the fair comparison holds network size fixed.

Q205 The lesson from the +678% retraction is:

- A) Never compare B) Before ranking groups, check the sub-units are equal (same number of stations); state the baseline in the same sentence as the improvement C) Use bigger numbers D) Avoid statistics

Answer: B. *Framework:* Fair-comparison principle; a recurring feedback theme.

Q206 Pair-starvation being ~unchanged across every correction round confirms:

- A) A coding bug B) The orbit is unstable C) It is a STRUCTURAL property (8 stations / 28 pairs), independent of the specific physics formulas D) Weather is wrong

Answer: C. *Framework:* Invariance under model changes → structural, not parametric.

Q207 The MILP optimality-gap sensitivity analysis (Annex A) checks:

- A) The orbit B) The weather C) QBER D) Whether the heuristic greedy result is far from the proven optimum

Answer: D. *Framework:* `alg2_mipgap_sensitivity`; validates that greedy $\approx$ optimal, so heuristic use is justified.

Q208 Why sweep the cloud-outage threshold (85%) in a sensitivity analysis?

- A) To change the orbit B) To show the headline conclusions are not artefacts of one arbitrary threshold choice C) To increase QBER D) For fun

Answer: B. *Framework:* `figA1_cloud_threshold_sensitivity`; robustness to a modelling parameter.

Q209 Jain’s fairness index and the Gini coefficient (EDA Layer 10) measure:

- A) Rain B) QBER C) Inequality/fairness of key distribution across pairs D) Orbit accuracy

Answer: C. *Framework:* Fairness metrics; relevant to the max-min vs throughput trade-off.

Q210 Why is a paired t-test preferred over Mann-Whitney here (if normality holds)?

- A) It ignores pairing B) Paired t-test has more power when the paired differences are approximately normal; Mann-Whitney is the fallback when they are not C) It is required D) It is older

Answer: B. *Framework:* Parametric vs non-parametric choice; check the differences’ distribution first.

Q211 Median (not mean) SF-relay latency is reported because:

- A) It is easier B) Latency distributions are skewed with long tails; the median is a more robust central summary C) To hide outliers D) The mean is undefined

Answer: B. *Framework:* `sf_latency_minutes` uses `np.median`; robust statistics for skewed data.

Q212 The correlation matrix should be checked for autocorrelation before treating days as independent because:

- A) It is a rule B) It increases QBER C) Correlated days reduce the effective sample size behind each correlation estimate, widening true CIs D) It changes the orbit

Answer: C. *Framework:* EDA Layer 10 note; independence is an assumption to test, not assume.

Q213 "Snapshot bias" is a statistical concern because:

- A) A single-instant sample may misrepresent the population across time-of-day, season, or deployment phase B) It blurs images C) It affects only weather D) It is a rounding error

Answer: A. *Framework:* EDA Layer 6; representativeness of the sampling frame.

Q214 Why does the project prefer a homogeneous single-shell template over the full 10k constellation for statistical cleanliness?

- A) One altitude/inclination gives a consistent, reproducible physical model; mixing shells introduces heterogeneous parameters that confound the channel model B) Laziness C) It is faster only D) Shell-1 is the only data

Answer: A. *Framework:* Controlling nuisance variation; a design-of-experiments principle.

Q215 The daytime background penalty is described as an "order-of-magnitude estimate," meaning:

- A) It is proven B) It is a plausibility bound (10.5–7.9 orders below noise across an FOV sweep), NOT a validated simulation C) It is exact D) It is ignored

Answer: B. *Framework:* daytime_background_fov_sweep; honest labelling of confidence level.

Q216 Why re-run the significance test after the v3 physics fixes (not just once)?

- A) The numbers changed, so a prior significance result at the old physics no longer certifies the new ones — a flagged outstanding task B) To waste time C) The orbit changed D) No need

Answer: A. *Framework:* IMPACT_EVALUATION.md "việc chưa làm"; statistics must track the current model.

Q217 The value of stating "K_{day} is not a formal metric" is:

- A) Transparency: a quantity that is mathematically valid but not a reported headline metric is labelled as such, preventing overinterpretation B) It changes availability C) It is wrong D) Hiding it

Answer: A. *Framework:* Metric hygiene; distinguish formal from auxiliary quantities.

Q218 Reporting availability as invariant to rain_fraction is a form of:

- A) Data leakage B) Overfitting C) Error D) Robustness demonstration (the headline metric does not move with a nuisance parameter)

Answer: D. *Framework:* compute_reliability_metrics; $A = 1 - P_{\text{cloud}}$ identically.

Q219 Cross-validating TLE from Celestrak vs Space-Track is an example of:

- A) Changing the orbit B) Sensor cross-validation (EDA Layer 6): comparing two independent measurements of the same quantity C) Wasting effort D) Fixing QBER

Answer: B. *Framework:* Data-quality practice; discrepancy flags provenance problems.

Q220 The statistician's headline judgement of this project is:

- A) It over-claims B) It compares fairly (full enumeration, paired, block-bootstrapped), controls for network size, and labels confidence levels — the small effects are credible precisely because the methodology is disciplined C) It is unreproducible D) It uses no statistics

Answer: B. *Framework:* Methodology synthesis; the angle Rosset (Planck statistics) would probe.

10.8 P8 — Systems Architect / Integrator

Q221 SIKD's core architectural idea is:

- A) Two satellites B) A fibre link C) One optical carrier multiplexing a classical data channel and a quantum key channel via subcarrier modulation D) Two separate lasers

Answer: C. *Framework:* channel_model/sikd_performance; simultaneous information + key on one beam.

Q222 The received photocurrent expression is:

- A) $i_r = m_k \cdot s_k$ B) $i_r = Re \cdot PT \cdot h$ C) $i_r = \frac{1}{2} \cdot Re \cdot PT \cdot h \cdot [1 + m_k \cdot s_k + m_a \cdot s_a] + n$ D) $i_r = n$

Answer: C. *Framework:* Eq. 3.1; the "1" DC term drives shot noise, the m_a term becomes cross-talk.

Q223 The fundamental architectural tension in SIKD is:

- A) Cost vs mass B) Data and key share one detector, so the data channel leaks into the key channel as cross-talk (m_a^2 coupling) C) Rain vs cloud D) Orbit vs weather

Answer: B. *Framework:* The coupling that makes power-split a real optimisation, not two independent links.

Q224 Why multiplex data and key on ONE carrier instead of two separate systems?

- A) To slow the link B) To raise QBER C) No reason D) To save a laser/aperture (SWaP) and reuse one beam/detector for both services simultaneously

Answer: D. *Framework:* SIKD motivation; efficiency for resource-constrained satellites (Lukens 2025 hybrid networks).

Q225 The assumed aperture (10 cm diameter) places this system closest to:

- A) Micius (0.3 m TX, 1.0 m RX) B) TBIRD (0.022 m TX, 0.40 m RX) — cheap, light, multi-station-friendly C) A fibre system D) GEO relay

Answer: B. *Framework:* SWaP comparison; small aperture $\leftrightarrow$ cheaper deployment $\leftrightarrow$ weaker key collection.

Q226 The SWaP trade-off of a small aperture is:

- A) Only drawbacks B) Cheaper/lighter/multi-station deployment, but collects a smaller fraction of the already-weak quantum signal C) Only benefits D) No effect

Answer: B. *Framework:* hg scales with aperture; an unresolved optimisation in the thesis.

Q227 The full computational pipeline order is:

- A) channel (hg, hl, $\sigma^2 X$) $\rightarrow$ noise $\rightarrow$ thresholds $\rightarrow$ Psift/QBER $\rightarrow$ SKR $\rightarrow$ BER_CC B) SKR $\rightarrow$ channel C) QBER $\rightarrow$ orbit D) weather $\rightarrow$ SKR $\rightarrow$ orbit

Answer: A. *Framework:* `compute_sikd_performance` wrapper; the dependency chain.

Q228 The system integrates weather into performance by:

- A) Only orbit B) Only QBER C) `get_city_params` $\rightarrow$ `compute_hl(V, R)` $\rightarrow$ `compute_sikd_performance` $\rightarrow$ `SKR_eff` $\times$ (1-P_cloud) or 3-state D) Ignoring it

Answer: C. *Framework:* `weather_model` integration; climatology drives the channel state.

Q229 Modules have no `__init__.py` and are imported via `sys.path` because:

- A) Python requires it B) A bug C) To hide them D) They are standalone .py files in a flat modules/ folder, imported by path insertion — a deliberate simple structure

Answer: D. *Framework:* Project convention; lightweight, test-friendly.

Q230 The design decision to make cloud a binary outage (not gradual) is:

- A) Wrong B) A modelling choice reflecting that thick cloud blocks an optical beam almost totally; it simplifies the state space to clear/rain/outage C) An oversight D) About rain

Answer: B. *Framework:* Three-state model; architecture-level simplification with physical basis.

Q231 The turbulence model is selected at runtime by:

- A) Always Gamma-Gamma B) `get_turbulence_model( $\sigma^2 R$ )`: log-normal if < 0.3 , else Gamma-Gamma C) Always log-normal D) Randomly

Answer: B. *Framework:* `weather_model.get_turbulence_model`; adaptive to turbulence regime (here always log-normal).

Q232 The Skyfield timescale is loaded once at import because:

- A) To slow startup B) For weather C) Python requires it D) It is a singleton (not per-call), avoiding repeated expensive initialisation

Answer: D. *Framework:* Performance design decision; `_TS = load.timescale()`.

Q233 From an architecture view, why separate `link_geometry` (symmetric) from `citypair_feasibility` (directional)?

- A) No reason B) Speed C) DUAL/SF classification depends only on distance (symmetric), while SF relay latency depends on satellite heading (directional) — different mathematical objects D) To duplicate code

Answer: C. *Framework:* Module separation reflects a real physical distinction.

Q234 The system's official reported metric is:

- A) Availability $A = 1 - P_{\text{cloud}}$ (plus the fair pair-key comparison); `K_day` was demoted B) QBER only C) `K_day` D) Throughput

Answer: A. *Framework:* Metric governance (Task 17); formal vs auxiliary metrics.

Q235 Extending to a multi-shell constellation would require:

- A) Nothing B) Altitude-dependent SKR tables and a more complex scheduling formulation — future work C) A new wavelength
D) Removing weather

Answer: B. *Framework:* Architecture limitation; the channel model is single-altitude.

Q236 The ISL relay was introduced to replace what architectural weakness?

- A) Bad weather B) Single-hop SF forcing a wait for the same satellite (up to ~711 min); ISL mesh delivers via whichever satellite next passes the destination C) Wrong orbit D) High QBER

Answer: B. *Framework:* `isl_relay`; a topology-level architectural upgrade.

Q237 The power-split module reuses `compute_sikd_performance` directly because:

- A) A copy-paste error B) It is faster C) To avoid weather D) That function already accepts m_k/m_a as parameters, so no refactor was needed and the 207+ existing tests stay valid

Answer: D. *Framework:* `sikd_powersplit` design; reuse over rewrite preserves test coverage.

Q238 The system parameter that most directly couples the two channels is:

- A) Orbit altitude B) Temperature C) Filter isolation Iso (finite isolation lets one subcarrier leak into the other) D) Wavelength

Answer: C. *Framework:* $\sigma^2_{CT} \propto 10^{(-Iso/10)}$; the architectural knob for cross-talk.

Q239 Why is the reverse (key→data) cross-talk usually negligible?

- A) It is night B) Filters remove it C) It scales as m_k^2 , and $m_k \ll m_a$, so the key's small modulation leaks little into the data receiver
D) It does not exist

Answer: C. *Framework:* `crosstalk_variance(mK)`; asymmetric coupling.

Q240 The architecture assumes trusted-relay security, which means the design must eventually:

- A) Remove the relay B) Ignore security C) Provide a composable proof and a trust model for the satellite node (future work) D) Use fibre

Answer: C. *Framework:* Trusted-relay is an architectural assumption with a security debt.

Q241 A systems view of the day/night tension is:

- A) Only about rain B) A cross-layer conflict: the meteorology layer says "operate in daytime," the receiver layer assumes "night for low background" — the architecture has not reconciled them C) Trivial D) About the orbit

Answer: B. *Framework:* Cross-layer integration gap; a genuine architectural open item.

Q242 The choice $R_b = 1$ Gbps sets:

- A) Weather B) The raw bit rate, hence the noise bandwidth $\Delta f = R_b/2$ and the absolute SKR scaling ($SKR_{bps} = SKR_{norm} \cdot R_b$)
C) QBER D) The orbit

Answer: B. *Framework:* `RB_DEFAULT`; a system-level rate parameter with broad impact.

Q243 The architecture's biggest unmodelled real-world effect is arguably:

- A) Daytime background noise (and residual pointing jitter) — assumed away, flagged as future validation B) Rain C) QBER D) Cloud

Answer: A. *Framework:* Scope boundary; the largest gap between model and deployment.

Q244 Why does the design fold acquisition time into a handover penalty rather than a separate timing sub-model?

- A) It captures the operationally relevant cost (lost key during re-acquisition) without a full PAT dynamics model — a deliberate modelling-scope choice B) Laziness C) To hide it D) It is more accurate

Answer: A. *Framework:* `schedulers` modeling scope; `c_h` abstracts PAT timing.

Q245 The system's headline architectural claim is:

- A) More lasers are needed B) Weather is negligible C) Single-link SKR is the right metric D) Tropical satellite quantum networks should be evaluated at the network level (fair scheduling under real weather and orbits), not by an idealised single-link clear-sky key rate

Answer: D. *Framework:* Central thesis claim; a shift in the unit of evaluation.

Q246 Micius vs this system, the key architectural difference is:

- A) No difference B) Orbit altitude only C) Wavelength D) Micius is one clean high-power link; this is a network of many cheap stations under real tropical cloud — a different, operational question

Answer: D. *Framework:* Positioning; scale and realism over single-link demonstration.

Q247 The SCM (subcarrier multiplexing) architecture places key and data at frequencies:

- A) Different wavelengths B) Different RF subcarriers f1 (key) and f2 (data), separated by filtering after detection C) Different satellites D) The same frequency

Answer: B. *Framework:* SCM design; separation is spectral, isolation finite.

Q248 From an integrator's view, the v3 corrections mattered because:

- A) They changed the architecture B) They changed the orbit C) They removed weather D) They corrected the numbers feeding every downstream decision (SKR tables, handover-cost calibration, 620-day enumeration) — physics errors cascade through the whole pipeline

Answer: D. *Framework:* Cascade integrity; a formula fix propagates system-wide.

Q249 The system reuses one canonical event sweep (sweep_segments) to:

- A) Change weather B) Slow down C) Use more memory D) Guarantee all three algorithm groups share one definition of segments/handover cost, preventing silent divergence between scripts

Answer: D. *Framework:* Integration discipline; single source of truth.

Q250 The architect's summary judgement:

- A) It is unfinished and broken B) The design is final and complete C) It ignores physics D) A coherent, testable, weather-and-orbit-integrated pipeline with clearly scoped simplifications (single altitude, night operation, trusted relay) and honest debts to repay in future work

Answer: D. *Framework:* Architecture synthesis; scoped correctness with documented limits.

10.9 P9 — Thesis Examiner / Reviewer Angle

Q251 The committee are best described as:

- A) Network engineers B) Cryptographers C) Space-science generalists (astrophysics/CMB, geophysics/remote sensing, particle-physics statistics), NOT QKD specialists D) QKD specialists

Answer: C. *Framework:* Patanchon (APC/CMB), Jacquemoud (IPGP/remote sensing), Rosset (IN2P3/Planck stats); define QKD terms, lean on their fields.

Q252 For examiner Jacquemoud (radiative transfer / remote sensing), the strongest bridge is:

- A) QBER theory B) "Beer-Lambert atmospheric attenuation is the same radiative-transfer/atmospheric-correction machinery you use; ERA5 is the same reanalysis" C) Orbital mechanics D) Hungarian algorithm

Answer: B. *Framework:* Committee analysis; speak his mother tongue (extinction, visibility, aerosols).

Q253 For examiner Patanchon (detector noise / component separation), the strongest bridge is:

- A) City pairs B) Weather C) Rain D) "The 4-term noise budget and the data-into-key cross-talk are a detector noise budget and a component-separation/leakage problem, like bandpass mismatch in CMB"

Answer: D. *Framework:* Committee analysis; cross-talk $\propto P^2$ parallels instrument systematics.

Q254 For examiner Rosset (statistics), the strongest bridge is:

- A) Orbits B) "The +2.5% is a paired significance test on a full enumeration, with a block bootstrap for autocorrelation" C) Lasers D) Wavelength

Answer: B. *Framework:* Committee analysis; paired, full-enumeration, autocorrelation-aware.

Q255 Jacquemoud's likely toughest question, "why is Beer-Lambert enough without multiple scattering?", is answered by:

- A) "The beam is wide" B) "Multiple scattering doesn't exist" C) "Cloud is treated as binary outage, not transmission; for clear/rain a narrow line-of-sight beam is extinction-dominated" D) Avoiding it

Answer: C. *Framework:* Prepared answer A-J1; concede it is first-order, justify the cloud=outage choice.

Q256 Rosset's likely toughest question, "is +2.5% significant or noise?", is answered by:

- A) "I didn't test it" B) "Full enumeration (no sampling error), paired day-by-day (removes between-day variance), same sign across all m_k swept; block bootstrap handles autocorrelation" C) "It's a big number" D) "It's obviously significant"

Answer: B. *Framework:* Prepared answer A-R1/R2; the pairing + bootstrap defence.

Q257 Patanchon’s likely question, “why does cross-talk dominate when the link is strong?”, is answered by:

- A) “It’s always small” B) “ $\sigma^2_{CT} \propto$ received power squared, so at high elevation the strong link brings proportionally more leakage; it ranges $0.5 \times -290 \times$ thermal” C) “It doesn’t” D) Avoiding it

Answer: B. *Framework:* Prepared answer A-P1; the P^2 insight.

Q258 The single most important honesty slide/point in the defense is:

- A) Hiding the negative result B) Claiming +678% C) Overstating SKR D) Openly presenting weather-info as a small COST, the day/night tension, and the borrowed BB84 threshold — proactively, not defensively

Answer: D. *Framework:* Committee prefers integrity over a big number; the maturity signal.

Q259 If asked “why not use the full 10k-satellite constellation?”, the honest answer is:

- A) “My channel model is parameterised for one altitude; multi-shell needs altitude-dependent SKR tables and complicates scheduling without changing the core finding — weather remains the bottleneck” B) “1k is special to ASEAN” C) “I was lazy” D) “10k is impossible”

Answer: A. *Framework:* Prepared defense; scope, not laziness or ASEAN-specificity.

Q260 If asked “isn’t a negative weather-info result a failure?”, the answer is:

- A) “It’s a bug” B) “No — it’s a measured, robust mechanism (handover cost > clearer-satellite benefit), and a genuine, non-obvious finding with engineering value” C) “Ignore it” D) “Yes, the project failed”

Answer: B. *Framework:* Reframe a null/negative result as a real finding; confidence without defensiveness.

Q261 The weakest defensible point (must be flagged proactively) is:

- A) The Hungarian algorithm B) The city list C) $H_{atm} = 20$ km (physical scale height is 1–2 km), the day/night tension, and the borrowed BB84 threshold D) The wavelength

Answer: C. *Framework:* Know your weak points before the examiner finds them.

Q262 The retracted +678% number should be handled in defense by:

- A) Hiding the correction B) Still claiming it C) Blaming the code D) Not presenting it; if asked, explain it conflated adding stations with weather-awareness and was replaced by a fair, fixed-network-size comparison

Answer: D. *Framework:* Scientific honesty; the correction itself is a credibility asset.

Q263 A reviewer from remote sensing might challenge the ERA5 cloud→outage mapping. The best response is:

- A) “I guessed” B) “It doesn’t matter” C) “ERA5 is perfect” D) “I map total cloud cover $\geq 85\%$ to a binary optical outage and analyse the threshold’s sensitivity; rain uses a real tropical-FSO calibration (Koné 2024)”

Answer: D. *Framework:* Defensible modelling + sensitivity + real calibration.

Q264 Compared with the nearest scheduling precedent (Polnik 2020), this thesis adds:

- A) A different wavelength B) Only more satellites C) Real 10-year tropical climatology, a fair equal-resource comparison, and separation of weather-value from coordination-value D) Nothing

Answer: C. *Framework:* Literature-gap statement; the contribution over prior scheduling work.

Q265 The thesis frames its work as “selection/scheduling, not routing” because:

- A) It routes packets B) Per the supervisor’s guidance, network-layer multi-hop routing is out of scope; the work stays at pass allocation on a shared relay C) No reason D) Routing is too hard

Answer: B. *Framework:* Scope framing (Dr. Minh); honest about the layer addressed.

Q266 A CMB/instrument examiner would find the most familiar territory in:

- A) The Hungarian algorithm B) Rain statistics C) City-pair geometry D) The receiver noise budget and the shared-detector cross-talk (a systematics/leakage problem)

Answer: D. *Framework:* Patanchon’s home ground; frame it as instrument systematics.

Q267 Which prepared “bridge” would misfire with this committee?

- A) Using orbital propagation analogies B) Assuming they know sifting/dual-threshold/BB84-vs-BBM92 without defining them C) Defining QKD terms D) Speaking their field’s language

Answer: B. *Framework:* Audience analysis; never assume QKD fluency.

Q268 The defense time format is:

- A) 10 min presentation + 10 min Q&A B) 5 min + 5 min C) 30 min talk D) 15 min + 15 min

Answer: A. *Framework:* Ms. Tra’s email; rehearse to the reduced 10-minute talk.

Q269 If challenged on the borrowed BB84 threshold, the strongest stance is:

A) "It's an explicit assumption I state openly; after the v3 corrections QBER is far below even the stricter BBM92 10^{-3} target, so the operating points are secure under either threshold" B) Deny it C) "I'll remove QKD" D) "It doesn't matter"

Answer: A. *Framework:* Turn the caveat into a strength — the corrected numbers satisfy the stricter bound anyway.

Q270 The examiner-facing summary of the whole thesis is one sentence:

A) "Weather is irrelevant" B) "Single-link SKR is high" C) "More satellites solve everything" D) "Judge tropical satellite quantum networks by fairly-compared, network-level scheduling outcomes under real weather and real orbits, not by an idealised clear-sky single-link rate"

Answer: D. *Framework:* The central claim; the line to end on.

Q271 Son Tong Si (USTH faculty, general) is most likely to ask about:

A) Motivation, application relevance, big-picture positioning, and the Vietnam/ASEAN angle B) Gauss-Hermite quadrature C) The Hungarian algorithm's complexity D) Cross-talk variance

Answer: A. *Framework:* Committee analysis; prepare concise, non-technical answers for the generalist.

Q272 The best one-sentence QKD definition for a non-specialist committee is:

A) "It's a satellite protocol" B) "It uses qubits" C) "It's encryption" D) "Two parties share a secret key whose secrecy follows from physics, not from assuming a computation is hard, so it stays secure even against a quantum computer"

Answer: D. *Framework:* Prepared crosscutting answer; physics-based secrecy in one line.

Q273 If asked "what is genuinely new here?", the answer is:

A) "A new orbit" B) "Nothing" C) "Evaluating a tropical satellite quantum network at the network level, fairly, on real weather and real orbits — instead of a single clear-sky link rate" D) "A new laser"

Answer: C. *Framework:* Novelty statement; the unit-of-evaluation shift.

Q274 The examiner-credible way to present the modest +2.5% is:

A) Call it +678% B) State it plainly with the baseline in the same sentence, then explain the structural min() ceiling — small but honest and mechanistically understood C) Hide it D) Inflate it

Answer: B. *Framework:* Fair-comparison + mechanism; credibility over spin.

Q275 A reviewer asking "did you validate against any real measurement?" should hear:

A) "No validation at all" B) "The rain-attenuation law is calibrated on real tropical FSO measurements (Koné 2024, Abidjan); weather is real ERA5; orbits are real TLEs — the channel physics follows Vu et al. 2022" C) "I used random numbers" D) "Everything is simulated with no basis"

Answer: B. *Framework:* Anchor each layer to a real data source or cited model.

Q276 The honest answer to "why is the daytime penalty only estimated, not simulated?" is:

A) "Daytime has no sun" B) "It doesn't matter" C) "It's fully simulated" D) "I give an order-of-magnitude FOV-sweep estimate suggesting it may be small for this PIN/DT-DD receiver; a full daytime-background simulation is stated future work"

Answer: D. *Framework:* Label confidence honestly; distinguish estimate from proof.

Q277 If Rosset probes the block bootstrap, the key detail to state is:

A) "I resampled individual days" B) "Bootstrap is unnecessary" C) "I didn't bootstrap" D) "I resampled multi-day BLOCKS to preserve autocorrelation; the qualitative conclusions survive and the CI widens as expected"

Answer: D. *Framework:* Block (not i.i.d.) resampling is the correct answer for correlated time series.

Q278 The most impressive thing a reviewer can find in this thesis is:

A) The self-correction history (three formula errors found and fixed, numbers openly revised) — evidence of rigour, not weakness B) No limitations C) Perfect results D) A big headline number

Answer: A. *Framework:* IMPACT_EVALUATION.md; finding your own errors is a strength.

Q279 The three "trap" questions to master the mechanism (not just the number) are:

A) Wavelength, altitude, city count B) Why cross-talk dominates a strong link (Patanchon); why +2.5% is significant though small (Rosset); why Beer-Lambert suffices without multiple scattering (Jacquemoud) C) Only statistical ones D) Only physics ones

Answer: B. *Framework:* Committee analysis; each maps to one examiner's home field.

Q280 The reviewer-angle takeaway is:

A) Present a flawless story B) Use big numbers C) Present fairly-compared, mechanistically-understood results, define QKD terms, bridge to each examiner's field, and flag limitations proactively — integrity is the winning strategy with this panel D) Avoid all weaknesses

Answer: C. *Framework*: Defense strategy synthesis.

10.10 P10 — Applied / Operations / Policy & EDA Breadth

Q281 Starlink officially launched service in Vietnam on:

A) 2020 B) 2030 C) Never D) 12 August 2026, after a legal process from March 2025

Answer: D. *Framework*: EDA conversation; four licensed gateway ground stations (Phú Thọ, Đà Nẵng, HCMC).

Q282 Starlink's own strategy for continuous internet coverage relies on multi-shell + ISL, which for QKD:

A) Helps pass frequency and matching (more satellites, denser mesh) but does NOT solve the cloud-outage bottleneck — no satellite sees through cloud at optical wavelengths B) Is irrelevant C) Raises QBER only D) Fully solves everything

Answer: A. *Framework*: EDA comparison table; the weather bottleneck is orbit-invariant.

Q283 Starlink petitioned the FCC to lower its elevation mask from 40° to 25°, which trades:

A) Rain vs cloud B) Larger per-satellite footprint / longer visibility vs higher atmospheric attenuation at low elevation C) Nothing D) Cost vs mass

Answer: B. *Framework*: The same footprint-vs-quality trade the thesis analyses (793 vs 573 km).

Q284 Why does the thesis's low "coverage" not contradict Starlink working in Vietnam?

A) It does contradict B) The thesis measures secret-key delivery under strict QBER + cloud constraints, not internet ping — a fundamentally harder service C) Starlink doesn't work D) Different orbit

Answer: B. *Framework*: EDA conversation; QKD ≠ IP connectivity.

Q285 EDA Layer 5 (ground/RF hardware) is currently:

A) Fully done B) Not done — the model is purely theoretical (no measured antenna/G-T/RFI data) C) Irrelevant D) Half done

Answer: B. *Framework*: EDA mapping table; a whole layer of future work.

Q286 EDA Layer 6 (data quality/provenance) is called the highest-ROI next step because:

A) The project already practised its spirit (epoch, snapshot, handover-cost sensitivity) but lacks a standardised checklist to avoid repeating those exact errors B) It changes QBER C) It is easy D) It is required

Answer: A. *Framework*: Một vài vấn đề...md; consolidate lessons into a methods document.

Q287 EDA Layer 7 (space environment) would add, e.g.:

A) Space-weather (Kp/solar flux) vs atmospheric-drag correlation affecting TLE accuracy; conjunction/debris statistics B) Weather only C) Nothing D) QBER

Answer: A. *Framework*: EDA breadth; relevant to long-term TLE reliability.

Q288 A cross-cutting statistical tool recommended for outage-duration analysis is:

A) QBER B) A single mean C) Fitting Weibull/log-normal distributions and reporting consecutive-outage statistics (for buffer/storage design) D) Linear regression only

Answer: C. *Framework*: EDA Layer 3/10; distribution fitting over point estimates.

Q289 For global spatial visualisation the recommended projections are:

A) Mollweide/Robinson for global, Plate Carrée for regional (ASEAN) B) Only Mercator C) None D) 3D only

Answer: A. *Framework*: EDA Layer 10; projection choice matters for coverage maps.

Q290 For site-diversity selection, the suggested unsupervised technique is:

A) SGP4 B) Linear regression C) k-means / hierarchical clustering on site characteristics D) A t-test

Answer: C. *Framework*: EDA Layer 10; clustering for station siting.

Q291 The recommended EDA execution order starts with:

A) A data-provenance audit (Layer 6) before trusting any number, then univariate, then cross-layer, hypothesis-driven, then a sensitivity pass B) Drawing every possible plot C) Publishing first D) Modelling first

Answer: A. *Framework*: Một vài vấn đề...md; provenance-first, hypothesis-driven.

Q292 "Each figure must answer one specific question; each number must have a mechanism" is a principle guarding against:

- A) Unexplained results and aimless "plot everything" EDA B) Too few plots C) Fast code D) Good statistics

Answer: A. *Framework*: EDA discipline; mechanism over correlation-hunting.

Q293 The Video_Assets folder supports communication with what kind of resources?

- A) Free CC-BY Earth textures (Solar System Scope), public-domain NASA 3D satellite models, and prototype visualisation videos
B) Paid stock only C) Only text D) None

Answer: A. *Framework*: Video_Assets/README; licensed assets for a thesis explainer video.

Q294 Why does the licensing of visualisation assets matter for a thesis?

- A) Public sharing/submission requires proper attribution (CC-BY) or public-domain status; using the wrong asset creates a licensing problem B) It never matters C) It affects the orbit D) It changes QBER

Answer: A. *Framework*: README_Sources_and_Licenses; provenance extends to media, not just data.

Q295 The applied "basic → advanced" development ladder for this project runs roughly:

- A) Only scheduling B) Only physics C) Physical link (Ch.3/6) → weather (Ch.4) → orbits (Ch.5) → scheduling (Ch.7/8) → ISL/power-split extensions → EDA breadth (hardware, space-env, economics, policy) D) Random

Answer: C. *Framework*: Thesis structure + EDA 10-layer map; simplest components up to system-level and beyond.

Q296 The policy/regulatory angle (EDA Layer 9) relevant to the Introduction is:

- A) QBER B) Nothing C) Starlink Vietnam licensing history and the elevation-mask regulatory trend ($40^\circ \rightarrow 25^\circ$), reflecting evolving constellation strategy D) Rain

Answer: C. *Framework*: EDA Layer 9; a short contextual paragraph for the intro.

Q297 The economics/fleet-ops angle (EDA Layer 8) would track:

- A) QBER B) Weather C) Launch cadence vs deorbit rate (net fleet growth), fleet age distribution, cost-per-key trends D) Orbits only

Answer: C. *Framework*: EDA Layer 8; long-term viability context (out of current technical scope).

Q298 The most defensible one-line applied justification for the whole project is:

- A) "QKD is trendy" B) "To use satellites" C) "To beat Starlink" D) "Secure non-terrestrial communication for tropical Vietnam/ASEAN needs its key-delivery evaluated under the region's real, cloud-dominated weather and real orbital geometry"

Answer: D. *Framework*: Applied motivation; regional secure-NTN relevance.

Q299 The clearest capacity-planning recommendation this project yields for an operator is:

- A) Change wavelength B) Use cleverer scheduling C) Add ground stations (and pursue site diversity across de-correlated weather regions), because starvation is structural, not algorithmic D) Increase laser power only

Answer: C. *Framework*: The operations conclusion; infrastructure over algorithms.

Q300 The single sentence that captures the entire project across all perspectives:

- A) "Satellite QKD is easy" B) "More satellites solve tropical QKD" C) "Under real tropical weather and real orbits, a satellite quantum network's usable key is governed at the network level — cloud sets availability, geometry sets feasibility, coordination helps modestly, weather-awareness barely pays for its handovers, and the binding limit is having enough ground stations, not smarter scheduling" D) "Weather is irrelevant to QKD"

Answer: C. *Framework*: Full synthesis across physics, weather, orbits, scheduling, statistics, and operations — the project in one line.

10.11 P11 — Figure, Diagram & Video Reading

Q301 fig06_channel_performance.png plots SKR and QBER vs elevation. Its SKR curve peaks near:

- A) 90° elevation B) 30° only C) $\sim 35\text{--}40^\circ$ elevation, then declines toward both 30° and 90° D) It is flat

Answer: C. *Framework*: 07_channel_performance_fig.py; SKR is non-monotonic — cross-talk ($\propto P^2$) bites at high elevation while attenuation bites at low elevation, giving an interior peak.

Q302 In fig06, the horizontal dashed line on the QBER panel marks:

- A) Zero B) The 11% BB84 security threshold C) The SKR maximum D) The rain rate

Answer: B. *Framework*: The reference line lets you read at a glance that all v3 operating points sit far below threshold.

Q303 `figA3_psift_qber_vs_mK.png` (log-log) shows, as m_k increases, that QBER:

- A) Rises B) Diverges C) Falls, opening a feasible region around $m_k \approx 0.1\text{--}0.15$ D) Is constant

Answer: C. *Framework:* `22_mK_mD_iso_sensitivity_figs.py`; this figure is the direct justification for resetting m_k to 0.15.

Q304 `figA4_min_mK_vs_iso.png` plots the minimum feasible m_k against filter isolation Iso. As Iso improves (larger dB), the minimum m_k :

- A) Stays fixed B) Increases C) Decreases (better isolation $\rightarrow$ less cross-talk $\rightarrow$ smaller m_k suffices) D) Becomes negative

Answer: C. *Framework:* Better filtering relaxes the key-power requirement; the curve is shown for several m_a values.

Q305 `figA5_skr_contour_mD_mK.png` is a 2-D contour of SKR over (m_a, m_k) . The infeasible region it hatches is:

- A) $m_k = 0$ B) $m_a = 0.5$ C) $\text{Iso} < 15$ D) $m_k + m_a > 1$ (modulation-budget violation)

Answer: D. *Framework:* `evaluate_split` raises `ValueError` beyond the budget; the figure marks that region as physically meaningless.

Q306 `figA2_ber_cc_vs_PT.png` plots classical-channel BER vs transmit power PT for several m_a . Higher m_a generally gives:

- A) Lower BER (stronger data signal), at the cost of more cross-talk into the key channel B) Higher BER C) No change D) Zero BER always

Answer: A. *Framework:* `compute_BER_CC`; the figure shows the data-side benefit that trades against the key-side cost.

Q307 `diagram_02_channel_model.png` depicts the chain:

- A) Weather $\rightarrow$ orbit $\rightarrow$ SKR B) Satellite $\rightarrow$ ground only C) QBER $\rightarrow$ SKR $\rightarrow$ key D) PT $\rightarrow$ geometric spreading $h_g \rightarrow$ atmospheric $h_l \rightarrow$ turbulence $h_a \rightarrow$ received power

Answer: D. *Framework:* The three multiplicative channel factors in sequence.

Q308 `diagram_03_sikd_pipeline.png` and `diagram_07_sikd_block_diagram.png` both show that key and data:

- A) Share one laser/carrier via subcarrier multiplexing and one photodetector, split by RF filtering after detection B) Use two separate lasers C) Are on different satellites D) Never mix

Answer: A. *Framework:* The SCM architecture; the shared detector is the cross-talk origin.

Q309 `diagram_08_handover_aperture_concept.png` illustrates:

- A) Rain B) QBER C) The acquisition/handover concept and the receive aperture geometry (why a handover costs acquisition time) D) City pairs

Answer: C. *Framework:* Motivates the 30-s handover penalty in scheduling.

Q310 Reading `fig06` together with `figA3`, the combined message is:

- A) m_k is irrelevant B) SKR is flat C) QBER is always 10% D) At the corrected $m_k = 0.15$ the whole $30^\circ\text{--}90^\circ$ range is secure and high-SKR, resolving the pre-v3 problem where low elevations breached threshold

Answer: D. *Framework:* Cross-figure synthesis; the v3 correction story in two plots.

Q311 `fig01_availability_heatmap.png` is an 8-city $\times$ 12-month grid. The darkest (lowest-availability) cell is around:

- A) Bangkok, February B) Hanoi, June C) Manila, March D) Kuala Lumpur, November ($\sim 10.7\%$)

Answer: D. *Framework:* `ASEAN_CLIMATE_DATA`; `P_cloud = 0.893` there.

Q312 In `fig01`, the brightest (highest-availability) cell is:

- A) Bangkok, February ($\sim 96.5\%$) B) Singapore, December C) Kuala Lumpur, November D) Jakarta, January

Answer: A. *Framework:* `P_cloud = 0.035`; the two extremes anchor the "10.7%–96.5%" range.

Q313 `fig02a_diurnal_cloud_heatmap.png` (month $\times$ local-hour) shows cloud cover is lowest around:

- A) Afternoon B) Mid-morning (9–12h local) C) Midnight D) It is uniform

Answer: B. *Framework:* `load_hourly_climatology`; nocturnal stratus dissipates mid-morning.

Q314 `fig02b_diurnal_window.png` overlays cloud and rain diurnal cycles to identify the best FSO window as:

- A) Afternoon B) Mid-morning ($\sim 05:00\text{--}10:00$), in daylight C) Overnight D) There is none

Answer: B. *Framework:* The figure that visualises the day/night tension.

Q315 fig02b shows rain peaking in the:

- A) Morning B) Midnight C) Afternoon (13–17h), out of phase with the cloud minimum D) Same time as cloud

Answer: C. *Framework:* load_hourly_rain_climatology; convective afternoon rain, 8/8 cities.

Q316 fig03a_correlation_matrix.png is an 8×8 heatmap of:

- A) Elevation B) SKR C) Distances D) Pearson correlation of daily cloud cover between cities

Answer: D. *Framework:* correlation_matrix; high correlation limits site-diversity benefit.

Q317 fig03b_joint_availability_dang2023.png compares the project's joint-clear probability against:

- A) ERA5 raw B) Micius C) Dang et al. 2023 (81.92 / 96.86 / 99.44%) D) Starlink

Answer: C. *Framework:* joint_clear_probability; a validation-against-literature figure.

Q318 diagram_04_weather_model.png shows the three states:

- A) Sunny/windy/dry B) Hot/cold/mild C) Clear (attenuate) / Rain (attenuate more) / Cloud (link OFF) D) Day/night/dawn

Answer: C. *Framework:* The three-state model; cloud is binary outage.

Q319 figA1_cloud_threshold_sensitivity.png demonstrates that the headline conclusions are:

- A) Highly sensitive to the 85% cloud threshold B) Undefined C) Robust to reasonable changes in the cloud-outage threshold D) Dependent on rain only

Answer: C. *Framework:* Sensitivity analysis; guards against a threshold artefact.

Q320 Reading fig01 + fig02b together, the meteorological headline is:

- A) All cities are identical B) Weather is negligible C) Rain dominates D) Cloud governs availability (10.7%–96.5% by city/month) AND the clearest window is daytime — creating the day/night tension

Answer: D. *Framework:* Cross-figure synthesis of the weather chapter.

Q321 fig05_citypair_feasibility_map.png colours the 28 city pairs into:

- A) Green = DUAL (14) and orange = SF/store-and-forward (14) B) Red/blue by distance C) All one colour D) By country

Answer: A. *Framework:* classify_pair; the geographic 14/14 split.

Q322 fig09a_asean_ground_tracks.png shows Starlink Shell-1 ground tracks over ASEAN as:

- A) Stationary dots B) Polar tracks C) Dense inclined (~53°) tracks with the 8 station markers D) A single track

Answer: C. *Framework:* SGP4 propagation of ~1,019 real satellites.

Q323 fig09a1_dual_example.png vs fig09a2_sf_example.png illustrate, respectively:

- A) Two SF cases B) Rain vs cloud C) A DUAL pair (one pass covers both) vs an SF pair (too far, needs relay) D) Day vs night

Answer: C. *Framework:* Concrete geometric examples of the two feasibility classes.

Q324 fig09b_elevation_vs_time.png plots elevation over time for passes; each triangular arc represents:

- A) A satellite orbit B) A day C) One pass (rise → peak → set), lasting ~204 s at ASEAN latitudes D) A city

Answer: C. *Framework:* extract_passes; the arc peak is the max-elevation instant.

Q325 fig04b_pass_hour_histogram.png shows the distribution of pass peak times across:

- A) City distance B) Elevation C) QBER D) Local hour of day

Answer: D. *Framework:* local_hour_peak; used to cross-check against the weather window.

Q326 fig09c_coverage_duration.png reports, per station, the:

- A) Coverage/visibility duration statistics (how long satellites are visible) B) SKR C) Rain total D) Correlation

Answer: A. *Framework:* Pass/coverage descriptive statistics.

Q327 05_Code_EDA/figures/starlink_altitude_inclination_clusters_...png uses PCA/clustering to show:

- A) A single blob B) Ground stations C) Random noise D) Distinct Starlink shells separated by altitude/inclination (Shell-1 as one cluster)

Answer: D. *Framework:* 03_starlink_altitude_pca_clustering.py; justifies filtering Shell-1 as a homogeneous set.

Q328 05_Code_EDA/figures/starlink_groundtrack_repeat_...png visualises:

- A) Rain B) City pairs C) QBER D) The ground-track repeat pattern of the constellation

Answer: D. *Framework:* EDA Layer 1; ground-track geometry.

Q329 05_Code_EDA/figures/starlink_coverage_snapshot_...png is a single-instant coverage view, which risks:

- A) Wrong wavelength B) Nothing C) Snapshot bias — one instant may not represent the full orbital cycle (an EDA Layer 6 caution)
D) Rain error

Answer: C. *Framework:* Data-quality lesson embedded in the EDA figure.

Q330 Reading fig05 + fig09a, the orbital-geometry headline is:

- A) Real 53° Shell-1 geometry splits the 28 pairs 14/14 into DUAL vs SF — a ceiling no scheduler can lift B) Orbits don't matter
C) All pairs are DUAL D) Coverage is continuous

Answer: A. *Framework:* Cross-figure synthesis of the geometry chapter.

Q331 fig16_fair_pairkey_comparison.png shows bars for ALG-0-feasible, ALG-1-blind, ALG-1-aware in dry and wet. The pattern is:

- A) All equal B) ALG-1-aware highest C) ALG-1-blind slightly above ALG-0 (+2.5%), ALG-1-aware slightly BELOW ALG-1-blind (weather-info costs) D) ALG-0 highest

Answer: C. *Framework:* 30_plot_fair_pairkey_comparison.py; the +2.47%/-0.29% story in one bar chart.

Q332 In fig16, the small arrow annotations between bars display:

- A) Elevation B) QBER C) The percentage gains (+2.47% matching, -0.29% weather-info) D) Rain

Answer: C. *Framework:* The figure labels the two isolated effects explicitly.

Q333 fig15_alg2_waterfall.png decomposes where network key is lost along the pair path. The largest drop is typically:

- A) Wavelength B) Rain C) The min()-bottleneck / matching + handover cost + fairness cost stages D) Orbit

Answer: C. *Framework:* pair_key_totals min(); the waterfall shows the structural ceiling.

Q334 fig14_hold_duration_distribution.png shows the distribution of segment/hold durations, with a median around:

- A) ~3.3 s (far shorter than the 30-s acquisition), motivating the multi-segment handover-cost fix B) 5 min C) 30 s D) 1 hour

Answer: A. *Framework:* _segments_with_handover_cost; the figure explains the ~2.7× undercount bug.

Q335 fig07_algorithm_comparison_cdf.png is a CDF of daily pair-key across days. Curves for the three strategies are:

- A) Identical to zero B) Very close together (small differences), consistent with the modest ~2.5% gains C) Crossing wildly D) Far apart

Answer: B. *Framework:* The CDF visualises how small the fair-comparison gaps are.

Q336 fig09d1_top_satellites_per_station.png and fig09d2_relay_hotspot_per_station.png were the subject of what earlier caution?

- A) Wrong weather B) Wrong orbit C) An N-vs-1 style comparison caution — check that sub-units (stations) are counted evenly before ranking D) None

Answer: C. *Framework:* The fair-comparison discipline extends to these per-station figures.

Q337 fig09e_top5_pair_exchanges.png highlights which city pairs exchange the most key, revealing:

- A) A few dominant pairs (hub effect), consistent with structural starvation of others B) Uniform distribution C) All pairs equal
D) No pattern

Answer: A. *Framework:* Hub stations concentrate exchanges; others starve.

Q338 figA2_mipgap_sensitivity.png validates that:

- A) The greedy heuristic is far from optimal B) Weather is wrong C) The greedy allocation is close to the proven MILP optimum (small gap), justifying heuristic use D) MILP is infeasible

Answer: C. *Framework:* 16_alg2_mipgap_sensitivity.py; optimality validation.

Q339 diagram_10_scheduling_pipeline.png shows the two-level flow:

- A) Orbit → rain B) Weather → SKR C) QBER → key D) Visibility → matching (Hungarian) → matched passes → pair-key allocation (greedy/ILP/maxmin)

Answer: D. *Framework:* The end-to-end scheduling architecture.

Q340 `diagram_09_ground_diversity_concept.png` conceptually argues that adding stations helps because:

- A) Geographically de-correlated weather means not all stations cloud over together — reducing starvation B) It raises QBER C) It has no effect D) It changes the orbit

Answer: A. *Framework:* Site-diversity concept; the capacity-planning lever.

Q341 Reading `fig16 + fig15`, the scheduling headline is:

- A) Weather-awareness is a big win B) Scheduling solves everything C) Coordination helps ~2.5%, weather-awareness slightly costs, and the min()-bottleneck plus starvation are structural — so more stations beat smarter scheduling D) Rain dominates

Answer: C. *Framework:* Cross-figure synthesis of the results chapter.

Q342 If a bar in `fig16` for ALG-1-aware were shown ABOVE ALG-1-blind, it would:

- A) Prove +678% B) Mean nothing C) Contradict the measured negative weather-info result — a red flag to recheck the figure's data source (v3 vs pre-v3) D) Confirm the thesis

Answer: C. *Framework:* Consistency check; the sign is a v3 signature (pre-v3 was positive).

Q343 `fig10_isl_relay_latency_heatmap.png` shows ISL relay latency between city pairs, converging to:

- A) One day B) Hundreds of minutes C) A few minutes (2.9–4.1 min across pairs), vs the ~711-min single-hop worst case D) Zero

Answer: C. *Framework:* `isl_relay`; the ISL upgrade collapses the single-hop latency.

Q344 `fig11_isl_graph_diagnostic.png` visualises the ISL mesh as:

- A) A tree B) A single line C) A "+Grid" graph connecting each satellite to plane and cross-plane neighbours D) Disconnected dots

Answer: C. *Framework:* `isl_topology.build_isl_graph`; BFS runs on this graph.

Q345 `fig12_powersplit_pareto.png` plots the Pareto frontier between:

- A) Orbit and weather B) Secret key rate (SKR) and classical data throughput over (m_k , m_a) C) Elevation and QBER D) Rain and cloud

Answer: B. *Framework:* `pareto_frontier`; non-dominated key-vs-data operating points.

Q346 Moving along the `fig12` Pareto frontier toward higher SKR means:

- A) Lower data throughput (you spend modulation budget on the key channel) B) Higher data throughput too C) Higher QBER only D) No change

Answer: A. *Framework:* The fundamental key-vs-data trade on a shared carrier.

Q347 `fig13_powersplit_adaptive_gain.png` shows the benefit of adaptive (per-pass) power-split over a fixed split as:

- A) A positive gain — allocating just enough m_k per pass frees power for data when the key demand is met B) Zero C) Negative always D) Random

Answer: A. *Framework:* `adaptive_split`; demand-driven allocation.

Q348 `diagram_05_satellite_routing.png` (renamed conceptually to "selection/scheduling") depicts:

- A) Pass selection / key allocation on a shared relay — deliberately NOT multi-hop network routing B) Rain routing C) Network-layer BGP routing D) Orbit selection

Answer: A. *Framework:* Dr. Minh's "selection, not routing" framing.

Q349 In `fig10`, the directionality ($i \rightarrow j \neq j \rightarrow i$) is most visible for pairs at:

- A) Zero distance B) The same city C) Different latitudes (ascending/descending heading changes visit order) D) The same latitude

Answer: C. *Framework:* `sf_latency_minutes` directionality (though ISL reduces its impact).

Q350 The capacity-optimal ISL relay (vs time-optimal), if plotted, would deliver:

- A) More key by choosing a higher-elevation drop-off within a latency budget B) Lower key C) The same key D) Zero key

Answer: A. *Framework:* `capacity_optimal_relay`; trades a little latency for a better-angled, higher-SKR pass.

Q351 Reading `fig10 + fig11`, the ISL headline is:

- A) ISL changes weather B) ISL raises QBER C) ISL is useless D) A real ISL mesh turns an intractable ~711-min single-hop relay into a few-minute multi-hop delivery, at a conservative 30-s/hop cost

Answer: D. *Framework:* Cross-figure synthesis of the ISL extension.

Q352 Video_Assets/prototype_8station_skyview.mp4 is intended to visualise:

- A) Orbits of GPS B) Rain radar C) The sky view from the 8 ground stations (satellites rising/setting), making pass geometry intuitive D) QBER curves

Answer: C. *Framework:* Satvis_Scenarios_and_Prototype.md; an explainer for pass dynamics.

Q353 prototype_dual_sf_isl_scenarios.mp4 and prototype_zoom_dual_sfisl.mp4 illustrate:

- A) Weather only B) Power-split C) QBER D) The DUAL vs store-and-forward vs ISL-relay scenarios visually

Answer: D. *Framework:* Visual counterpart to fig05/fig09a1/fig09a2/fig10.

Q354 prototype_footprint_dome_beam.mp4 shows:

- A) The satellite coverage footprint / beam dome geometry (link_geometry made visual) B) QBER C) Statistics D) Rain

Answer: A. *Framework:* Visualises R_cov and the footprint the mask defines.

Q355 prototype_pass_formula_demo.mp4 is a visual demo of:

- A) Pass extraction (rise/peak/set, elevation over time) — the geometry behind extract_passes B) Rain C) The SKR formula D) The Hungarian algorithm

Answer: A. *Framework:* Animates the pass-analysis concept.

Q356 The Earth textures in Video_Assets/earth_textures_solarsystemscope/ are licensed:

- A) Public domain with no attribution B) Proprietary, no reuse C) Secret D) CC BY 4.0 — usable including commercially, with attribution to Solar System Scope

Answer: D. *Framework:* README_Sources_and_Licenses.md; attribution required.

Q357 The NASA 3D satellite models in Video_Assets/nasa_3d_satellite_models/ are:

- A) CC BY-NC B) Copyrighted C) Public domain (NASA "free and without copyright") D) Paid

Answer: C. *Framework:* README_Sources_and_Licenses.md; e.g. the LLCD laser-comm model is closest to the FSO/SIKD concept.

Q358 Why does licensing of these video assets matter for the thesis?

- A) Public submission/sharing requires correct attribution (CC BY) or public-domain status — provenance extends to media, not just data B) It affects the orbit C) It changes QBER D) It never matters

Answer: A. *Framework:* Data-quality/provenance discipline applied to media.

Q359 The most concept-appropriate NASA model to represent this system's optical link is:

- A) The Lunar Laser Communications Demonstration (LLCD) — a real laser-comm satellite B) A weather balloon C) A generic CubeSat D) A GPS satellite

Answer: A. *Framework:* README_Sources_and_Licenses.md; LLCD is the nearest real analogue to FSO/SIKD.

Q360 Across all figures and videos, the single visual takeaway of the project is:

- A) Rain is the enemy B) One clean link works C) Satellites are enough D) Under real weather (heatmaps), real orbits (ground tracks/footprints), and fair scheduling (bar/waterfall), the usable network key is cloud-limited and station-count-limited — not fixed by cleverer algorithms

Answer: D. *Framework:* The whole visual narrative in one line — the same conclusion as Q300, seen through the figures.

Part IV

Thesis Annexes (transcribed)

11 Reproducibility and code structure

Every numeric result traces to a numbered script writing two artifacts per run: a `verify` file (one key number per line) and a `provenance` file (exact parameters, input hashes, formulas). The codebase (`05_Code_v3/`) holds 13 standalone modules, 47 numbered analysis scripts, 13 test files (325 tests, all passing), and four data subdirectories (`raw/`, `intermediate/`, `verify/` — 35 files, `provenance/`). A single orchestrator `scripts/run_all.py` runs every step in dependency order and cross-checks every `verify` number against the compiled text so the two cannot silently drift. The slowest step is the 620-day full-enumeration comparison (12–45 min); everything

else is seconds to minutes.

11.1 Core simulation modules

Module	Responsibility
<code>orbital_mechanics.py</code>	TLE parsing, SGP4 propagation, ground-station coordinates, visible-satellite queries
<code>channel_model.py</code>	Geometric spreading h_g , atmospheric attenuation h_l , turbulence σ_X^2
<code>weather_model.py</code>	Three-state model; $P_{\text{cloud}}/f_{\text{rain}}/V/R$ by city and month
<code>weather_stats.py</code>	Hourly climatology, diurnal decomposition, inter-city correlation
<code>sikd_performance.py</code>	Noise model, DT/DD thresholds, $P_{\text{sift}}/\text{QBER}$ (Gauss–Hermite), SKR
<code>schedulers.py</code>	ALG-0/1/2, Hungarian matching, greedy and MILP pair allocation
<code>routing.py</code>	Orchestration: per-pass SKR + scheduling pipeline
<code>link_geometry.py</code>	Coverage radius, slant range, symmetric dual-downlink feasibility
<code>citypair_feasibility.py</code>	Directed single-hop store-and-forward latency
<code>isl_topology.py</code>	ISL mesh graph (RAAN plane-clustering, “+Grid” 4-neighbor)
<code>isl_relay.py</code>	Multi-hop relay: time-optimal (BFS) and capacity-optimal routing
<code>sikd_powersplit.py</code>	Adaptive m_K/m_D power-split, Pareto frontier, real-pass gain

12 Extended SIKD signal model

Why subcarrier multiplexing. One optical carrier has far more bandwidth than either a low-rate key channel or a high-rate data channel needs alone; SCM assigns them distinct RF subcarriers f_1 (key) and $f_2 \gg f_1$ (data) before intensity-modulating the same carrier. The transmitted intensity is $P_s(t) = P_T[1 + m_K s_K(t) + m_D s_D(t)]$ with $m_K \ll m_D$: the quantum channel only needs enough depth to sift a key near the noise floor, while the data channel is provisioned for throughput. Direct detection recovers exactly the photocurrent $i_r(t)$; an RF bandpass filter at f_1 isolates the key subcarrier before the dual-threshold stage.

Why the classical channel is not free. A bandpass filter at f_1 has finite stopband rejection I_{so} against energy at f_2 , so a fraction of the data power reaches the key receiver as cross-talk $\sigma_{\text{CT}}^2 = (R_e P_T m_D h_g h_l)^2 10^{-I_{\text{so}}/10}/8$, growing with the *square* of received power. Raising data throughput (m_D) therefore directly raises the key channel’s QBER — the mechanism by which cross-talk dominates precisely when the link is otherwise strongest. The noise model, dual-threshold rule, sift probability and QBER are all reproducible from Vu et al. (2022); only the system-level choice to multiplex a classical payload alongside QKD on one carrier rests on the internal manuscript (Vu, 2025).

13 Numerical results: full robustness detail

13.1 Inter-city correlation

Site diversity helps only if weather states are not perfectly correlated. Same-monsoon neighbours correlate positively (Hanoi–Da Nang $r = 0.499$ in July; Singapore–Kuala Lumpur $r = 0.754$ dry, 0.689 wet — the strongest). Cities on opposite sides of the monsoon reversal can correlate *negatively* (Hanoi–Jakarta $r = -0.231$ in July), which is more valuable for diversity than a weak positive correlation.

13.2 Joint availability vs. the closest prior model

Sites	This work	Nguyen et al. (2023)	Difference
Hanoi alone	53.2%	81.9%	−28.7 pp
+ Da Nang	78.2%	96.9%	−18.7 pp
+ Ho Chi Minh City	85.6%	99.4%	−13.8 pp

The absolute gap comes largely from the binary 85% cloud threshold vs. a continuous Gamma-attenuation model. Sweeping the threshold narrows but does not close the gap (1-site gap: 45.3 pp at 70%, 34.3 at 80%, 28.7 at 85%, 22.1 at 90%), an honest, quantified limitation of the binary-outage approach.

13.3 Statistical significance and autocorrelation

Paired significance at the final $m_K = 0.15$ operating point (raw network key, 310 days/season):

Comparison	Season	Mean	Std.	95% CI (bootstrap)
Matching gain	Dry	−6.658%	3.195 pp	[−6.799, −6.514]
Matching gain	Wet	−7.983%	3.944 pp	[−8.133, −7.827]
Weather-info	Dry	−1.400%	0.557 pp	[−1.426, −1.374]
Weather-info	Wet	−1.310%	0.674 pp	[−1.341, −1.278]

A paired t -test rejects zero gain at $p < 10^{-106}$; a paired bootstrap (10,000 resamples) gives CIs excluding zero. These are *raw-network-key* figures (handover charged directly against total throughput), which differ in sign from the fair pair-key metric — a genuine difference in what each metric measures. Lag-1 autocorrelation of the daily gain series (+0.453/ +0.185 matching dry/wet; +0.114/ +0.213 weather-info) exceeds the ± 0.113 white-noise band, so a 7-day moving-block and a year-cluster bootstrap were run; they widen intervals by ~ 15 –60% but all four still exclude zero. The $p < 10^{-106}$ should be read as overstated (it assumes 310 independent days), but the qualitative claim — both gains real, non-zero, correctly signed — survives the most conservative check.

13.4 Handover-cost characterization

Algorithm	Handovers/day		Key lost	
	Dry	Wet	Dry	Wet
ALG-0	5,695	5,695	28.6%	28.6%
ALG-1-blind	6,176	6,176	31.4%	31.8%
ALG-1-aware	6,375	6,368	32.3%	32.5%

Visibility segments have median duration 3.3 s, far shorter than a 30 s handover, so a handover’s delay spans several segments; carry-over accounting charges the full delay across all of them. Weather-awareness raises handovers (6,176 $\rightarrow$ 6,375/day), the mechanism behind its small net cost.

13.5 ALG-2 optimality benchmark

Quantity	Value
Days proven optimal within 60 s / 2%	30 of 30
Mean MILP solve time	2.3 s
Mean gap, greedy vs. optimum	1.03%
Worst-case gap	4.56%
Starved pair-instances (greedy / MILP / max-min)	373 / 642 / 332 of 840

Greedy is within 1.03% of the proven MILP optimum but leaves *fewer* pairs starved, because exact throughput-maximization concentrates key into already-well-served pairs. The three algorithms line up exactly as their objectives predict (642 MILP-exact, 373 greedy, 332 max-min starved), so the reported pair-starvation is a conservative estimate, not a heuristic artifact.

13.6 Development history of the weather-awareness estimate

Methodology	Reported gain
Hand-estimated weather, synthetic TLE	+25 to +40%
Real ERA5, synthetic TLE	+52 to +56%
Real ERA5, real TLE, “sticky” baseline	+86 to +109% (artifact)
Real ERA5, real TLE, per-step baseline, 1 station	+0% (identity)
Naive multi-station vs. single-station	+85 to +678% (retracted)
Fair, fixed-network-size, pair-key metric	−1.20 to −0.78%
Corrected SKR formula, $m_K = 0.05$	sign flips
Corrected SKR and $m_K = 0.15$	+1.92/ + 1.96% match; −1.35/ − 1.21% w-info
+ geometric (ν_R, spherical) and cross-talk fixes (final)	+2.47/ + 2.48% match; −0.29/ − 0.23% w-info

The retracted +678% conflated two independent effects: the value of *having more stations* (raises throughput almost regardless of policy) and the value of *weather-aware scheduling*. The general lesson, applied throughout: before ranking groups, verify they are equivalent in every dimension except the one under test — including which of a scheduler’s own capabilities each side may use.

13.7 Physical-layer corrections

SKR formula. The printed $\text{SKR}_{\text{norm}} = P_{\text{sift}}[1 - H_2(\text{QBER}) - \chi_E]$ omitted the error-correction leakage and reached zero only at $\text{QBER} = 50\%$, inconsistent with the asserted 11% threshold; the corrected $P_{\text{sift}}[1 - 2H_2(\text{QBER}) - \chi_E]$ (ideal reconciliation) vanishes at $\approx 11\%$. **Shot noise** previously scaled with m_K rather than the total DC current (negligible, since cross-talk dominates by ~ 3 orders). **Rytov coefficient** $0.56 \rightarrow 2.25$ (fourfold scintillation, still weak-regime). m_K $0.05 \rightarrow 0.15$: the source manuscript’s own feasibility sweep shows the low-QBER region opens only for $m_K \gtrsim 0.1$ at $I_{\text{so}} = 15$ dB; raising it recovers the full 30–90° range and the tens-of-Mbps SKR. All corrections are in 05_Code_v3; see IMPACT_EVALUATION.md for the before/after list.

14 Equatorial-season robustness and publications

Equatorial robustness. Availability was re-checked for three near-equatorial cities (Singapore, Kuala Lumpur, Jakarta) across January/April/July/October; the “cloud-not-rain” conclusion and the 10.7–96.5% span hold across the seasonal cycle, confirming the finding is not an artifact of the two proxy months.

Publication plan. The work targets (i) a conference paper (GLOBECOM-class workshop) on reliability and service continuity for secure NTN under tropical weather, and (ii) a journal extension on the joint key–data trade-off (Pareto, adaptive power-split, scheduling), with the corrected v3 numbers as the canonical results.

Sources

Curated formulas and derivations: 05_Code_v3/modules/ and Tu-hoc/SIKD_Satellite_QKD/Study_Materials/. Thesis annexes: 10_Thesis_Shorten_v2/sections/annex_a..e. Question bank and glossary generated from symbols.json, exercises.json, questions.json. Live site: <https://tatcataitn.github.io/for-ttn-USTH/Thesis-M1>.